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ADENO-ASSOCIATED VIRUS COMPOSITIONS FOR THE TREATMENT OF DUCHENNE MUSCULAR DYSTROPHY

Abstract

The present invention provides novel gene therapy compositions for use in treatment of Duchenne Muscular Dystrophy (DMD).

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Background/Summary

CLAIM OF PRIORITY [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/561,130, filed Mar. 4, 2024; U.S. Provisional Patent Application No. 63/666,899, filed Jul. 2, 2024; U.S. Provisional Patent Application No. 63/676,780, filed Jul. 29, 2024; and U.S. Provisional Patent Application No. 63/688,024, filed Aug. 28, 2024; each of which is incorporated by reference herein in its entirety.

SEQUENCE LISTING

[0002] This instant application contains a Sequence Listing which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. The XML copy, created on Mar. 3, 2025, is named PAT059900-US-NP_SL.xml and is 2,260,685 bytes in size.

FIELD OF DISCLOSURE

[0003] This disclosure relates to compositions for the treatment of Duchenne Muscular Dystrophy (DMD).

BACKGROUND

[0004] Duchenne muscular dystrophy (DMD) is a severe type of muscular dystrophy, a group of rare genetic neuromuscular diseases, affecting approximately one in 3,600 to 9,200 live male births. DMD is the most common type of muscular dystrophy predominantly affecting men.

[0005] DMD causes progressive muscle weakness due to muscle fiber disarray, replacement with connective tissue or fat, and death. Voluntary muscles are affected first, especially those of the hips, pelvic area, thighs, calves, eventually progressing to the shoulders and neck, followed by arms, respiratory muscles, and other areas. Symptoms usually appear before age five, expressing as difficulty with motor skills including walking. As a result, falls may be frequent with walking becoming increasingly difficult and frequently impossible before age 13. Most men affected with DMD become essentially paralyzed from the neck down by the age of 21. Cardiomyopathy, particularly dilated cardiomyopathy, is also commonly seen in half of 18-year-olds with DMD. In late stages of the disease, respiratory impairment and swallowing impairment can occur. As a result, median life expectancy for men suffering from DMD is just 28-30 years, however comprehensive care may allow those suffering to live into their 30s or 40s.

[0006] DMD is caused by mutations and/or deletions in any of the 79 exons encoding the large dystrophin protein. While mutations causing DMD can affect any exon, exons 2-20 and 45-55 are common hotspots for large deletion and duplication mutations. The disorder follows an X-linked recessive inheritance pattern, with approximately two-thirds of cases inherited from the mother and one-third resulting from a new mutation.

[0007] Typical therapies for DMD are limited to management strategies, such as physical therapy, braces, and corrective surgery to alleviate symptoms. Assisted ventilation may be required in those with weakness of breathing muscles.

[0008] Gene therapies have been proposed for the treatment of DMD, including delandistrogene moxeparvovec-rokl, marketed under the brand name ELEVIDYS® by Sarepta Therapeutics. However, expression of the microdystrophin protein in skeletal muscles of patients injected with DMD gene therapies has been low and not uniform.

[0009] As a result, there exists a need for improved DMD gene therapies.

SUMMARY

[0010] The present disclosure is directed to compositions (e.g., adeno-associated viral (AAV) compositions) and methods of use thereof for treating Duchenne Muscular Dystrophy (DMD).

[0011] Accordingly, aspects of the invention provide a formulation comprising an AAV particle comprising (e.g., encapsidating) a nucleic acid comprising a promoter and a sequence encoding microdystrophin.

[0012] Aspects of the invention provide a composition comprising an adeno-associated virus (AAV) particle comprising an engineered capsid protein comprising the amino acid sequence RGDR (SEQ ID NO: 2440). The capsid protein encapsidates a nucleic acid molecule (e.g., a cargo) comprising a promoter, a codon optimized microdystrophin sequence, for example a sequence having at least 95% sequence identity with a sequence selected from SEQ ID NO: 2405-2407. The nucleic acid may further comprise a dorsal root ganglia (DRG) de-targeting miRNA binding site. In aspects of the invention, the nucleic acid molecule may comprise a sequence having the having at least 95% sequence identity with a sequence selected from SEQ ID NO: 2405-2407 that encodes a codon optimized microdystrophin sequence.

[0013] The DRG de-targeting miRNA binding site may comprise a sequence selected from among SEQ ID NOs: 2399-2401.

[0014] The promoter may be a novel promoter that is exhibits specificity (e.g. high expression in) for muscle cells, for example myocytes. The promoter may comprise a sequence having at least 95% sequence identity a sequence selected from among SEQ ID NOs: 2402-2404.

[0015] For example, the nucleic acid molecule (e.g. cargo) may comprise, in order: [0016] a first inverted terminal repeats (ITR) region; [0017] a promoter; [0018] an intron; [0019] a sequence having at least 95% sequence identity with a sequence selected from any one of SEQ ID NO: 2405-2407; [0020] a first DRG de-targeting miRNA binding site and preferably, a second DRG de-targeting miRNA binding site; [0021] a poly-adenylation signal, preferably a shortened poly-adenylation signal; and [0022] a second ITR region.

[0023] The ITR regions may be AAV2 ITR regions derived from an AAV2.

[0024] For example, the full nucleic acid molecule may comprise a sequence selected from any one of SEQ ID NOs: 2412-2438.

[0025] Advantageously, when administered to a subject with a mutation in the DMD gene that results in low or a lack of dystrophin protein expression, the composition is effective at restoring a healthy phenotype.

[0026] The engineered capsid protein may advantageously exhibit decreased liver tropism and/or increased muscle tropism.

[0027] For example, the capsid protein may comprise an amino acid sequence selected from Tables 1a in hypervariable region IV (HVR IV) relative to wild type AAV9. The capsid protein may comprise substitutions at amino acids 451-455 relative to a wild type AAV9 vector capsid selected from column 1 of Table 1b and a 7-mer insert selected from column 2 of Table 1b inserted after amino acid 455 relative to a wild type AAV9 vector. The capsid protein may further comprise a deletion of G267 relative to wild type AAV9 capsid or equivalent position in another AAV capsid.

[0028] Accordingly, aspects of the invention further provide methods and uses of the compositions of the invention (for example in the preparation of a medicament) for treating a subject suffering from DMD. The methods and uses may comprise providing to a subject a composition of the invention as described above.

[0029] Aspects of the present disclosure provide a recombinant nucleic acid comprising a muscle specific promoter; a sequence encoding microdystrophin; and at least one dorsal root ganglia (DRG) de-targeting miRNA binding site.

[0030] In some embodiments, the muscle specific promoter comprises a CK8 promoter, a desmin promoter, a Mb promoter, a MCK promoter, a MHCK7 promoter, a skeletal muscle alpha actin promoter, or a TTNI2 promoter. In some embodiments, the promoter comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of SEQ ID NO: 2402. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of SEQ ID NO: 2403. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of SEQ ID NO: 2404.

[0031] In some embodiments, the sequence encoding microdystrophin is a sequence encoding human microdystrophin. In some embodiments, the sequence encoding microdystrophin encodes a microdystrophin protein comprising or consisting of an amino acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to SEQ ID NO: 2439. In some embodiments, the sequence encoding microdystrophin encodes a microdystrophin protein comprising or consisting of the amino acid sequence of SEQ ID NO: 2439.

[0032] In some embodiments, the sequence encoding microdystrophin comprises or consists of a codon optimized sequence encoding microdystrophin. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2405. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2406. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2407.

[0033] In some embodiments, the at least one DRG de-targeting miRNA binding site comprises a binding site for miR-338, miR-138, or miR-9. In some embodiments, the at least one DRG de-targeting miRNA binding site comprises a binding site for miR-338-3p, miR-138-5p, or miR-9-5p. In some embodiments, the at least one DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401. In some embodiments, the at least one DRG de-targeting miRNA binding site comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401.

[0034] In some embodiments, the at least one DRG de-targeting miRNA binding site comprises a first and a second DRG de-targeting miRNA binding site. In some embodiments, each of the first and the second DRG de-targeting miRNA binding sites comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401. In some embodiments, each of the first and the second DRG de-targeting miRNA binding sites comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401. In some embodiments, each of the first and the second DRG de-targeting miRNA binding sites comprises or consists of the nucleic acid sequence of SEQ ID NO: 2399. In some embodiments, each of the first and the second DRG de-targeting miRNA binding sites comprises or consists of the nucleic acid sequence of SEQ ID NO: 2400. In some embodiments, each of the first and the second DRG de-targeting miRNA binding sites comprises or consists of the nucleic acid sequence of SEQ ID NO: 2401.

[0035] In some embodiments, the recombinant nucleic acid comprises, optionally from 5′ to 3′, a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401.

[0036] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0037] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0038] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0039] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0040] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0041] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0042] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0043] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0044] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0045] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0046] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0047] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0048] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.
[0049] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.
[0050] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.
[0051] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.
[0052] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.
[0053] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.
[0054] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic

acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0055] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0056] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0057] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0058] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0059] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0060] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least

92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0061] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0062] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0063] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0064] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0065] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0066] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0067] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0068] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0069] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0070] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0071] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic

acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0072] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0073] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0074] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0075] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0076] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0077] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0078] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0079] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0080] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0081] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0082] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0083] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0084] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0085] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0086] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0087] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0088] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0089] In some embodiments, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0090] In some embodiments, the recombinant nucleic acid further comprises an intron. In some embodiments, the intron comprises a simian virus 40 (SV40) intron sequence, a minute virus of mice (MVM) intron, a RK intron, a β-globin intron, and/or a chicken β-actin (CBA) intron, including functional variants or fragments thereof. In some embodiments, the intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409. In some embodiments, the intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409.

[0091] In some embodiments, the recombinant nucleic acid further comprises a polyadenylation (PolyA) signal. In some embodiments, the Poly A signal comprises a bovine growth hormone polyadenylation (bgh-Poly A) signal, a synthetic PolyA signal, or an SV40 Poly A signal. In some embodiments, the PolyA signal comprises the bgh-Poly A signal, and the bgh-Poly A signal comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410. In some embodiments, the PolyA signal comprises the bgh-Poly A signal, and the bgh-Poly A signal comprises or consists of the nucleic acid sequence of SEQ ID NO: 2410.

[0092] In some embodiments, the recombinant nucleic acid further comprises a 5′ inverted terminal repeat (ITR) and a 3′ ITR. In some embodiments, the 5′ ITR and/or the 3′ ITR is an adeno-associated virus 2 (AAV2) ITR, or a variant or fragment thereof.

[0093] In some embodiments, the 5′ ITR comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408. In some embodiments, the 5′ ITR comprises or consists of the nucleic acid sequence of SEQ ID NO: 2408.

[0094] In some embodiments, the 3′ ITR comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411. In some embodiments, the 3′ ITR comprises or consists of the nucleic acid sequence of SEQ ID NO: 2411.

[0095] In some embodiments, the recombinant nucleic acid comprises, optionally from 5′ to 3′, a 5′ ITR; a promoter; an intron; a sequence encoding microdystrophin; a first DRG de-targeting miRNA binding site; a second DRG de-targeting miRNA binding site; a PolyA signal; and a 3′ ITR.

[0096] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399;

a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0097] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO:

[0098] 2411.

[0099] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0100] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ

ID NO: 2402; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0101] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: [0102] 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0103] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least

88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0104] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO:

[0105] 2411.

[0106] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0107] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0108] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: [0109] 2402; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0110] In some embodiments, the recombinant nucleic acid comprises a 5′ comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least

97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0111] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0112] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0113] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0114] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0115] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least

90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0116] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO:

[0117] 2411.

[0118] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0119] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least

87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0120] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: [0121] 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0122] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0123] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO:

[0124] 2411.

[0125] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0126] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least

96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0127] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: [0128] 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0129] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0130] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO:

[0131] 2411.

[0132] In some embodiments, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0133] In some embodiments, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0134] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0135] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0136] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0137] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0138] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0139] In some embodiments, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0140] In some embodiments, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0141] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0142] In some embodiments, the recombinant nucleic acid comprises a 5′ comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0143] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0144] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0145] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0146] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0147] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0148] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0149] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0150] In some embodiments, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0151] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0152] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0153] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0154] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence of SEQ ID NO: 2411.

[0155] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0156] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0157] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0158] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0159] In some embodiments, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2412-2438. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2412-2438.

[0160] In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2412. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2413. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2414. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2415. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2416. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2417. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2418. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2419. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2420. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2421. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2422. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2423. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2424. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2425. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2426. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2427. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2428. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2429. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2430. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2431. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2432. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2433. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2434. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2435. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2436. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2437. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2438.

[0161] Aspects of the present disclosure provide a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of SEQ ID NO: 2402. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of SEQ ID NO: 2403. In some embodiments, the promoter comprises or consists of the nucleic acid sequence of SEQ ID NO: 2404. In some embodiments, the recombinant nucleic acid comprising the promoter exhibits enhanced expression in skeletal muscle cells and/or cardiac muscle cells compared to non-skeletal muscle cells and/or non-cardiac muscle cells.

[0162] Aspects of the present disclosure provide a recombinant nucleic acid comprising a codon optimized sequence encoding microdystrophin that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2405. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2406. In some embodiments, the codon optimized sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2407.

[0163] Aspects of the present disclosure provide an adeno-associated virus (AAV) particle comprising any one of the recombinant nucleic acids described herein and a capsid protein. In some embodiments, the capsid protein is derived from a serotype selected from AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV6.2, AAV7, AAV8, AAV9, AAV10, AAV11, AAV12, AAV13, AAVrh, AAVrh10, AAVrh74, AAV-DJ, AAV-DJ/8, Anc80, and AAV7m8, or a derivative, hybrid or chimeric serotype thereof.

[0164] In some embodiments, the AAV particle has increased muscle tropism compared to a wild type AAV9 particle. In some embodiments, the AAV particle has reduced liver tropism compared to a wild type AAV9 particle.

[0165] In some embodiments, the capsid protein comprises an amino acid sequence of formula (I): X.sub.1NX.sub.2X.sub.3X.sub.4RGDRX.sub.5X.sub.6L (formula I; SEQ ID NO: 1), wherein each of X.sub.1-X.sub.6 is any amino acid.

[0166] In some embodiments, the amino acid sequence of formula (I) is in a hypervariable region IV (HVR IV) relative to a wild type AAV9 capsid. In some embodiments, relative to a wild type AAV9 capsid, X.sub.1 is a substitution at amino acid 451, X.sub.2 is a substitution at amino acid 453, X.sub.3 is a substitution at amino acid 454, X.sub.4 is a substitution at amino acid 455, and RGDRX.sub.5X.sub.6L (SEQ ID NO: 2441) is inserted after amino acid 455. In some embodiments, X.sub.1 is A, I, F, G, H, L, M, Q, S, T, or V; X.sub.2 is A, G, S, T, or Y; X.sub.3 is S, N, G, or P; X.sub.4 is A, G, H, I, M, S, T, or V; X.sub.5 is A, G, or Q; and X.sub.6 is A, I, L, M, N, S, or Y. In some embodiments, the amino acid sequence of formula (I) comprises of any one of SEQ ID NOs: 801-1599. In some embodiments, the amino acid sequence of formula (I) comprises of any one of SEQ ID NOs: 1600-2398. In some embodiments, the amino acid sequence of formula (I) comprises or consists of any one of SEQ ID NOs: 2-800.
[0167] In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NOs: 191, 198, 199, 215, 256, 379, 544, 550, or 748. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 191. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 198. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 199. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 215. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 256. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 379. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 544. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 550. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 748.
[0168] In some embodiments, the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 191 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 198 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 199 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 215 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 256 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 379 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 544 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 550 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector. In some embodiments, the amino acid sequence of formula (I) comprises or consists of SEQ ID NO: 748 and the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector.
[0169] Aspects of the present disclosure provide a pharmaceutical composition comprising any of the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), or the AAV particles described herein, and a pharmaceutically acceptable carrier.
[0170] Aspects of the present disclosure provide a method of treating DMD in a subject in need thereof, the method comprising administering to the subject an effective amount of any of the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), the AAV particles, or the pharmaceutical compositions described herein.
[0171] In some embodiments, the subject is a human subject. In some embodiments, the subject has at least one mutation in the dystrophin gene.
[0172] In some embodiments, the administering comprises intramuscular administration, subcutaneous injection, intravenous injection, or intravenous (IV) administration.
[0173] Aspects of the present disclosure provide uses of any of the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), the AAV particles, or the pharmaceutical compositions described herein for the manufacture of a medicament for treating DMD.
[0174] Aspects of the present disclosure provide any of the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), the AAV particles, or the pharmaceutical compositions described herein for use in treating DMD.
[0175] Aspects of the present disclosure provide a kit comprising any of the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), the AAV particles, or the pharmaceutical compositions described herein. In some embodiments, the kit further comprises instructions for use in treating Duchenne muscular dystrophy (DMD) in a subject in need thereof.
[0176] Aspects of the nucleic acid molecules (e.g., cargo) and capsid proteins are described in further detail below.
[0177] For sequences disclosed throughout this application, it is understood that nucleic acid molecules and peptides may comprise one or more substitutions, for example conservative substitutions, that allow sequences to continue to function. Accordingly, sequences may have at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% sequence identity to the sequence disclosed.

[0178] A conservative substitution refers to amino acid substitutions that do not significantly affect or alter binding characteristics of a particular protein. Generally, conservative substitutions are ones in which a substituted amino acid residue is replaced with an amino acid residue having a similar side chain. For example, conservative substitutions may include a substitution found in one of the following groups: Group 1: Alanine (Ala or A), Glycine (Gly or G), Serine (Ser or S), Threonine (Thr or T); Group 2: Aspartic acid (Asp or D), Glutamic acid (Glu or Z); Group 3: Asparagine (Asn or N), Glutamine (Gln or Q); Group 4: Arginine (Arg or R), Lysine (Lys or K), Histidine (His or H); Group 5: Isoleucine (Ile or I), Leucine (Leu or L), Methionine (Met or M), Valine (Val or V); and Group 6: Phenylalanine (Phe or F), Tyrosine (Tyr or Y), Tryptophan (Trp or W). Additionally, or alternatively, amino acids can be grouped into conservative substitution groups by similar function, chemical structure, or composition (e.g., acidic, basic, aliphatic, aromatic, or sulfur-containing). For example, an aliphatic grouping may include, for purposes of substitution, Gly, Ala, Val, Leu, and Ile. Other conservative substitutions groups include sulfur-containing: Met and Cysteine (Cys or C); acidic: Asp, Glu, Asn, and Gln; small aliphatic, nonpolar, or slightly polar residues: Ala, Ser, Thr, Pro, and Gly; polar, negatively charged residues and their amides: Asp, Asn, Glu, and Gln; polar, positively charged residues: His, Arg, and Lys; large aliphatic, nonpolar residues: Met, Leu, Ile, Val, and Cys; and large aromatic residues: Phe, Tyr, and Trp.

Codon Optimized Microdystrophin

[0179] The present invention provides novel engineered nucleic acids encoding microdystrophin proteins for use in gene therapy. The microdystrophin proteins of the invention are optimized to provide increased expression in muscle cells. Advantageously, microdystrophin proteins of the invention are substantially shorter than dystrophin, allowing for optimized packaging in expressing cassettes, including viral vectors.

[0180] Microdystrophin refers to a biologically active fragment of dystrophin, e.g., ΔR4-R23/ΔCT microdystrophin. Microdystrophin can be from any source, e.g., a human microdystrophin. The microdystrophin can comprise or consist of an amino acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to SEQ ID NO: 2439. In some embodiments, microdystrophin comprises or consists of the amino acid sequence of SEQ ID NO: 2439.

[0181] Recombinant nucleic acids described herein can include a microdystrophin protein described herein or a variant thereof. For example, the recombinant nucleic acid comprises a microdystrophin comprising or consisting of an amino acid sequence of SEQ ID NO: 2439, or a variant thereof comprising 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid substitutions.

TABLE-US-00001 Microdystrophin protein sequence (SEQ ID NO: 2439)

MLWWEEVEDCYEREDVQKKTFTKWVNAQFSKFGKQHIENLFSDLQDGRRLLDLLEGLTGQKLPKEKGSTRVHALN
NVNKALRVLQNNNVDLVNIGSTDIVDGNHKLTLGLIWNIILHWQVKNVMKNIMAGLQQTNSEKILLSWVRQSTRN
YPQVNVINFTTSWSDGLALNALIHSHRPDLFDWNSVVCQQSATQRLEHAFNIARYQLGIEKLLDPEDVDTTYPDK
KSILMYITSLFQVLPQQVSIEAIQEVEMLPRPPKVTKEEHFQLHHQMHYSQQITVSLAQGYERTSSPKPRFKSYA
YTQAAYVTTSDPTRSPFPSQHLEAPEDKSFGSSLMESEVNLDRYQTALEEVLSWLLSAEDTLQAQGEISNDVEVV
KDQFHTHEGYMMDLTAHQGRVGNILQLGSKLIGTGKLSEDEETEVQEQMNLLNSRWECLRVASMEKQSNLHRVLM
DLQNQKLKELNDWLTKTEERTRKMEEEPLGPDLEDLKRQVQQHKVLQEDLEQEQVRVNSLTHMVVVVDESSGDHA
TAALEEQLKVLGDRWANICRWTEDRWVLLQDILLKWQRLTEEQCLFSAWLSEKEDAVNKIHTTGFKDQNEMLSSL
QKLAVLKADLEKKKQSMGKLYSLKQDLLSTLKNKSVTQKTEAWLDNFARCWDNLVQKLEKSTAQISQAVTTTQPS
LTQTTVMETVTTVTTREQILVKHAQEELPPPPPQKKRTLERLQELQEATDELDLKLRQAEVIKGSWQPVGDLLID
SLQDHLEKVKALRGEIAPLKENVSHVNDLARQLTTLGIQLSPYNLSTLEDLNTRWKLLQVAVEDRVRQLHEAHRD
FGPASQHELSTSVQGPWERAISPNKVPYYINHETQTTCWDHPKMTELYQSLADLNNVRESAYRTAMKLRRLQKAL
CLDLLSLSAACDALDQHNLKQNDQPMDILQIINCLTTIYDRLEQEHNNLVNVPLCVDMCLNWLLNVYDTGRTGRI
RVLSFKTGIISLCKAHLEDKYRYLFKQVASSTGFCDQRRLGLLLHDSIQIPRQLGEVASFGGSNIEPSVRSCFQF
ANNKPEIEAALFLDWMRLEPQSMVWLPVLHRVAAAETAKHQAKCNICKECPIIGFRYRSLKHFNYDICQSCFFSG
RVAKGHKMHYPMVEYCTPTTSGEDVRDFAKVLKNKERTKRYFAKHPRMGYLPVQTVLEGDNMETDTM

[0182] Aspects of the invention provide a composition comprising a nucleic acid molecule comprising a sequence having at least 95% sequence identity with a sequence selected from SEQ ID NOs: 2405-2407, provided below:

TABLE-US-00002 Name, SEQ ID NO Optimized sequence v1,

ATGCTGTGGTGGGAGGAAGTGGAGGACTGCTACGAGAGAGAGGACGTGCAGAAGAAGACCTTCACCA
SEQ ID
AGTGGGTGAACGCCCAGTTCAGCAAGTTCGGCAAGCAGCACATCGAGAACCTGTTCAGCGACCTGCA NO:
GGACGGCAGAAGACTGCTGGACCTGCTGGAGGGCCTGACCGGCCAGAAGCTGCCTAAGGAGAAGGGC 2405
AGCACCAGAGTGCACGCTCTCAACAACGTGAACAAGGCCCTGAGAGTGCTGCAGAACAACAACGTGG
ACCTTGTGAACATCGGAAGCACCGACATCGTGGATGGCAACCACAAGCTGACCCTGGGACTGATCTG
GAACATCATCCTGCACTGGCAGGTGAAGAACGTCATGAAGAACATCATGGCCGGCCTGCAGCAGACC
AACAGCGAAAAGATCCTGCTGAGCTGGGTCAGACAGAGCACCAGAAACTACCCTCAGGTGAACGTGA
TCAACTTCACCACCAGCTGGAGCGACGGCCTGGCCCTGAACGCCCTGATCCACAGCCACAGACCCGA
CCTGTTCGACTGGAACAGCGTGGTGTGCCAGCAGAGCGCCACCCAGAGACTGGAGCACGCCTTCAAC
ATCGCCAGATACCAGCTGGGCATCGAGAAGCTGCTGGACCCCGAGGACGTGGATACCACCTACCCCG
ACAAGAAGAGCATCCTGATGTACATCACCAGCCTGTTCCAGGTGCTGCCCCAGCAAGTGAGCATCGA
GGCCATCCAGGAGGTGGAGATGCTGCCCAGACCTCCAAAGGTGACCAAGGAGGAGCACTTCCAGCTG

CACCACCAGATGCACTACAGCCAGCAGATCACCGTGAGCCTGGCCCAGGGCTACGAGAGAACCAGCA
GCCCCAAGCCCAGATTCAAGAGCTACGCCTACACCCAGGCCGCCTACGTGACCACCAGCGACCCCAC
CAGAAGCCCCTTCCCCAGCCAGCACCTGGAGGCCCCCGAGGATAAGAGCTTCGGCAGCAGCCTGATG
GAGAGCGAGGTGAACTTGGACAGATACCAGACCGCCCTGGAGGAGGTGCTGAGCTGGCTGCTGAGCG
CCGAGGACACCCTTCAGGCACAGGGCGAGATCAGCAACGACGTGGAGGTGGTGAAGGACCAGTTCCA
CACCCACGAGGGCTACATGATGGACCTGACCGCCCACCAGGGCAGAGTGGGTAACATCCTCCAGCTG
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AGAACCAGAAAGATGGAGGAGGAGCCCCTGGGCCCGGACCTGGAGGACCTGAAGCGGCAGGTGCAGC
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TGTGAACAAGATCCACACCACCGGCTTCAAGGACCAGAACGAGATGCTGAGCAGCCTGCAGAAGCTG
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ACCTGCTGAGCACCCTGAAGAACAAGAGCGTGACCCAGAAGACCGAGGCCTGGCTGGACAACTTCGC
CAGATGCTGGGACAACCTGGTTCAGAAGCTGGAGAAGAGCACCGCCCAGATCAGCCAGGCCGTGACC
ACCACCCAGCCCAGCCTGACCCAGACCACCGTGATGGAGACCGTGACCACCGTGACCACCAGAGAGC
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CATCATCAGCCTGTGCAAGGCCCACCTGGAGGACAAGTACAGATACTTATTCAAGCAGGTGGCCAGC
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GAACCAAGAGATACTTCGCTAAGCACCCCAGAATGGGCTACCTGCCCGTGCAGACCGTGCTGGAGGG
CGATAACATGGAGACCGACACCATGTGA v2,
ATGCTGTGGTGGGAAGAGGTCGAGGACTGCTACGAGCGCGAAGATGTGCAGAAGAAAACTTTCACTA SEQ
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CGCACCAGGAAGATGGAGGAGGAGCCACTGGGCCCCGATCTGGAGGACCTGAAAAGACAGGTGCAGC
AGCATAAAGTGCTGCAGGAGGACCTGGAGCAGGAACAAGTGAGAGTGAATTCTCTGACACACATGGT
GGTGGTGGTGGACGAGAGCAGCGGCGACCACGCAACAGCCGCCCTGGAGGAGCAGCTGAAAGTGCTG
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CGTGAATAAGATCCACACCACCGGGTTCAAGGACCAGAATGAGATGCTGTCCAGCCTGCAGAAGCTG
GCCGTGCTGAAGGCCGATCTGGAAAAGAAGAAGCAGAGCATGGGAAAGCTGTATAGCCTGAAGCAGG
ACCTGCTGTCTACTCTGAAAAACAAGTCAGTGACCCAGAAGACCGAGGCCTGGCTGGACAACTTCGC
CCGCTGCTGGGACAACCTGGTGCAGAAGCTGGAAAAGAGCACCGCCCAGATCAGCCAGGCAGTGACT
ACCACACAGCCCTCCCTGACACAGACTACTGTGATGGAGACCGTGACAACAGTGACCACAAGGGAGC
AGATCCTGGTGAAGCACGCCCAGGAAGAACTGCCCCCTCCACCCCCTCAGAAGAAGAGAACTCTGGA
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AAGCCCTGCGGGGCGAGATCGCCCCTCTGAAGGAGAACGTCAGCCATGTGAATGACCTGGCCAGACA
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CAAGGTGCCATACTACATCAACCACGAGACCCAGACCACCTGTTGGGATCACCCCAAGATGACAGAG
CTGTACCAGTCTCTGGCTGACCTGAACAACGTGCGGTTCTCAGCCTACAGGACCGCCATGAAGCTGC
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AGCACCGGCTTTTGCGACCAGCGCAGGCTGGGCCTGCTGCTCCACGACTCCATCCAGATCCCTAGAC
AGCTGGGCGAGGTGGCTTCTTTTGGCGGATCCAATATCGAGCCCAGTGTGCGGAGCTGTTTTCAGTT
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GGACAAAGAGATACTTTGCAAAACACCCAAGAATGGGCTACCTGCCAGTGCAGACCGTGCTGGAGGG
GGACAATATGGAGACAGACACAATGTGA

[0183] In aspects of the invention, the nucleic acid molecule comprises a sequence having the sequence identity of SEQ ID NO: 2405, SEQ ID NO: 2406, or SEQ ID NO: 2407. In aspects of the invention, the nucleic acid molecule may comprise a sequence having at least 85%, 86%, 87%, 88% 89%, 90%, 91%, 92%, 93%, 94%, or 95% of the sequence identity of SEQ ID NO: 2405, SEQ ID NO: 2406, or SEQ ID NO: 2407.

[0184] Advantageously, microdystrophin proteins are substantially shorter than dystrophin, allowing for optimized packaging in expressing cassettes, including viral vectors.

[0185] Typical AAV vectors have a packaging capacity of ˜4.7 kb. Nucleic acids are almost universally packaged with two terminal ITRs, each about 0.14 kb in length. Additionally regulatory elements, for example promoter, intron, and poly (A) elements, may be about 0.6-1 kb. As a result, the maximal allowable cDNA length must be less than 3.8 kb. Gene therapies are restricted by the packaging capacity limit of adeno-associated viral (AAV) vectors. As a consequence, the size of the native dystrophin coding sequence prevents packaging of the transgene into traditional AAV serotypes, including AAV9.

[0186] Accordingly, in aspects of the invention, the nucleic acid molecule of the present invention may be suitably packaged in a vector.

[0187] For example, the vector may be an adeno-associated viral (AAV) vector. The nucleic acid molecule may comprise in order an inverted terminal repeats (ITR) region, a promoter, an intron, the sequence having at least 95% sequence identity with a sequence selected from SEQ ID NOs: 2405-2407, a poly adenylation signal, and an ITR region.

[0188] The AAV vector may comprise a capsid protein having at least one modification that results in reduced liver-tropism of the AAV vector and/or preferential targeting of the AAV vector to muscle tissue.

[0189] Advantageously, when administered to a subject with a mutation in the DMD gene, resulting in lack of dystrophin protein expression, the composition is effective at restoring a healthy phenotype.

[0190] Further advantageously, the microdystrophin protein expressed by the nucleic acid molecule may be smaller than 137.5 kilodaltons (for example, 137.32 kilodaltons). Aspects of the invention also provide methods of treating subjects suffering from DMD using compositions of the invention.

[0191] Methods of the invention comprise providing to the subject compositions of the invention described above, comprising a nucleic acid molecule comprising a sequence having at least 95% sequence identity with a sequence selected from SEQ ID NOs: 2405-2407.

DRG De-Targeting miRNAs

[0192] The present invention provides recombinant nucleic acids and recombinant expression vectors (e.g., AAV vectors) that encode a transgene (e.g., a microdystrophin) and the binding site(s) for specific micro RNAs (miRs) that direct the miRs to selectively inhibit expression of the same transgene (e.g., the microdystrophin). By the present invention, it was discovered that miR-338-3p, miR-138-5p, and miR-9-5p are highly expressed in dorsal root ganglial (DRG) cells and not expressed, or expressed at substantially low levels, in muscle cells (including skeletal and heart muscle cells).

[0193] Accordingly, aspects of the invention provide a recombinant nucleic acid and an engineered vector comprising a first nucleic acid sequence encoding a transgene (e.g., sequence encoding a microdystrophin) and a second nucleic acid sequence encoding for the binding site(s) of one or more miRs selected from among miR-338-3p, miR-138-5p, or miR-9-5p. In such instances, the second nucleic acid sequence can be referred to as DRG de-targeting miRNA binding site. Thus, as used herein, the term “DRG de-targeting binding site” refers to a nucleic acid sequence encoding for a binding site of a miR (e.g., miR-338-3p, miR-138-5p, and miR-9-5p) that is highly expressed in DRG cells but not expressed, or expressed at significantly lower levels, in muscle cells. The nucleic acid sequence encoding a binding site of a miR is sufficiently complementary with the nucleic acid sequence encoding the miR to effect base pairing (binding) between the DRG de-targeting binding site and the miR. Advantageously, the second nucleic acid sequence directs the miR, when present, to repress expression of the transgene. As a result, the selected miRs (miR-338-3p, miR-138-5p, miR-9-5p) in cells transduced by the vectors of the invention will be directed to inhibit expression of the transgene.

[0194] For example, the transgene (e.g., microdystrophin) may provide a therapeutic effect when expressed in skeletal and/or heart muscle but provide a toxic effect when expressed in DRG cells. Without being bound to a mechanism of action, transduction of both muscle cells and DRG cells by vectors of the invention results in transcription of both transgene mRNA and the binding site(s) of the select miR. In DRG cells with substantial endogenous expression of miR-338-3p, miR-138-5p, and miR-9-5p, transgene expression (e.g., microdystrophin expression) will be substantially repressed. In muscle cells with limited to no expression of miR-338-3p, miR-138-5p, miR-9-5p, transgene expression will not be substantially repressed. As a result, the therapeutic effect of the transgene will continue in muscle cells and the toxic effect in DRG cells will be abated.

[0195] In aspects of the invention, the second nucleic acid sequence may be operably linked to the first nucleic acid sequence, for example, the second nucleic acid sequence may be operably linked to the 3′ or 5′ end of the first nucleic acid sequence. Advantageously, this may result in increased transcription of both nucleic acid sequences.

[0196] The engineered vector may be a viral vector. For example, the vector may be an adeno-associated viral (AAV) vector. Further advantageously, the engineered vector may comprise a capsid protein with modified tropism for skeletal and/or heart muscle in comparison with the wild type AAV vector. For example, transduction and expression of the transgene may be increased in muscle tissue while limited expression of the transgene in off-target tissue types, for example, DRG may be observed.

[0197] The first nucleic acid sequence and second nucleic acid sequence may also be operably linked to tissue specific promoters. Advantageously, the first nucleic acid sequence may be operably linked to a muscle specific promoter. In aspects of the invention, the second nucleic acid is 3′ or 5′ of the first nucleic acid. For example, the second nucleic acid may be incorporated into a 3′ or 5′ untranslated region (UTR) of the transgene mRNA when transcribed.

[0198] In muscle tissue, where miR-338-3p, miR-138-5p, miR-9-5p expression is limited, the muscle specific promoter promotes transcription of the transgene and the miR binding site has limited to no effect due to the absence of the requisite miR. In cells expressing miR-338-3p, miR-138-5p, miR-9-5p, for example DRG, the muscle specific promoter limits transcription of the transgene while the miR binding site is still present in the transcript, inhibiting translation of any transgene mRNA transcribed.

[0199] Aspects of the invention also provide methods and uses of the recombinant nucleic acids and the recombinant expression vectors of the invention, for example, in therapeutic compositions. Accordingly, aspects of the invention provide a method or use of a recombinant nucleic acid or a recombinant expression vector, the method or use comprising providing to a subject a recombinant nucleic acid or a recombinant expression vector (e.g., an engineered vector (e.g., an AAV vector)) comprising a first nucleic acid sequence encoding a transgene (e.g., sequence encoding a microdystrophin protein) and a second nucleic acid sequence expressing the binding site of a miR selected from among miR-338-3p, miR-138-5p, or miR-9-5p. The second nucleic acid sequence thereby directs the miR, when present (for example endogenously), to repress expression of the transgene.

[0200] The transgene (e.g., microdystrophin) may be expressed in heart and/or skeletal muscle, thereby providing a therapeutic effect. The second nucleic acid sequence may be expressed in dorsal root ganglia cells and direct endogenous miR-338-3p, miR-138-5p, and/or miR-9-5p to repress expression of the transgene in dorsal root ganglia cells. For example, the transgene may provide a therapeutic effect when expressed in skeletal and/or heart muscle but provide a toxic effect when expressed in DRG cells.

[0201] Accordingly, in aspects of the invention, the DRG de-targeting miRNA may comprise a sequence have at least 90% or at least 95% of the sequence identity of one or more of:

TABLE-US-00003 (SEQ ID NO: 2399; miR-338-3p) CAACAAAATCACTGATGCTGGA; (SEQ ID NO: 2400; mir-138-5p) CGGCCTGATTCACAACACCAGCT; or (SEQ ID NO: 2401; miR-9-5p) TCATACAGCTAGATAACCAAAGA.

Novel Promoters

[0202] The present invention provides novel promoters, that can be linked to a nucleic acid sequence encoding microdystrophin so that the transcription preferably occurs within myocytes and cardiomyocytes. Promoter regions enable the host cells to replicate the AAV delivered nucleic acid only in those cell types and tissues or organs in which the desired protein should be created. Here, the muscle specific promoter is included because it is principally desired that the proteins only be translated in myocytes and cardiomyocytes. Specificity of the cell type into which the nucleic acid is delivered and thus the protein translated is desired because of the adverse effects that may ensue from delivering the nucleic acid and having it translated in cells in which that nucleic acid and thus protein is not needed or not endogenously expressed.

[0203] Accordingly, aspects of the present invention provide a promoter having the sequence of any one of SEQ ID NOs: 2402-2404.

TABLE-US-00004 Promoter 1 (SEQ ID NO: 2402)
TAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATA
ACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCC
CATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGT
ACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGC
CCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGCTAGA
CTAGCATGCTGCCCATGTAAGGAGGCAAGGCCTGGGGACACCCGAGATGCCTGGTTATAATTAACCCAGACATGT
GGCTGCCCCCCCCCCCCAACACCTGCTGCCTCTAAAAATAACCCTGCATGCCATGTTCCCGGCGAAGGGCCAGCT
GTCCCCCGCCAGCTAGACTCAGCTCTCAGGGGCCCCTCCCTGGGGACAGCCCCTCCTGGCTAGTCACACCCTGTA
GGCTCCTCTATATAACCCAGGGGCACAGGGGCTGCCCTCATTCTACCACCACCTCCACAGCACAGACAGACACTC
AGGAGCCAGCCAGC Promoter 2 (SEQ ID NO: 2403)
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTGGCCTATTCTCTGAGAGGA
CGCTGGGCTAATAATAGGAGGGGAGATTAGGGAGGGATTTGACTGCATTGTTATCTCACAGGAATGGCGTTTGAG
AGCCTTGCTTCCAAGCTCTTCCAAGAGTCATGCCACCCCCCTCCAATCTCTGTGGAAGTGCCCACTGCCTTTCCC
GCTTCGTTCTCTTCTCTCTAAATCTGAGCCCACCCCACTCAGGGGTACAGGATCTGGTTCATTTATCTCCCAGAA
TGATGCCTTGGGTTGCGTCCCCTCTAAACTCTGGCACATTCCAGCCCCTCTTTGGGAGAAAGAACATCTCTTCTT
GCCCAGGCCCTAGTGAGACATGCATCGCAAGACATGCTCCCCCCTTCCTCCCCGCTGGGCTCCAGCTGCTGCCAG
CCCTCCAAGAAAGGAGAGGCCCTGAGTTGGGCTATTTTGGTATCTGGGGTGGGCACCCGCAGGGCTAAGGTTACC
TTGGTATGTTAAGGGCTCCCTGGGGCAGGACTATATAACCCCAGAGGGACTGCCCCATGCTGACTCCTTGCTTCT
TTCCTGCTCGAGCCGCTGCCTTGCCCCCCGGAGACTGAAGACA Promoter 3 (SEQ ID NO: 2404)
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTGGCCTATTCTCTGAGAGGA
CGCTGGGCTAATAATAGGAGGGGAGATTAGGGAGGGATTTGACTGCATTGTTATCTCACAGGAATGGCGTTTGAG
AGCCTTGCTTCCAAGCTCTTCCAAGAGTCATGCCACCCCCCTCCAATCTCTGTGGAAGTGCCCACTGCCTTTCCC
GCTTCGTTCTCTTCTCTCGGTCCCATCCCTGACCCACTAGCCTGGTGCCGGGGGGAGACCAATTTCTGGAGGAAG
CTCTGGGTACCCAGTCTTGCCCTAAGCCTGCCTCTGAACAGGCTGTTCTTGTTTATGACCCCCGTCCCAGCAGAA
TAGTGATTGCTCATCCCTGCATGTGATTTGAGAGGGCCAGTCTTTCAGGCCTGGAGGAAACTCAAGCCCCACCAG
CCGCTTGGCATCGGAATGCTCAGCTGGTCCCCTCCTGGGGCTCATGTGACCAAGATCTAGCTATAAATATCAGAG
GGGCCTCAGACCAAGACAGAATTCGGGCACCAGGAGAAGGAAGCCAACAGGATCCGACCCGGTGTTTTGTGACAA
AGGCAAGACCCCCAGGTCTACTTAGAGCAACGTTAGTAGAGGA

[0204] In some embodiments, the recombinant nucleic acid comprises a muscle specific promoter. Non-limiting examples of muscle specific promoters include a CK8 promoter, a desmin promotor, a Mb promoter, a MCK promoter, a MHCK7 promoter, a skeletal muscle alpha actin promoter, and a TTNI2 promoter.

[0205] In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2404-2407. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404.

[0206] In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2402. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2403. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2404.

Engineered Capsid Proteins

[0207] The present invention also provides novel capsid protein variants for viral vectors that de-target liver tissue and target skeletal muscle and heart tissue at the same time. Aspects of the present invention provide adeno-associated virus (AAV) vectors comprising a capsid protein comprising an insert comprising the amino acid sequence RGDR (SEQ ID NO: 2440). In the capsid protein, RGDR (SEQ ID NO: 2440) can be inserted after amino acid 455 in reference to an AAV9 capsid or an equivalent position in another AAV capsid. AAV vectors can comprise the amino acid sequence X.sub.1NX.sub.2X.sub.3X.sub.4RGDRX.sub.5X.sub.6L (SEQ ID NO: 1), wherein each of X.sub.1, X.sub.2, X.sub.3, X.sub.4, X.sub.5, and X.sub.6 is any amino acid.

[0208] In aspects of the invention, X.sub.1 is an amino acid selected from the group consisting of: A, I, F, G, H, L, M, Q, S, T, and V. In some aspects of the invention, X.sub.1 may be an amino acid selected from the group consisting of: A, I, L, M, S, and V.

[0209] In aspects of the invention, X.sub.2 is an amino acid selected from the group consisting of A, G, S, T, and Y.

[0210] In aspects of the invention, X.sub.3 is an amino acid selected from the group consisting of S, N, G, and P. In some aspects of the invention, X.sub.3 is S.

[0211] In aspects of the invention, X.sub.4 is an amino acid selected from the group consisting of A, G, H, I, M, S, T, and V.

[0212] In aspects of the invention, X.sub.5 is an amino acid selected from the group consisting of A, G, and Q.
[0213] In aspects of the invention, X.sub.6 is an amino acid selected from the group consisting of A, I, L, M, N, Y. In some aspects of the invention, X.sub.6 is selected from the group consisting of A, S, and Y.
[0214] In aspects of the invention, X.sub.1 is located at amino acid 451, X.sub.2 is located at amino acid 453, X.sub.3 is located at amino acid 454, X.sub.4 is located amino acid 455, and RGDRX.sub.5X.sub.6L (SEQ ID NO: 2441) is inserted after amino acid 455 in reference to an AAV9 capsid or an equivalent position in another AAV capsid.
[0215] In aspects of the invention, two amino acids from X.sub.1, X.sub.2, X.sub.3, and X.sub.4 are wild type amino acids in reference to an AAV9 capsid or an equivalent position in another AAV capsid and two amino acids from X.sub.1, X.sub.2, X.sub.3, and X.sub.4 are not wild type amino acids.
[0216] For example, capsid proteins variants of the invention can comprise a sequence as set forth in Table 1a, e.g., a capsid protein can comprise any one of SEQ ID NOs: 2-800. Notably, capsid protein variants on the invention comprise deletions, substitutions, and/or insertions relative to wild type viral vector capsids.
[0217] In aspects of the invention, the capsid protein comprises an amino acid sequence selected from Table 1a and the amino acid sequence is in hypervariable region IV (HVR IV) relative to wild type AAV9. The capsid protein variant can comprise substitutions at amino acids 451-455 relative to a wild type AAV9 capsid. For example, the substitutions at amino acids 451-455 relative to a wild type AAV9 capsid can be an amino acid sequence selected from column 1 of Table 1b, e.g., the substitutions relative to a wild type AAV9 capsid can be an amino acid sequence selected from any one of SEQ ID NOs: 801-1599. In aspects of the invention, the capsid protein variant can further comprise an insert. For example, the capsid protein can comprise a 7-mer insert selected from column 2 of Table 1b, e.g., the capsid protein variant comprises a 7-mer insert comprising an amino acid sequence selected from any one of SEQ ID NOs: 1600-2398. The insert can be in the location after amino acid 455 relative to a wild type AAV9 vector.
[0218] Advantageously, viral vectors comprising an amino acid sequence of the invention exhibit muscle tropism as compared to a wild type AAV vector (e.g., a wild type AAV9 vector).
[0219] In aspects of the invention, the capsid protein further comprises a deletion of G267 in reference to an AAV9 capsid or an equivalent position in another AAV capsid. Advantageously, the capsid protein comprising a deletion of G267 may exhibit reduced liver tropism as compared to a wild type AAV capsid protein.
[0220] As described, for the HVR IV variants, the 5 amino acids upstream are at positions 451-455, shown in column 1 of Table 1b. The 7-mer insert for HVR IV variants starts with “RGDR” (SEQ ID NO: 2440) and is inserted after amino acid 455, shown in column 2 of Table 1b.
[0221] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of LNASI (SEQ ID NO: 990) at positions 451-455, the amino acid sequence of RGDRQLL (SEQ ID NO: 1789) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0222] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of LNASM (SEQ ID NO: 997) at positions 451-455, the amino acid sequence of RGDRQAL (SEQ ID NO: 1796) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0223] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of LNASM (SEQ ID NO: 998) at positions 451-455, the amino acid sequence of RGDRQSL (SEQ ID NO: 1797) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0224] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of LNAST (SEQ ID NO: 1014) at positions 451-455, the amino acid sequence of RGDRQYL (SEQ ID NO: 1813) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0225] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of LNGST (SEQ ID NO: 1055) at positions 451-455, the amino acid sequence of RGDRQYL (SEQ ID NO: 1854) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0226] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of LNYST (SEQ ID NO: 1178) at positions 451-455, the amino acid sequence of RGDRQAL (SEQ ID NO: 1977) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0227] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of SNAST (SEQ ID NO: 1343) at positions 451-455, the amino acid sequence of RGDRQYL (SEQ ID NO: 2142) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0228] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of SNASV (SEQ ID NO: 1349) at positions 451-455, the amino acid sequence of RGDRQAL (SEQ ID NO: 2148) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.

[0229] In some embodiments, the capsid protein comprises, relative to the wild type AAV9 capsid protein, the amino acid sequence of VNGST (SEQ ID NO: 1547) at positions 451-455, the amino acid sequence of RGDRQYL (SEQ ID NO: 2346) inserted after amino acid 455, a deletion of the amino acid G267, and the remainder of the capsid is otherwise identical to wild type AAV9.

TABLE-US-00005 TABLE 1a HVR IV Capsid Variants Amino Acid SEQ ID Sequence NO:
ANASARGDRAAL 2 ANASARGDRGSL 3 ANASARGDRGYL 4 ANASARGDRQAL 5 ANASARGDRQSL 6
ANASARGDRQYL 7 ANASGRGDRAAL 8 ANASGRGDRASL 9 ANASGRGDRAYL 10 ANASGRGDRGAL 11
ANASGRGDRGSL 12 ANASGRGDRGYL 13 ANASGRGDRQAL 14 ANASGRGDRQSL 15 ANASGRGDRQYL 16
ANASMRGDRAAL 17 ANASMRGDRASL 18 ANASMRGDRAYL 19 ANASMRGDRQAL 20 ANASMRGDRQSL 21
ANASSRGDRGSL 22 ANASSRGDRGYL 23 ANASSRGDRQAL 24 ANASSRGDRQSL 25 ANASSRGDRQYL 26
ANASTRGDRASL 27 ANASTRGDRAYL 28 ANASTRGDRGAL 29 ANASTRGDRGSL 30 ANASTRGDRGYL 31
ANASTRGDRQAL 32 ANASTRGDRQSL 33 ANASTRGDRQYL 34 ANASVRGDRASL 35 ANASVRGDRGYL 36
ANASVRGDRQAL 37 ANASVRGDRQSL 38 ANASVRGDRQYL 39 ANGNFRGDRQNL 40 ANGNVRGDRGML 41
ANGSARGDRAYL 42 ANGSARGDRGSL 43 ANGSARGDRGYL 44 ANGSARGDRQAL 45 ANGSARGDRQYL 46
ANGSGRGDRAAL 47 ANGSGRGDRAYL 48 ANGSGRGDRGAL 49 ANGSGRGDRGSL 50 ANGSGRGDRGYL 51
ANGSGRGDRQAL 52 ANGSGRGDRQSL 53 ANGSGRGDRQYL 54 ANGSMRGDRGYL 55 ANGSMRGDRQAL 56
ANGSMRGDRQSL 57 ANGSMRGDRQYL 58 ANGSSRGDRAYL 59 ANGSSRGDRGSL 60 ANGSSRGDRGYL 61
ANGSSRGDRQAL 62 ANGSSRGDRQSL 63 ANGSSRGDRQYL 64 ANGSTRGDRAAL 65 ANGSTRGDRAYL 66
ANGSTRGDRGAL 67 ANGSTRGDRGYL 68 ANGSTRGDRQAL 69 ANGSTRGDRQSL 70 ANGSTRGDRQYL 71
ANGSVRGDRAYL 72 ANGSVRGDRGYL 73 ANGSVRGDRQAL 74 ANGSVRGDRQYL 75 ANSGMRGDRGAL 76
ANSSARGDRGAL 77 ANSSARGDRGSL 78 ANSSARGDRGYL 79 ANSSARGDRQAL 80 ANSSARGDRQSL 81
ANSSARGDRQYL 82 ANSSGRGDRAAL 83 ANSSGRGDRASL 84 ANSSGRGDRAYL 85 ANSSGRGDRGAL 86
ANSSGRGDRGSL 87 ANSSGRGDRGYL 88 ANSSGRGDRQAL 89 ANSSGRGDRQSL 90 ANSSGRGDRQYL 91
ANSSMRGDRAYL 92 ANSSMRGDRQAL 93 ANSSMRGDRQSL 94 ANSSMRGDRQYL 95 ANSSSRGDRGSL 96
ANSSSRGDRGYL 97 ANSSSRGDRQAL 98 ANSSSRGDRQSL 99 ANSSSRGDRQYL 100 ANSSTRGDRAAL 101
ANSSTRGDRGSL 102 ANSSTRGDRQAL 103 ANSSTRGDRQSL 104 ANSSTRGDRQYL 105 ANSSVRGDRAYL 106
ANSSVRGDRGYL 107 ANSSVRGDRQAL 108 ANSSVRGDRQSL 109 ANSSVRGDRQYL 110 ANTSARGDRAYL
111 ANTSARGDRGAL 112 ANTSARGDRGSL 113 ANTSARGDRGYL 114 ANTSARGDRQAL 115
ANTSARGDRQYL 116 ANTSGRGDRASL 117 ANTSGRGDRGAL 118 ANTSGRGDRGSL 119 ANTSGRGDRGYL
120 ANTSGRGDRQAL 121 ANTSGRGDRQSL 122 ANTSGRGDRQYL 123 ANTSMRGDRAYL 124
ANTSMRGDRQAL 125 ANTSMRGDRQSL 126 ANTSMRGDRQYL 127 ANTSSRGDRGSL 128 ANTSSRGDRQAL
129 ANTSSRGDRQYL 130 ANTSTRGDRAAL 131 ANTSTRGDRQAL 132 ANTSTRGDRQSL 133
ANTSTRGDRQYL 134 ANTSVRGDRAYL 135 ANTSVRGDRGSL 136 ANTSVRGDRQAL 137 ANTSVRGDRQSL
138 ANTSVRGDRQYL 139 ANYPHRGDRGAL 140 ANYSARGDRAAL 141 ANYSARGDRASL 142
ANYSARGDRGAL 143 ANYSARGDRGSL 144 ANYSARGDRQAL 145 ANYSARGDRQSL 146 ANYSGRGDRAAL
147 ANYSGRGDRASL 148 ANYSGRGDRGAL 149 ANYSGRGDRGSL 150 ANYSGRGDRQAL 151
ANYSGRGDRQSL 152 ANYSGRGDRQYL 153 ANYSMRGDRAAL 154 ANYSMRGDRASL 155 ANYSMRGDRAYL
156 ANYSMRGDRQAL 157 ANYSMRGDRQSL 158 ANYSSRGDRGAL 159 ANYSSRGDRGSL 160
ANYSSRGDRQAL 161 ANYSSRGDRQSL 162 ANYSSRGDRQYL 163 ANYSTRGDRQAL 164 ANYSTRGDRQSL
165 ANYSTRGDRQYL 166 ANYSVRGDRQAL 167 ANYSVRGDRQSL 168 ANYSVRGDRQYL 169
FNSGMRGDRGAL 170 GNSGMRGDRGAL 171 HNSGMRGDRGAL 172 LNASARGDRAAL 173 LNASARGDRASL
174 LNASARGDRAYL 175 LNASARGDRGAL 176 LNASARGDRGSL 177 LNASARGDRGYL 178
LNASARGDRQAL 179 LNASARGDRQSL 180 LNASARGDRQYL 181 LNASGRGDRAAL 182 LNASGRGDRASL
183 LNASGRGDRAYL 184 LNASGRGDRGAL 185 LNASGRGDRGSL 186 LNASGRGDRGYL 187
LNASGRGDRQAL 188 LNASGRGDRQSL 189 LNASGRGDRQYL 190 LNASIRGDRQLL 191 LNASMRGDRAAL
192 LNASMRGDRASL 193 LNASMRGDRAYL 194 LNASMRGDRGAL 195 LNASMRGDRGSL 196
LNASMRGDRGYL 197 LNASMRGDRQAL 198 LNASMRGDRQSL 199 LNASMRGDRQYL 200 LNASSRGDRAAL
201 LNASSRGDRASL 202 LNASSRGDRGAL 203 LNASSRGDRGSL 204 LNASSRGDRQAL 205 LNASSRGDRQSL
206 LNASSRGDRQYL 207 LNASTRGDRAAL 208 LNASTRGDRASL 209 LNASTRGDRGAL 210
LNASTRGDRGSL 211 LNASTRGDRGYL 212 LNASTRGDRQAL 213 LNASTRGDRQSL 214 LNASTRGDRQYL
215 LNASVRGDRAAL 216 LNASVRGDRASL 217 LNASVRGDRGAL 218 LNASVRGDRGSL 219
LNASVRGDRQAL 220 LNASVRGDRQSL 221 LNASVRGDRQYL 222 LNGNFRGDRQNL 223 LNGNVRGDRGML
224 LNGSARGDRAAL 225 LNGSARGDRAYL 226 LNGSARGDRGSL 227 LNGSARGDRGYL 228
LNGSARGDRQAL 229 LNGSARGDRQSL 230 LNGSARGDRQYL 231 LNGSGRGDRAAL 232 LNGSGRGDRAYL
233 LNGSGRGDRGAL 234 LNGSGRGDRGSL 235 LNGSGRGDRGYL 236 LNGSGRGDRQAL 237
LNGSGRGDRQYL 238 LNGSMRGDRASL 239 LNGSMRGDRGYL 240 LNGSMRGDRQAL 241 LNGSMRGDRQSL
242 LNGSMRGDRQYL 243 LNGSSRGDRAAL 244 LNGSSRGDRAYL 245 LNGSSRGDRGAL 246
LNGSSRGDRQYL 247 LNGSTRGDRAAL 248 LNGSTRGDRASL 249 LNGSTRGDRAYL 250 LNGSTRGDRGAL
251 LNGSTRGDRGSL 252 LNGSTRGDRGYL 253 LNGSTRGDRQAL 254 LNGSTRGDRQSL 255
LNGSTRGDRQYL 256 LNGSVRGDRASL 257 LNGSVRGDRAYL 258 LNGSVRGDRGYL 259 LNGSVRGDRQAL
260 LNGSVRGDRQSL 261 LNGSVRGDRQYL 262 LNSGMRGDRGAL 263 LNSSARGDRASL 264

LNSSARGDRAYL 265 LNSSARGDRGAL 266 LNSSARGDRGIL 267 LNSSARGDRGSL 268 LNSSARGDRGYL 269 LNSSARGDRQAL 270 LNSSARGDRQSL 271 LNSSARGDRQYL 272 LNSSGRGDRAAL 273 LNSSGRGDRASL 274 LNSSGRGDRAYL 275 LNSSGRGDRGAL 276 LNSSGRGDRGSL 277 LNSSGRGDRGYL 278 LNSSGRGDRQAL 279 LNSSGRGDRQSL 280 LNSSGRGDRQYL 281 LNSSMRGDRAAL 282 LNSSMRGDRASL 283 LNSSMRGDRAYL 284 LNSSMRGDRGAL 285 LNSSMRGDRGSL 286 LNSSMRGDRQAL 287 LNSSMRGDRQSL 288 LNSSMRGDRQYL 289 LNSSSRGDRAAL 290 LNSSSRGDRGAL 291 LNSSSRGDRGSL 292 LNSSSRGDRQAL 293 LNSSSRGDRQSL 294 LNSSSRGDRQYL 295 LNSSTRGDRAAL 296 LNSSTRGDRASL 297 LNSSTRGDRGAL 298 LNSSTRGDRGSL 299 LNSSTRGDRQAL 300 LNSSTRGDRQSL 301 LNSSTRGDRQYL 302 LNSSVRGDRAAL 303 LNSSVRGDRASL 304 LNSSVRGDRGAL 305 LNSSVRGDRGSL 306 LNSSVRGDRGYL 307 LNSSVRGDRQAL 308 LNSSVRGDRQSL 309 LNSSVRGDRQYL 310 LNTSARGDRAAL 311 LNTSARGDRASL 312 LNTSARGDRAYL 313 LNTSARGDRGAL 314 LNTSARGDRGSL 315 LNTSARGDRGYL 316 LNTSARGDRQAL 317 LNTSARGDRQSL 318 LNTSARGDRQYL 319 LNTSGRGDRAAL 320 LNTSGRGDRASL 321 LNTSGRGDRAYL 322 LNTSGRGDRGAL 323 LNTSGRGDRGSL 324 LNTSGRGDRGYL 325 LNTSGRGDRQAL 326 LNTSGRGDRQSL 327 LNTSGRGDRQYL 328 LNTSMRGDRAAL 329 LNTSMRGDRASL 330 LNTSMRGDRAYL 331 LNTSMRGDRGAL 332 LNTSMRGDRGSL 333 LNTSMRGDRQAL 334 LNTSMRGDRQSL 335 LNTSMRGDRQYL 336 LNTSSRGDRAAL 337 LNTSSRGDRGSL 338 LNTSSRGDRQAL 339 LNTSSRGDRQSL 340 LNTSSRGDRQYL 341 LNTSTRGDRAAL 342 LNTSTRGDRASL 343 LNTSTRGDRGAL 344 LNTSTRGDRGSL 345 LNTSTRGDRQAL 346 LNTSTRGDRQSL 347 LNTSTRGDRQYL 348 LNTSVRGDRAAL 349 LNTSVRGDRASL 350 LNTSVRGDRAYL 351 LNTSVRGDRGAL 352 LNTSVRGDRGSL 353 LNTSVRGDRQAL 354 LNTSVRGDRQSL 355 LNYPHRGDRGAL 356 LNYSARGDRGAL 357 LNYSARGDRGSL 358 LNYSARGDRQAL 359 LNYSARGDRQSL 360 LNYSARGDRQYL 361 LNYSGRGDRAAL 362 LNYSGRGDRASL 363 LNYSGRGDRGAL 364 LNYSGRGDRGSL 365 LNYSGRGDRQAL 366 LNYSGRGDRQSL 367 LNYSGRGDRQYL 368 LNYSMRGDRQAL 369 LNYSMRGDRQSL 370 LNYSSRGDRGAL 371 LNYSSRGDRGSL 372 LNYSSRGDRQAL 373 LNYSSRGDRQSL 374 LNYSSRGDRQYL 375 LNYSTRGDRASL 376 LNYSTRGDRGAL 377 LNYSTRGDRGSL 378 LNYSTRGDRQAL 379 LNYSTRGDRQSL 380 LNYSTRGDRQYL 381 LNYSVRGDRQAL 382 LNYSVRGDRQSL 383 MNASARGDRAAL 384 MNASARGDRAYL 385 MNASARGDRGSL 386 MNASARGDRGYL 387 MNASARGDRQAL 388 MNASARGDRQSL 389 MNASARGDRQYL 390 MNASGRGDRAAL 391 MNASGRGDRASL 392 MNASGRGDRAYL 393 MNASGRGDRGSL 394 MNASGRGDRQAL 395 MNASGRGDRQSL 396 MNASGRGDRQYL 397 MNASIRGDRQLL 398 MNASMRGDRAAL 399 MNASMRGDRAYL 400 MNASMRGDRGAL 401 MNASMRGDRQAL 402 MNASMRGDRQSL 403 MNASSRGDRAAL 404 MNASTRGDRAAL 405 MNASTRGDRASL 406 MNASTRGDRAYL 407 MNASTRGDRGAL 408 MNASTRGDRGSL 409 MNASTRGDRQAL 410 MNASTRGDRQSL 411 MNASTRGDRQYL 412 MNASVRGDRAAL 413 MNASVRGDRASL 414 MNASVRGDRAYL 415 MNASVRGDRQAL 416 MNASVRGDRQSL 417 MNASVRGDRQYL 418 MNGNFRGDRQNL 419 MNGSARGDRAYL 420 MNGSARGDRGYL 421 MNGSARGDRQAL 422 MNGSARGDRQSL 423 MNGSARGDRQYL 424 MNGSGRGDRAAL 425 MNGSGRGDRAYL 426 MNGSGRGDRGAL 427 MNGSGRGDRGSL 428 MNGSGRGDRGYL 429 MNGSGRGDRQAL 430 MNGSGRGDRQSL 431 MNGSGRGDRQYL 432 MNGSMRGDRQAL 433 MNGSMRGDRQSL 434 MNGSMRGDRQYL 435 MNGSSRGDRQYL 436 MNGSTRGDRAYL 437 MNGSTRGDRGAL 438 MNGSTRGDRGYL 439 MNGSTRGDRQAL 440 MNGSTRGDRQSL 441 MNGSTRGDRQYL 442 MNGSVRGDRGYL 443 MNGSVRGDRQAL 444 MNGSVRGDRQYL 445 MNSGMRGDRGAL 446 MNSSARGDRAAL 447 MNSSARGDRAYL 448 MNSSARGDRGSL 449 MNSSARGDRGYL 450 MNSSARGDRQAL 451 MNSSARGDRQSL 452 MNSSARGDRQYL 453 MNSSGRGDRAAL 454 MNSSGRGDRASL 455 MNSSGRGDRAYL 456 MNSSGRGDRGAL 457 MNSSGRGDRGSL 458 MNSSGRGDRQAL 459 MNSSGRGDRQSL 460 MNSSGRGDRQYL 461 MNSSMRGDRAAL 462 MNSSMRGDRASL 463 MNSSMRGDRAYL 464 MNSSMRGDRGSL 465 MNSSMRGDRQAL 466 MNSSMRGDRQSL 467 MNSSMRGDRQYL 468 MNSSSRGDRAAL 469 MNSSTRGDRAAL 470 MNSSTRGDRASL 471 MNSSTRGDRQAL 472 MNSSTRGDRQSL 473 MNSSTRGDRQYL 474 MNSSVRGDRAAL 475 MNSSVRGDRAYL 476 MNSSVRGDRQAL 477 MNSSVRGDRQSL 478 MNSSVRGDRQYL 479 MNTSARGDRAYL 480 MNTSGRGDRAAL 481 MNTSGRGDRQAL 482 MNTSGRGDRQYL 483 MNTSMRGDRAYL 484 MNTSTRGDRQYL 485 MNYPHRGDRGAL 486 MNYSARGDRAAL 487 MNYSARGDRGYL 488 MNYSARGDRQAL 489 MNYSARGDRQSL 490 MNYSGRGDRAAL 491 MNYSGRGDRASL 492 MNYSGRGDRGSL 493 MNYSGRGDRQAL 494 MNYSGRGDRQSL 495 MNYSSRGDRQAL 496 MNYSTRGDRAAL 497 MNYSTRGDRQAL 498 MNYSTRGDRQSL 499 MNYSTRGDRQYL 500 NNSGMRGDRGAL 501 QNSGMRGDRGAL 502 SNASARGDRAAL 503 SNASARGDRASL 504 SNASARGDRAYL 505 SNASARGDRGAL 506 SNASARGDRGSL 507 SNASARGDRGYL 508 SNASARGDRQAL 509 SNASARGDRQSL 510 SNASARGDRQYL 511 SNASGRGDRAAL 512 SNASGRGDRASL 513 SNASGRGDRAYL 514 SNASGRGDRGAL 515 SNASGRGDRGSL 516 SNASGRGDRGYL 517 SNASGRGDRQAL 518 SNASGRGDRQSL 519 SNASGRGDRQYL 520 SNASIRGDRQLL 521 SNASMRGDRAAL 522 SNASMRGDRASL 523 SNASMRGDRGAL 524 SNASMRGDRGSL 525 SNASMRGDRQAL 526 SNASMRGDRQSL 527 SNASSRGDRAAL 528 SNASSRGDRASL 529 SNASSRGDRAYL 530 SNASSRGDRGAL 531 SNASSRGDRGSL 532 SNASSRGDRQAL 533 SNASSRGDRQSL 534

SNASSRGDRQYL 535 SNASTRGDRAAL 536 SNASTRGDRASL 537 SNASTRGDRAYL 538 SNASTRGDRGAL 539 SNASTRGDRGSL 540 SNASTRGDRGYL 541 SNASTRGDRQAL 542 SNASTRGDRQSL 543 SNASTRGDRQYL 544 SNASVRGDRAAL 545 SNASVRGDRASL 546 SNASVRGDRAYL 547 SNASVRGDRGSL 548 SNASVRGDRGYL 549 SNASVRGDRQAL 550 SNASVRGDRQSL 551 SNASVRGDRQYL 552 SNGNFRGDRQNL 553 SNGNVRGDRGML 554 SNGSARGDRGYL 555 SNGSARGDRQAL 556 SNGSARGDRQSL 557 SNGSARGDRQYL 558 SNGSGRGDRASL 559 SNGSGRGDRAYL 560 SNGSGRGDRGAL 561 SNGSGRGDRGSL 562 SNGSGRGDRGYL 563 SNGSGRGDRQAL 564 SNGSGRGDRQSL 565 SNGSGRGDRQYL 566 SNGSMRGDRAYL 567 SNGSMRGDRGYL 568 SNGSMRGDRQAL 569 SNGSMRGDRQSL 570 SNGSMRGDRQYL 571 SNGSSRGDRGYL 572 SNGSSRGDRQYL 573 SNGSTRGDRAAL 574 SNGSTRGDRASL 575 SNGSTRGDRGAL 576 SNGSTRGDRGYL 577 SNGSTRGDRQAL 578 SNGSTRGDRQSL 579 SNGSTRGDRQYL 580 SNGSVRGDRGYL 581 SNGSVRGDRQAL 582 SNGSVRGDRQSL 583 SNGSVRGDRQYL 584 SNSGMRGDRGAL 585 SNSSARGDRAYL 586 SNSSARGDRGAL 587 SNSSARGDRGIL 588 SNSSARGDRGSL 589 SNSSARGDRGYL 590 SNSSARGDRQAL 591 SNSSARGDRQSL 592 SNSSARGDRQYL 593 SNSSGRGDRAAL 594 SNSSGRGDRAYL 595 SNSSGRGDRGAL 596 SNSSGRGDRGSL 597 SNSSGRGDRGYL 598 SNSSGRGDRQAL 599 SNSSGRGDRQSL 600 SNSSGRGDRQYL 601 SNSSMRGDRAAL 602 SNSSMRGDRASL 603 SNSSMRGDRAYL 604 SNSSMRGDRGSL 605 SNSSMRGDRQAL 606 SNSSMRGDRQSL 607 SNSSMRGDRQYL 608 SNSSSRGDRAAL 609 SNSSSRGDRGSL 610 SNSSSRGDRQAL 611 SNSSSRGDRQSL 612 SNSSSRGDRQYL 613 SNSSTRGDRAAL 614 SNSSTRGDRASL 615 SNSSTRGDRGSL 616 SNSSTRGDRGYL 617 SNSSTRGDRQAL 618 SNSSTRGDRQSL 619 SNSSTRGDRQYL 620 SNSSVRGDRQAL 621 SNSSVRGDRQSL 622 SNSSVRGDRQYL 623 SNTSARGDRAAL 624 SNTSARGDRGAL 625 SNTSARGDRGSL 626 SNTSARGDRGYL 627 SNTSARGDRQAL 628 SNTSARGDRQSL 629 SNTSARGDRQYL 630 SNTSGRGDRAAL 631 SNTSGRGDRASL 632 SNTSGRGDRGAL 633 SNTSGRGDRGSL 634 SNTSGRGDRGYL 635 SNTSGRGDRQAL 636 SNTSGRGDRQSL 637 SNTSGRGDRQYL 638 SNTSMRGDRAYL 639 SNTSMRGDRGSL 640 SNTSMRGDRQAL 641 SNTSMRGDRQSL 642 SNTSSRGDRQAL 643 SNTSTRGDRQAL 644 SNTSTRGDRQSL 645 SNTSTRGDRQYL 646 SNTSVRGDRAYL 647 SNTSVRGDRQAL 648 SNTSVRGDRQSL 649 SNTSVRGDRQYL 650 SNYPHRGDRGAL 651 SNYSARGDRAAL 652 SNYSARGDRASL 653 SNYSARGDRGAL 654 SNYSARGDRGSL 655 SNYSARGDRQAL 656 SNYSARGDRQSL 657 SNYSARGDRQYL 658 SNYSGRGDRAAL 659 SNYSGRGDRGAL 660 SNYSGRGDRGSL 661 SNYSGRGDRQAL 662 SNYSGRGDRQSL 663 SNYSMRGDRAAL 664 SNYSMRGDRASL 665 SNYSMRGDRGSL 666 SNYSMRGDRQAL 667 SNYSMRGDRQSL 668 SNYSSRGDRGAL 669 SNYSSRGDRGSL 670 SNYSSRGDRQAL 671 SNYSSRGDRQSL 672 SNYSTRGDRAAL 673 SNYSTRGDRGAL 674 SNYSTRGDRGSL 675 SNYSTRGDRQAL 676 SNYSTRGDRQSL 677 SNYSTRGDRQYL 678 SNYSVRGDRQAL 679 SNYSVRGDRQSL 680 SNYSVRGDRQYL 681 TNSGMRGDRGAL 682 VNASARGDRAAL 683 VNASARGDRGAL 684 VNASARGDRGSL 685 VNASARGDRGYL 686 VNASARGDRQAL 687 VNASARGDRQYL 688 VNASGRGDRAAL 689 VNASGRGDRAYL 690 VNASGRGDRGAL 691 VNASGRGDRGSL 692 VNASGRGDRGYL 693 VNASGRGDRQAL 694 VNASGRGDRQYL 695 VNASMRGDRAYL 696 VNASMRGDRQAL 697 VNASMRGDRQSL 698 VNASSRGDRGSL 699 VNASSRGDRQAL 700 VNASSRGDRQYL 701 VNASTRGDRAAL 702 VNASTRGDRGAL 703 VNASTRGDRGSL 704 VNASTRGDRGYL 705 VNASTRGDRQAL 706 VNASTRGDRQSL 707 VNASTRGDRQYL 708 VNASVRGDRGSL 709 VNASVRGDRGYL 710 VNASVRGDRQAL 711 VNASVRGDRQSL 712 VNASVRGDRQYL 713 VNGNFRGDRQNL 714 VNGNVRGDRGML 715 VNGSARGDRAYL 716 VNGSARGDRGAL 717 VNGSARGDRGSL 718 VNGSARGDRGYL 719 VNGSARGDRQAL 720 VNGSARGDRQSL 721 VNGSARGDRQYL 722 VNGSGRGDRAAL 723 VNGSGRGDRASL 724 VNGSGRGDRAYL 725 VNGSGRGDRGAL 726 VNGSGRGDRGSL 727 VNGSGRGDRGYL 728 VNGSGRGDRQAL 729 VNGSGRGDRQSL 730 VNGSGRGDRQYL 731 VNGSMRGDRGYL 732 VNGSMRGDRQAL 733 VNGSMRGDRQSL 734 VNGSMRGDRQYL 735 VNGSSRGDRAYL 736 VNGSSRGDRGAL 737 VNGSSRGDRGSL 738 VNGSSRGDRQAL 739 VNGSSRGDRQSL 740 VNGSSRGDRQYL 741 VNGSTRGDRAAL 742 VNGSTRGDRAYL 743 VNGSTRGDRGAL 744 VNGSTRGDRGYL 745 VNGSTRGDRQAL 746 VNGSTRGDRQSL 747 VNGSTRGDRQYL 748 VNGSVRGDRAYL 749 VNGSVRGDRQAL 750 VNGSVRGDRQSL 751 VNGSVRGDRQYL 752 VNSGMRGDRGAL 753 VNSSARGDRGAL 754 VNSSARGDRGSL 755 VNSSARGDRGYL 756 VNSSARGDRQAL 757 VNSSARGDRQSL 758 VNSSARGDRQYL 759 VNSSGRGDRAAL 760 VNSSGRGDRAYL 761 VNSSGRGDRGAL 762 VNSSGRGDRQAL 763 VNSSGRGDRQSL 764 VNSSGRGDRQYL 765 VNSSMRGDRAYL 766 VNSSMRGDRQAL 767 VNSSMRGDRQSL 768 VNSSMRGDRQYL 769 VNSSSRGDRGSL 770 VNSSSRGDRQAL 771 VNSSSRGDRQSL 772 VNSSTRGDRGAL 773 VNSSTRGDRQAL 774 VNSSTRGDRQSL 775 VNSSTRGDRQYL 776 VNSSVRGDRAYL 777 VNSSVRGDRQAL 778 VNSSVRGDRQSL 779 VNTSARGDRAYL 780 VNTSGRGDRAAL 781 VNTSGRGDRGAL 782 VNTSGRGDRGSL 783 VNTSGRGDRGYL 784 VNTSGRGDRQAL 785 VNTSGRGDRQSL 786 VNTSGRGDRQYL 787 VNTSTRGDRAYL 788 VNYPHRGDRGAL 789 VNYSARGDRGAL 790 VNYSARGDRGSL 791 VNYSARGDRQAL 792 VNYSARGDRQSL 793 VNYSGRGDRAAL 794 VNYSGRGDRGAL 795 VNYSGRGDRQAL 796 VNYSGRGDRQSL 797 VNYSGRGDRQYL 798 VNYSSRGDRGSL 799 VNYSSRGDRGYL 800

TABLE-US-00006 TABLE 1b Split Amino Acid Sequences from Table 1a 5 aa SEQ ID SEQ ID Substitution NO: 7 aa Insert NO: ANASA 801 RGDRAAL 1600 ANASA 802 RGDRGSL 1601 ANASA 803

RGDRGYL 1602 ANASA 804 RGDRQAL 1603 ANASA 805 RGDRQSL 1604 ANASA 806 RGDRQYL 1605 ANASG 807 RGDRAAL 1606 ANASG 808 RGDRASL 1607 ANASG 809 RGDRAYL 1608 ANASG 810 RGDRGAL 1609 ANASG 811 RGDRGSL 1610 ANASG 812 RGDRGYL 1611 ANASG 813 RGDRQAL 1612 ANASG 814 RGDRQSL 1613 ANASG 815 RGDRQYL 1614 ANASM 816 RGDRAAL 1615 ANASM 817 RGDRASL 1616 ANASM 818 RGDRAYL 1617 ANASM 819 RGDRQAL 1618 ANASM 820 RGDRQSL 1619 ANASS 821 RGDRGSL 1620 ANASS 822 RGDRGYL 1621 ANASS 823 RGDRQAL 1622 ANASS 824 RGDRQSL 1623 ANASS 825 RGDRQYL 1624 ANAST 826 RGDRASL 1625 ANAST 827 RGDRAYL 1626 ANAST 828 RGDRGAL 1627 ANAST 829 RGDRGSL 1628 ANAST 830 RGDRGYL 1629 ANAST 831 RGDRQAL 1630 ANAST 832 RGDRQSL 1631 ANAST 833 RGDRQYL 1632 ANASV 834 RGDRASL 1633 ANASV 835 RGDRGYL 1634 ANASV 836 RGDRQAL 1635 ANASV 837 RGDRQSL 1636 ANASV 838 RGDRQYL 1637 ANGNF 839 RGDRQNL 1638 ANGNV 840 RGDRGML 1639 ANGSA 841 RGDRAYL 1640 ANGSA 842 RGDRGSL 1641 ANGSA 843 RGDRGYL 1642 ANGSA 844 RGDRQAL 1643 ANGSA 845 RGDRQYL 1644 ANGSG 846 RGDRAAL 1645 ANGSG 847 RGDRAYL 1646 ANGSG 848 RGDRGAL 1647 ANGSG 849 RGDRGSL 1648 ANGSG 850 RGDRGYL 1649 ANGSG 851 RGDRQAL 1650 ANGSG 852 RGDRQSL 1651 ANGSG 853 RGDRQYL 1652 ANGSM 854 RGDRGYL 1653 ANGSM 855 RGDRQAL 1654 ANGSM 856 RGDRQSL 1655 ANGSM 857 RGDRQYL 1656 ANGSS 858 RGDRAYL 1657 ANGSS 859 RGDRGSL 1658 ANGSS 860 RGDRGYL 1659 ANGSS 861 RGDRQAL 1660 ANGSS 862 RGDRQSL 1661 ANGSS 863 RGDRQYL 1662 ANGST 864 RGDRAAL 1663 ANGST 865 RGDRAYL 1664 ANGST 866 RGDRGAL 1665 ANGST 867 RGDRGYL 1666 ANGST 868 RGDRQAL 1667 ANGST 869 RGDRQSL 1668 ANGST 870 RGDRQYL 1669 ANGSV 871 RGDRAYL 1670 ANGSV 872 RGDRGYL 1671 ANGSV 873 RGDRQAL 1672 ANGSV 874 RGDRQYL 1673 ANSGM 875 RGDRGAL 1674 ANSSA 876 RGDRGAL 1675 ANSSA 877 RGDRGSL 1676 ANSSA 878 RGDRGYL 1677 ANSSA 879 RGDRQAL 1678 ANSSA 880 RGDRQSL 1679 ANSSA 881 RGDRQYL 1680 ANSSG 882 RGDRAAL 1681 ANSSG 883 RGDRASL 1682 ANSSG 884 RGDRAYL 1683 ANSSG 885 RGDRGAL 1684 ANSSG 886 RGDRGSL 1685 ANSSG 887 RGDRGYL 1686 ANSSG 888 RGDRQAL 1687 ANSSG 889 RGDRQSL 1688 ANSSG 890 RGDRQYL 1689 ANSSM 891 RGDRAYL 1690 ANSSM 892 RGDRQAL 1691 ANSSM 893 RGDRQSL 1692 ANSSM 894 RGDRQYL 1693 ANSSS 895 RGDRGSL 1694 ANSSS 896 RGDRGYL 1695 ANSSS 897 RGDRQAL 1696 ANSSS 898 RGDRQSL 1697 ANSSS 899 RGDRQYL 1698 ANSST 900 RGDRAAL 1699 ANSST 901 RGDRGSL 1700 ANSST 902 RGDRQAL 1701 ANSST 903 RGDRQSL 1702 ANSST 904 RGDRQYL 1703 ANSSV 905 RGDRAYL 1704 ANSSV 906 RGDRGYL 1705 ANSSV 907 RGDRQAL 1706 ANSSV 908 RGDRQSL 1707 ANSSV 909 RGDRQYL 1708 ANTSA 910 RGDRAYL 1709 ANTSA 911 RGDRGAL 1710 ANTSA 912 RGDRGSL 1711 ANTSA 913 RGDRGYL 1712 ANTSA 914 RGDRQAL 1713 ANTSA 915 RGDRQYL 1714 ANTSG 916 RGDRASL 1715 ANTSG 917 RGDRGAL 1716 ANTSG 918 RGDRGSL 1717 ANTSG 919 RGDRGYL 1718 ANTSG 920 RGDRQAL 1719 ANTSG 921 RGDRQSL 1720 ANTSG 922 RGDRQYL 1721 ANTSM 923 RGDRAYL 1722 ANTSM 924 RGDRQAL 1723 ANTSM 925 RGDRQSL 1724 ANTSM 926 RGDRQYL 1725 ANTSS 927 RGDRGSL 1726 ANTSS 928 RGDRQAL 1727 ANTSS 929 RGDRQYL 1728 ANTST 930 RGDRAAL 1729 ANTST 931 RGDRQAL 1730 ANTST 932 RGDRQSL 1731 ANTST 933 RGDRQYL 1732 ANTSV 934 RGDRAYL 1733 ANTSV 935 RGDRGSL 1734 ANTSV 936 RGDRQAL 1735 ANTSV 937 RGDRQSL 1736 ANTSV 938 RGDRQYL 1737 ANYPH 939 RGDRGAL 1738 ANYSA 940 RGDRAAL 1739 ANYSA 941 RGDRASL 1740 ANYSA 942 RGDRGAL 1741 ANYSA 943 RGDRGSL 1742 ANYSA 944 RGDRQAL 1743 ANYSA 945 RGDRQSL 1744 ANYSG 946 RGDRAAL 1745 ANYSG 947 RGDRASL 1746 ANYSG 948 RGDRGAL 1747 ANYSG 949 RGDRGSL 1748 ANYSG 950 RGDRQAL 1749 ANYSG 951 RGDRQSL 1750 ANYSG 952 RGDRQYL 1751 ANYSM 953 RGDRAAL 1752 ANYSM 954 RGDRASL 1753 ANYSM 955 RGDRAYL 1754 ANYSM 956 RGDRQAL 1755 ANYSM 957 RGDRQSL 1756 ANYSS 958 RGDRGAL 1757 ANYSS 959 RGDRGSL 1758 ANYSS 960 RGDRQAL 1759 ANYSS 961 RGDRQSL 1760 ANYSS 962 RGDRQYL 1761 ANYST 963 RGDRQAL 1762 ANYST 964 RGDRQSL 1763 ANYST 965 RGDRQYL 1764 ANYSV 966 RGDRQAL 1765 ANYSV 967 RGDRQSL 1766 ANYSV 968 RGDRQYL 1767 FNSGM 969 RGDRGAL 1768 GNSGM 970 RGDRGAL 1769 HNSGM 971 RGDRGAL 1770 LNASA 972 RGDRAAL 1771 LNASA 973 RGDRASL 1772 LNASA 974 RGDRAYL 1773 LNASA 975 RGDRGAL 1774 LNASA 976 RGDRGSL 1775 LNASA 977 RGDRGYL 1776 LNASA 978 RGDRQAL 1777 LNASA 979 RGDRQSL 1778 LNASA 980 RGDRQYL 1779 LNASG 981 RGDRAAL 1780 LNASG 982 RGDRASL 1781 LNASG 983 RGDRAYL 1782 LNASG 984 RGDRGAL 1783 LNASG 985 RGDRGSL 1784 LNASG 986 RGDRGYL 1785 LNASG 987 RGDRQAL 1786 LNASG 988 RGDRQSL 1787 LNASG 989 RGDRQYL 1788 LNASI 990 RGDRQLL 1789 LNASM 991 RGDRAAL 1790 LNASM 992 RGDRASL 1791 LNASM 993 RGDRAYL 1792 LNASM 994 RGDRGAL 1793 LNASM 995 RGDRGSL 1794 LNASM 996 RGDRGYL 1795 LNASM 997 RGDRQAL 1796 LNASM 998 RGDRQSL 1797 LNASM 999 RGDRQYL 1798 LNASS 1000 RGDRAAL 1799 LNASS 1001 RGDRASL 1800 LNASS 1002 RGDRGAL 1801 LNASS 1003 RGDRGSL 1802 LNASS 1004 RGDRQAL 1803 LNASS 1005 RGDRQSL 1804 LNASS 1006 RGDRQYL 1805 LNAST 1007 RGDRAAL 1806 LNAST 1008 RGDRASL 1807 LNAST 1009 RGDRGAL 1808 LNAST 1010 RGDRGSL 1809 LNAST 1011 RGDRGYL 1810 LNAST 1012 RGDRQAL 1811 LNAST 1013 RGDRQSL 1812 LNAST 1014 RGDRQYL 1813 LNASV 1015 RGDRAAL 1814 LNASV 1016 RGDRASL 1815 LNASV 1017 RGDRGAL 1816 LNASV 1018 RGDRGSL 1817 LNASV 1019 RGDRQAL 1818 LNASV 1020 RGDRQSL 1819 LNASV 1021 RGDRQYL 1820 LNGNF 1022 RGDRQNL 1821 LNGNV 1023 RGDRGML 1822 LNGSA 1024 RGDRAAL 1823 LNGSA 1025 RGDRAYL 1824 LNGSA 1026 RGDRGSL 1825 LNGSA 1027 RGDRGYL 1826 LNGSA 1028 RGDRQAL 1827 LNGSA 1029 RGDRQSL 1828 LNGSA 1030 RGDRQYL 1829 LNGSG 1031 RGDRAAL 1830

LNGSG 1032 RGDRAYL 1831 LNGSG 1033 RGDRGAL 1832 LNGSG 1034 RGDRGSL 1833 LNGSG 1035 RGDRGYL 1834 LNGSG 1036 RGDRQAL 1835 LNGSG 1037 RGDRQYL 1836 LNGSM 1038 RGDRASL 1837 LNGSM 1039 RGDRGYL 1838 LNGSM 1040 RGDRQAL 1839 LNGSM 1041 RGDRQSL 1840 LNGSM 1042 RGDRQYL 1841 LNGSS 1043 RGDRAAL 1842 LNGSS 1044 RGDRAYL 1843 LNGSS 1045 RGDRGAL 1844 LNGSS 1046 RGDRQYL 1845 LNGST 1047 RGDRAAL 1846 LNGST 1048 RGDRASL 1847 LNGST 1049 RGDRAYL 1848 LNGST 1050 RGDRGAL 1849 LNGST 1051 RGDRGSL 1850 LNGST 1052 RGDRGYL 1851 LNGST 1053 RGDRQAL 1852 LNGST 1054 RGDRQSL 1853 LNGST 1055 RGDRQYL 1854 LNGSV 1056 RGDRASL 1855 LNGSV 1057 RGDRAYL 1856 LNGSV 1058 RGDRGYL 1857 LNGSV 1059 RGDRQAL 1858 LNGSV 1060 RGDRQSL 1859 LNGSV 1061 RGDRQYL 1860 LNSGM 1062 RGDRGAL 1861 LNSSA 1063 RGDRASL 1862 LNSSA 1064 RGDRAYL 1863 LNSSA 1065 RGDRGAL 1864 LNSSA 1066 RGDRGIL 1865 LNSSA 1067 RGDRGSL 1866 LNSSA 1068 RGDRGYL 1867 LNSSA 1069 RGDRQAL 1868 LNSSA 1070 RGDRQSL 1869 LNSSA 1071 RGDRQYL 1870 LNSSG 1072 RGDRAAL 1871 LNSSG 1073 RGDRASL 1872 LNSSG 1074 RGDRAYL 1873 LNSSG 1075 RGDRGAL 1874 LNSSG 1076 RGDRGSL 1875 LNSSG 1077 RGDRGYL 1876 LNSSG 1078 RGDRQAL 1877 LNSSG 1079 RGDRQSL 1878 LNSSG 1080 RGDRQYL 1879 LNSSM 1081 RGDRAAL 1880 LNSSM 1082 RGDRASL 1881 LNSSM 1083 RGDRAYL 1882 LNSSM 1084 RGDRGAL 1883 LNSSM 1085 RGDRGSL 1884 LNSSM 1086 RGDRQAL 1885 LNSSM 1087 RGDRQSL 1886 LNSSM 1088 RGDRQYL 1887 LNSSS 1089 RGDRAAL 1888 LNSSS 1090 RGDRGAL 1889 LNSSS 1091 RGDRGSL 1890 LNSSS 1092 RGDRQAL 1891 LNSSS 1093 RGDRQSL 1892 LNSSS 1094 RGDRQYL 1893 LNSST 1095 RGDRAAL 1894 LNSST 1096 RGDRASL 1895 LNSST 1097 RGDRGAL 1896 LNSST 1098 RGDRGSL 1897 LNSST 1099 RGDRQAL 1898 LNSST 1100 RGDRQSL 1899 LNSST 1101 RGDRQYL 1900 LNSSV 1102 RGDRAAL 1901 LNSSV 1103 RGDRASL 1902 LNSSV 1104 RGDRGAL 1903 LNSSV 1105 RGDRGSL 1904 LNSSV 1106 RGDRGYL 1905 LNSSV 1107 RGDRQAL 1906 LNSSV 1108 RGDRQSL 1907 LNSSV 1109 RGDRQYL 1908 LNTSA 1110 RGDRAAL 1909 LNTSA 1111 RGDRASL 1910 LNTSA 1112 RGDRAYL 1911 LNTSA 1113 RGDRGAL 1912 LNTSA 1114 RGDRGSL 1913 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RGDRQSL 2050 MNSSA 1252 RGDRQYL 2051 MNSSG 1253 RGDRAAL 2052 MNSSG 1254 RGDRASL 2053 MNSSG 1255 RGDRAYL 2054 MNSSG 1256 RGDRGAL 2055 MNSSG 1257 RGDRGSL 2056 MNSSG 1258 RGDRQAL 2057 MNSSG 1259 RGDRQSL 2058 MNSSG 1260 RGDRQYL 2059 MNSSM 1261 RGDRAAL 2060 MNSSM 1262 RGDRASL 2061 MNSSM 1263 RGDRAYL 2062 MNSSM 1264 RGDRGSL 2063 MNSSM 1265 RGDRQAL 2064 MNSSM 1266 RGDRQSL 2065 MNSSM 1267 RGDRQYL 2066 MNSSS 1268 RGDRAAL 2067 MNSST 1269 RGDRAAL 2068 MNSST 1270 RGDRASL 2069 MNSST 1271 RGDRQAL 2070 MNSST 1272 RGDRQSL 2071 MNSST 1273 RGDRQYL 2072 MNSSV 1274 RGDRAAL 2073 MNSSV 1275 RGDRAYL 2074 MNSSV 1276 RGDRQAL 2075 MNSSV 1277 RGDRQSL 2076 MNSSV 1278 RGDRQYL 2077 MNTSA 1279 RGDRAYL 2078 MNTSG 1280 RGDRAAL 2079 MNTSG 1281 RGDRQAL 2080 MNTSG 1282 RGDRQYL 2081 MNTSM 1283 RGDRAYL 2082 MNTST 1284 RGDRQYL 2083 MNYPH 1285 RGDRGAL 2084 MNYSA 1286 RGDRAAL 2085 MNYSA 1287 RGDRGYL 2086 MNYSA 1288 RGDRQAL 2087 MNYSA 1289 RGDRQSL 2088 MNYSG 1290 RGDRAAL 2089 MNYSG 1291 RGDRASL 2090 MNYSG 1292 RGDRGSL 2091 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SNGST 1376 RGDRGYL 2175 SNGST 1377 RGDRQAL 2176 SNGST 1378 RGDRQSL 2177 SNGST 1379 RGDRQYL 2178 SNGSV 1380 RGDRGYL 2179 SNGSV 1381 RGDRQAL 2180 SNGSV 1382 RGDRQSL 2181 SNGSV 1383 RGDRQYL 2182 SNSGM 1384 RGDRGAL 2183 SNSSA 1385 RGDRAYL 2184 SNSSA 1386 RGDRGAL 2185 SNSSA 1387 RGDRGIL 2186 SNSSA 1388 RGDRGSL 2187 SNSSA 1389 RGDRGYL 2188 SNSSA 1390 RGDRQAL 2189 SNSSA 1391 RGDRQSL 2190 SNSSA 1392 RGDRQYL 2191 SNSSG 1393 RGDRAAL 2192 SNSSG 1394 RGDRAYL 2193 SNSSG 1395 RGDRGAL 2194 SNSSG 1396 RGDRGSL 2195 SNSSG 1397 RGDRGYL 2196 SNSSG 1398 RGDRQAL 2197 SNSSG 1399 RGDRQSL 2198 SNSSG 1400 RGDRQYL 2199 SNSSM 1401 RGDRAAL 2200 SNSSM 1402 RGDRASL 2201 SNSSM 1403 RGDRAYL 2202 SNSSM 1404 RGDRGSL 2203 SNSSM 1405 RGDRQAL 2204 SNSSM 1406 RGDRQSL 2205 SNSSM 1407 RGDRQYL 2206 SNSSS 1408 RGDRAAL 2207 SNSSS 1409 RGDRGSL 2208 SNSSS 1410 RGDRQAL 2209 SNSSS 1411 RGDRQSL 2210 SNSSS 1412 RGDRQYL 2211 SNSST 1413 RGDRAAL 2212 SNSST 1414 RGDRASL 2213 SNSST 1415 RGDRGSL 2214 SNSST 1416 RGDRGYL 2215 SNSST 1417 RGDRQAL 2216 SNSST 1418 RGDRQSL 2217 SNSST 1419 RGDRQYL 2218 SNSSV 1420 RGDRQAL 2219 SNSSV 1421 RGDRQSL 2220 SNSSV 1422 RGDRQYL 2221 SNTSA 1423 RGDRAAL 2222 SNTSA 1424 RGDRGAL 2223 SNTSA 1425 RGDRGSL 2224 SNTSA 1426 RGDRGYL 2225 SNTSA 1427 RGDRQAL 2226 SNTSA 1428 RGDRQSL 2227 SNTSA 1429 RGDRQYL 2228 SNTSG 1430 RGDRAAL 2229 SNTSG 1431 RGDRASL 2230 SNTSG 1432 RGDRGAL 2231 SNTSG 1433 RGDRGSL 2232 SNTSG 1434 RGDRGYL 2233 SNTSG 1435 RGDRQAL 2234 SNTSG 1436 RGDRQSL 2235 SNTSG 1437 RGDRQYL 2236 SNTSM 1438 RGDRAYL 2237 SNTSM 1439 RGDRGSL 2238 SNTSM 1440 RGDRQAL 2239 SNTSM 1441 RGDRQSL 2240 SNTSS 1442 RGDRQAL 2241 SNTST 1443 RGDRQAL 2242 SNTST 1444 RGDRQSL 2243 SNTST 1445 RGDRQYL 2244 SNTSV 1446 RGDRAYL 2245 SNTSV 1447 RGDRQAL 2246 SNTSV 1448 RGDRQSL 2247 SNTSV 1449 RGDRQYL 2248 SNYPH 1450 RGDRGAL 2249 SNYSA 1451 RGDRAAL 2250 SNYSA 1452 RGDRASL 2251 SNYSA 1453 RGDRGAL 2252 SNYSA 1454 RGDRGSL 2253 SNYSA 1455 RGDRQAL 2254 SNYSA 1456 RGDRQSL 2255 SNYSA 1457 RGDRQYL 2256 SNYSG 1458 RGDRAAL 2257 SNYSG 1459 RGDRGAL 2258 SNYSG 1460 RGDRGSL 2259 SNYSG 1461 RGDRQAL 2260 SNYSG 1462 RGDRQSL 2261 SNYSM 1463 RGDRAAL 2262 SNYSM 1464 RGDRASL 2263 SNYSM 1465 RGDRGSL 2264 SNYSM 1466 RGDRQAL 2265 SNYSM 1467 RGDRQSL 2266 SNYSS 1468 RGDRGAL 2267 SNYSS 1469 RGDRGSL 2268 SNYSS

1470 RGDRQAL 2269 SNYSS 1471 RGDRQSL 2270 SNYST 1472 RGDRAAL 2271 SNYST 1473 RGDRGAL 2272 SNYST 1474 RGDRGSL 2273 SNYST 1475 RGDRQAL 2274 SNYST 1476 RGDRQSL 2275 SNYST 1477 RGDRQYL 2276 SNYSV 1478 RGDRQAL 2277 SNYSV 1479 RGDRQSL 2278 SNYSV 1480 RGDRQYL 2279 TNSGM 1481 RGDRGAL 2280 VNASA 1482 RGDRAAL 2281 VNASA 1483 RGDRGAL 2282 VNASA 1484 RGDRGSL 2283 VNASA 1485 RGDRGYL 2284 VNASA 1486 RGDRQAL 2285 VNASA 1487 RGDRQYL 2286 VNASG 1488 RGDRAAL 2287 VNASG 1489 RGDRAYL 2288 VNASG 1490 RGDRGAL 2289 VNASG 1491 RGDRGSL 2290 VNASG 1492 RGDRGYL 2291 VNASG 1493 RGDRQAL 2292 VNASG 1494 RGDRQYL 2293 VNASM 1495 RGDRAYL 2294 VNASM 1496 RGDRQAL 2295 VNASM 1497 RGDRQSL 2296 VNASS 1498 RGDRGSL 2297 VNASS 1499 RGDRQAL 2298 VNASS 1500 RGDRQYL 2299 VNAST 1501 RGDRAAL 2300 VNAST 1502 RGDRGAL 2301 VNAST 1503 RGDRGSL 2302 VNAST 1504 RGDRGYL 2303 VNAST 1505 RGDRQAL 2304 VNAST 1506 RGDRQSL 2305 VNAST 1507 RGDRQYL 2306 VNASV 1508 RGDRGSL 2307 VNASV 1509 RGDRGYL 2308 VNASV 1510 RGDRQAL 2309 VNASV 1511 RGDRQSL 2310 VNASV 1512 RGDRQYL 2311 VNGNF 1513 RGDRQNL 2312 VNGNV 1514 RGDRGML 2313 VNGSA 1515 RGDRAYL 2314 VNGSA 1516 RGDRGAL 2315 VNGSA 1517 RGDRGSL 2316 VNGSA 1518 RGDRGYL 2317 VNGSA 1519 RGDRQAL 2318 VNGSA 1520 RGDRQSL 2319 VNGSA 1521 RGDRQYL 2320 VNGSG 1522 RGDRAAL 2321 VNGSG 1523 RGDRASL 2322 VNGSG 1524 RGDRAYL 2323 VNGSG 1525 RGDRGAL 2324 VNGSG 1526 RGDRGSL 2325 VNGSG 1527 RGDRGYL 2326 VNGSG 1528 RGDRQAL 2327 VNGSG 1529 RGDRQSL 2328 VNGSG 1530 RGDRQYL 2329 VNGSM 1531 RGDRGYL 2330 VNGSM 1532 RGDRQAL 2331 VNGSM 1533 RGDRQSL 2332 VNGSM 1534 RGDRQYL 2333 VNGSS 1535 RGDRAYL 2334 VNGSS 1536 RGDRGAL 2335 VNGSS 1537 RGDRGSL 2336 VNGSS 1538 RGDRQAL 2337 VNGSS 1539 RGDRQSL 2338 VNGSS 1540 RGDRQYL 2339 VNGST 1541 RGDRAAL 2340 VNGST 1542 RGDRAYL 2341 VNGST 1543 RGDRGAL 2342 VNGST 1544 RGDRGYL 2343 VNGST 1545 RGDRQAL 2344 VNGST 1546 RGDRQSL 2345 VNGST 1547 RGDRQYL 2346 VNGSV 1548 RGDRAYL 2347 VNGSV 1549 RGDRQAL 2348 VNGSV 1550 RGDRQSL 2349 VNGSV 1551 RGDRQYL 2350 VNSGM 1552 RGDRGAL 2351 VNSSA 1553 RGDRGAL 2352 VNSSA 1554 RGDRGSL 2353 VNSSA 1555 RGDRGYL 2354 VNSSA 1556 RGDRQAL 2355 VNSSA 1557 RGDRQSL 2356 VNSSA 1558 RGDRQYL 2357 VNSSG 1559 RGDRAAL 2358 VNSSG 1560 RGDRAYL 2359 VNSSG 1561 RGDRGAL 2360 VNSSG 1562 RGDRQAL 2361 VNSSG 1563 RGDRQSL 2362 VNSSG 1564 RGDRQYL 2363 VNSSM 1565 RGDRAYL 2364 VNSSM 1566 RGDRQAL 2365 VNSSM 1567 RGDRQSL 2366 VNSSM 1568 RGDRQYL 2367 VNSSS 1569 RGDRGSL 2368 VNSSS 1570 RGDRQAL 2369 VNSSS 1571 RGDRQSL 2370 VNSST 1572 RGDRGAL 2371 VNSST 1573 RGDRQAL 2372 VNSST 1574 RGDRQSL 2373 VNSST 1575 RGDRQYL 2374 VNSSV 1576 RGDRAYL 2375 VNSSV 1577 RGDRQAL 2376 VNSSV 1578 RGDRQSL 2377 VNTSA 1579 RGDRAYL 2378 VNTSG 1580 RGDRAAL 2379 VNTSG 1581 RGDRGAL 2380 VNTSG 1582 RGDRGSL 2381 VNTSG 1583 RGDRGYL 2382 VNTSG 1584 RGDRQAL 2383 VNTSG 1585 RGDRQSL 2384 VNTSG 1586 RGDRQYL 2385 VNTST 1587 RGDRAYL 2386 VNYPH 1588 RGDRGAL 2387 VNYSA 1589 RGDRGAL 2388 VNYSA 1590 RGDRGSL 2389 VNYSA 1591 RGDRQAL 2390 VNYSA 1592 RGDRQSL 2391 VNYSG 1593 RGDRAAL 2392 VNYSG 1594 RGDRGAL 2393 VNYSG 1595 RGDRQAL 2394 VNYSG 1596 RGDRQSL 2395 VNYSG 1597 RGDRQYL 2396 VNYSS 1598 RGDRGSL 2397 VNYSS 1599 RGDRGYL 2398

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0230] FIGS. **1**A-**1**B show results from analysis of microdystrophin expression in quadriceps from B10-mdx mice dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **1**A shows images of FLAG immunofluorescence of microdystrophin expression in B10-mdx mouse quadriceps following administration of Composition 1 or Reference AAV composition. FIG. **1**B shows results from capillary electrophoresis quantifying microdystrophin expression in B10-mdx mouse quadriceps following administration of Composition 1 or Reference AAV composition.

[0231] FIGS. **2**A-**2**B show results from analysis of microdystrophin expression in triceps from B10-mdx mice dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **2**A shows images of FLAG immunofluorescence of microdystrophin expression in B10-mdx mouse triceps following administration of Composition 1 or Reference AAV composition. FIG. **2**B shows results from capillary electrophoresis quantifying microdystrophin expression in B10-mdx mouse triceps following administration of Composition 1 or Reference AAV composition.

[0232] FIGS. **3**A-**3**B show results from analysis of microdystrophin expression in heart tissue from B10-mdx mice dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **3**A shows images of FLAG immunofluorescence of microdystrophin expression in B10-mdx mouse hearts following administration of Composition 1 or Reference AAV composition. FIG. **3**B shows results from capillary electrophoresis quantifying microdystrophin expression in B10-mouse hearts following administration of Composition 1 or Reference AAV composition.

[0233] FIGS. **4**A-**4**C show results from analysis of microdystrophin expression and muscle physiology in B10-mdx mice dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **4**A shows a graph of specific force of muscles from B10-mice following administration of Composition 1 or Reference AAV composition.

[0234] FIG. **4**B shows a graph of percent force drop after eccentric damage, which demonstrates protection against eccentric damage, in muscles from B10-mice following administration of Composition 1 or Reference AAV composition. FIG. **4**C shows results from capillary electrophoresis quantifying microdystrophin expression in mouse EDL muscle following administration of Composition 1 or Reference AAV composition.

[0235] FIGS. **5**A-**5**B show results from analysis of microdystrophin expression in heart left ventricles from NHP dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **5**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP heart left ventricles following administration of Composition 1 or Reference AAV composition. FIG. **5**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP heart left ventricles following administration of Composition 1 or Reference AAV composition.

[0236] FIGS. **6**A-**6**B show results from analysis of microdystrophin expression in heart right ventricles from NHP dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **6**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP heart right ventricles following administration of Composition 1 or Reference AAV composition. FIG. **6**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP heart right ventricles following administration of Composition 1 or Reference AAV composition.

[0237] FIGS. **7**A-**7**B show results from analysis of microdystrophin expression in heart interventricular septum from NHP dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **7**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP heart interventricular septum following administration of Composition 1 or Reference AAV composition. FIG. **7**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP heart interventricular septum following administration of Composition 1 or Reference AAV composition.

[0238] FIGS. **8**A-**8**B show results from analysis of microdystrophin expression in gastrocnemius from NHP dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **8**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP gastrocnemius following administration of Composition 1 or Reference AAV composition. FIG. **8**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP gastrocnemius following administration of Composition 1 or Reference AAV composition.

[0239] FIGS. **9**A-**9**B show results from analysis of microdystrophin expression in pectoralis major from NHP dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **9**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP pectoralis major following administration of Composition 1 or Reference AAV composition. FIG. **9**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP pectoralis major following administration of Composition 1 or Reference AAV composition.

[0240] FIGS. **10**A-**10**B show results from analysis of microdystrophin expression in diaphragm from NHP dosed with either Composition 1 (4E13 vg/kg) or Reference AAV composition (1.33E14 vg/kg). FIG. **10**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP diaphragm following administration of Composition 1 or Reference AAV composition. FIG. **10**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP diaphragm following administration of Composition 1 or Reference AAV composition.

[0241] FIG. **11** shows a graph of vector genome detected in the liver of NHPs.

[0242] FIG. **12** shows representative images of FLAG immunofluorescence of diaphragm, quadriceps, heart, and extensor digitorum longus (EDL) tissue of D2-mdx mice treated with Composition 1. Mice were dosed with either a low (1E13 vg/kg), a mid (2E13 vg/kg), or high (4E13 vg/kg) dose of Composition 1. WT mice and D2-mdx mice were treated with vehicle as a control. Tissues were analyzed 7 weeks post-injection.

[0243] FIGS. **13**A-**13**B show results from capillary electrophoresis analysis of microdystrophin expression in D2-mdx mouse heart at 7 weeks (FIG. **13**A) and 22 weeks (FIG. **13**B) post-injection of Composition 1. Mice were dosed with either a low (1E13 vg/kg), a mid (2E13 vg/kg), or high (4E13 vg/kg) dose of Composition 1. WT mice and D2-mdx mice were treated with vehicle as a control.

[0244] FIGS. **14**A-**14**B show results from capillary electrophoresis analysis of microdystrophin expression in D2-mdx mouse skeletal muscle at 7 weeks (FIG. **13**A) and 22 weeks (FIG. **13**B) post-injection of Composition 1 or Reference AAV composition. Mice were dosed with either a low (1E13 vg/kg), a mid (2E13 vg/kg), or high (4E13 vg/kg) dose of Composition 1. WT mice and D2-mdx mice were treated with vehicle as a control.

[0245] FIGS. **15**A-**15**B show a graph of body weight of D2-mdx mice dosed with either a low (1E13 vg/kg), a mid (2E13 vg/kg), or high (4E13 vg/kg) dose of Composition 1. Body weight was measured pre-dose and monitored twice weekly throughout the 7 weeks (FIG. **15**A) and 22 weeks (FIG. **15**B) study cohorts. WT mice and D2-mdx mice were treated with vehicle as a control. Error bars are mean±standard deviation.

[0246] FIGS. **16**A-**16**B show a graph of body weight and tibialis anterior (TA) muscle weight of D2-mdx mice dosed with either a low (1E13 vg/kg), a mid (2E13 vg/kg), or high (4E13 vg/kg) dose of Composition 1. Body weight and TA weight was measured pre-dose and monitored twice weekly throughout the 7 weeks (FIG. **16**A) and 22 weeks (FIG. **16**B) study cohorts. WT mice and D2-mdx mice were treated with vehicle as a control. Error bars are mean±standard deviation. Asterisks indicate statistical differences from mdx-vehicle mice (*$p<0.05$; ** $p<0.01$; **** $p<0.0001$).

[0247] FIGS. **17**A-**17**E show results from physiological analysis of D2-mdx mice dosed with either a low (1E13 vg/kg), a mid (2E13 vg/kg), or high (4E13 vg/kg) dose of Composition 1. WT mice and D2-mdx mice were treated with vehicle as a control. Tissues were analyzed 7 weeks post-injection. Error bars are mean±standard deviation. Asterisks, indicate statistical differences from mdx-vehicle mice (*$p<0.05$; ** $P<0.01$; *** $p<0.001$; **** $p<0.0001$; lack of asterisks

indicates non-significance). (#), indicate WT-vehicle statistical differences from mdx-vehicle mice. (*), indicate WT-vehicle and Composition 1 statistical differences from mdx-vehicle mice. FIG. **17**A shows graphs of anatomical properties of EDL muscle, including EDL weight (left panel), optimal length (Lo; middle panel), and cross-sectional area (CSA; right panel). FIG. **17**B shows graphs of quantitative measurements of muscle contractility in EDL muscle, including specific twitch force (sPt; left panel) and specific tetanic force (sPo; right panel). FIG. **17**C shows a graph of eccentric contraction of EDL muscle. FIG. **17**D shows a graph of the stress-strain relationship to evaluate the elastic properties of EDL muscle. FIG. **17**E shows a graph of the stress relaxation rate to evaluate the viscous properties of EDL muscle.
[0248] FIGS. **18**A-**18**E show results from physiological analysis of D2-mdx mice dosed with either a low (1E13 vg/kg), a mid (2E13 vg/kg), or high (4E13 vg/kg) dose of Composition 1. WT mice and D2-mdx mice were treated with vehicle as a control. Tissues were analyzed 22 weeks post-injection. Error bars are mean±standard deviation. Asterisks, indicate statistical differences from mdx-vehicle mice (*p<0.05; ** P<0.01; *** p<0.001; **** p<0.0001; lack of asterisks indicates non-significance). (#), indicate WT-vehicle statistical differences from mdx-vehicle mice. (†), indicate WT-vehicle and Composition 1 statistical differences from mdx-vehicle mice. FIG. **18**A shows graphs of anatomical properties of EDL muscle, including EDL weight (left panel), optimal length (Lo; middle panel), and cross-sectional area (CSA; right panel). FIG. **18**B shows graphs of quantitative measurements of muscle contractility in EDL muscle, including specific twitch force (sPt; left panel) and specific tetanic force (sPo; right panel). FIG. **18**C shows a graph of eccentric contraction of EDL muscle. FIG. **18**D shows a graph of the stress-strain relationship to evaluate the elastic properties of EDL muscle. FIG. **18**E shows a graph of the stress relaxation rate to evaluate the viscous properties of EDL muscle.
[0249] FIGS. **19**A-**19**C show graphs of mRNA levels of microdystrophin-FLAG in heart (FIG. **19**A), skeletal muscle (FIG. **19**B), and off-target tissues (FIG. **19**C) from NHPs dosed with 4E13 vg/kg of Composition 1. Tissues were analyzed 28 days post-injection. The transcript copy was normalized to nhpRPL30 (n=3/group). Error bars are mean±standard deviation. Avg.: average value; Sk.: skeletal; DRG: dorsal root ganglion.
[0250] FIG. **20** shows representative images of FLAG immunofluorescence of heart and skeletal muscle from NHPs dosed with 4E13 vg/kg of Composition 1. Tissues were analyzed 28 days post-injection.
[0251] FIGS. **21**A-**21**B show results from capillary electrophoresis analysis of microdystrophin expression in heart (FIG. **21**A) and skeletal muscle (FIG. **21**B) from NHPs dosed with 4E13 vg/kg of Composition 1. Tissues were analyzed at 28 days or 90-days post-injection. The amount of the microdystrophin expressed in tissues was calculated by interpolation to the full-length dystrophin standard curve derived from normal human heart or skeletal muscle samples.
[0252] FIGS. **22**A-**22**B show results from capillary electrophoresis analysis of microdystrophin expression from an endogenous coding sequence or a codon optimized microdystrophin sequence (v1, v2, or v3) in C2C12 derived myotubes.
[0253] FIGS. **23**A-**23**B show results from capillary electrophoresis analysis of microdystrophin expression from an endogenous coding sequence or a codon optimized microdystrophin sequence (v1, v2, or v3) in human primary myotubes.
[0254] FIGS. **24**A-**24**C show results from analysis of microdystrophin expression in gastrocnemius from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **24**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP gastrocnemius following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG.
[0255] FIG. **24**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP gastrocnemius following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **24**C shows a graph of mRNA levels of microdystrophin in gastrocnemius from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG.]
[0256] FIGS. **25**A-**25**C show results from analysis of microdystrophin expression in triceps from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **25**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP triceps following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **25**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP triceps following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **25**C shows a graph of mRNA levels of microdystrophin in triceps from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG.
[0257] FIGS. **26**A-**26**C show results from analysis of microdystrophin expression in biceps brachii from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **26**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP biceps brachii following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **26**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP biceps brachii following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **26**C shows a graph of mRNA levels of microdystrophin in biceps brachii from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG.
[0258] FIGS. **27**A-**27**C show results from analysis of microdystrophin expression in pectoralis major from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **27**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP pectoralis major following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **27**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP pectoralis major brachii following administration of MyoAAV-LD A- or AAV9-CBh-uDys-FLAG. FIG. **27**C shows a graph of mRNA levels of microdystrophin in pectoralis major from NHP dosed with MyoAAV-LD A- or AAV9-CBh-uDys-FLAG.
[0259] FIGS. **28**A-**28**B show results from analysis of muscle physiology in B10-mdx mice dosed with DMD tool compound (2E13 vg/kg). FIG. **28**A shows a graph of specific force following administration of the DMD tool compound to WT and mdx mice. FIG. **28**B shows a graph of protection against eccentric damage following administration of the DMD tool compound to WT and mdx mice. *: WT significantly different from mdx, #: Mdx significantly different from mdx+tool compound, ‡: WT significantly different from mdx+tool compound.

DETAILED DESCRIPTION

[0260] The present invention provides novel engineered nucleic acids encoding microdystrophin proteins for use in gene therapy.

[0261] Specifically, aspects of the invention provide a formulation comprising an AAV particle comprising (e.g., encapsidating) a nucleic acid comprising a promoter and a sequence encoding microdystrophin.

[0262] Aspects of the invention provide a composition comprising an adeno-associated virus (AAV) particle comprising an engineered capsid protein comprising the amino acid sequence RGDR (SEQ ID NO: 2440). The capsid protein encapsidates a nucleic acid molecule (e.g., a cargo) comprising a promoter, a codon optimized microdystrophin sequence, for example, a sequence having at least 95% sequence identity with a sequence selected from any one of SEQ ID NOs: 2405-2407. The nucleic acid may further comprise a dorsal root ganglia (DRG) de-targeting miRNA binding site.

DMD Gene

[0263] The DMD gene is the largest known human gene. The most common mutations that cause DMD are large deletion mutations of one or more exons (60-70%), however duplication mutations (5-10%), and single nucleotide variants (25-35%), can also cause pathogenic dystrophin variants. In DMD, mutations often lead to a frame shift resulting in a premature stop codon and a truncated, non-functional or unstable protein. Nonsense point mutations can also result in premature termination codons with the same result. While mutations causing DMD can affect any exon, exons 2-20 and 45-55 are common hotspots for large deletion and duplication mutations.

[0264] Full-length dystrophin is a large (427 kDa) protein comprising a number of subdomains that contribute to its function. These subdomains include, in order from the amino-terminus toward the carboxy-terminus, the N-terminal actin-binding domain, a central so-called “rod” domain, a cysteine-rich domain and lastly a carboxy-terminal domain or region. The rod domain is comprised of 4 proline-rich hinge domains (abbreviated H), and 24 spectrin-like repeats (abbreviated R) in the following order: a first hinge domain (H1), 3 spectrin-like repeats (R1, R2, R3), a second hinge domain (H2), 16 more spectrin-like repeats (R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19), a third hinge domain (H3), 5 more spectrin-like repeats (R20, R21, R22, R23, R24), and a fourth hinge domain (H4) (including the WW domain). Following the rod domain are the cysteine-rich domain, and the COOH (C)-terminal (CT) domain.

Optimized Nucleic Acid Sequences

[0265] Optimized nucleic acid sequences of the invention may include codon-optimization, which refers to a gene coding sequence that has been optimized to increase expression by substituting one or more codons normally present in a coding sequence with a codon for the same amino acid. In this manner, the protein encoded by the gene is identical, but the underlying nucleobase sequence of the gene or corresponding mRNA is different. In some embodiments, the optimization substitutes one or more rare codons (that is, codons for tRNA that occur relatively infrequently in cells from a particular species) with synonymous codons that occur more frequently to improve the efficiency of translation. For example, in human codon-optimization one or more codons in a coding sequence are replaced by codons that occur more frequently in human cells for the same amino acid. Codon optimization can also increase gene expression through other mechanisms that can improve efficiency of transcription and/or translation. Strategies include, without limitation, increasing total GC content (that is, the percent of guanines and cytosines in the entire coding sequence), decreasing CpG content (that is, the number of CG or GC dinucleotides in the coding sequence), removing cryptic splice donor or acceptor sites, and/or adding or removing ribosomal entry sites, such as Kozak sequences. Desirably, a codon-optimized gene exhibits improved protein expression, for example, the protein encoded thereby is expressed at a detectably greater level in a cell compared with the level of expression of the protein provided by the wild type gene in an otherwise similar cell.

[0266] Sequence identity for a nucleic acid or amino acid sequence, for example an optimized nucleic acid, refers has the standard meaning in the art.

[0267] A percent nucleic acid sequence identity refers to the percentage of nucleotide residues in the candidate sequence that are identical with the nucleotides in the polynucleotide specifically disclosed herein. The alignment may include the introduction of gaps in the sequences to be aligned. In addition, for sequences which contain either more or fewer nucleotides than the polynucleotides specifically disclosed herein, it is understood that in one embodiment, the percentage of sequence identity will be determined based on the number of identical nucleotides in relation to the total number of nucleotides. Thus, for example, sequence identity of sequences shorter than a sequence specifically disclosed herein, will be determined using the number of nucleotides in the shorter sequence, in one embodiment. In percent identity calculations relative weight is not assigned to various manifestations of sequence variation, such as insertions, deletions, substitutions, etc.

[0268] Aa number of different programs can be used to identify whether a polynucleotide or polypeptide has sequence identity or similarity to a known sequence. Sequence identity or similarity may be determined using standard techniques known in the art, including, but not limited to the BLAST algorithm, described in Altschul et al., J Mol. Biol. 215:403 (1990) and Karlin et al., Proc. Natl. Acad. Sci. USA 90:5873 (1993).

Micro RNA

[0269] MicroRNAs are small, single-stranded, non-coding RNA molecules. miRNAs base-pair to complementary sequences in mRNA molecules, thereby producing post-transcriptional regulation of gene expression. Typically, miRNA molecules silence messenger RNA (mRNA) translation by facilitating a process that leads to cleavage of mRNA strand into two pieces or destabilization of the mRNA by shortening its poly (A) tail. miRNAs are typically ˜22 nt and are expressed in a cell and tissue type specific manner.

[0270] miRNAs resemble small interfering RNAs (siRNAs), however miRNAs derive from regions of RNA transcripts that fold back on themselves to form short hairpins. Animal miRNAs are initially transcribed as part of one arm of an RNA stem-loop that in turn forms part of a several hundred nucleotide-long miRNA precursor termed a pri-miRNA. A single pri-miRNA may contain from one to six miRNA precursors. These hairpin loop structures are typically composed of about 70 nucleotides each. Each hairpin is flanked by sequences necessary for efficient processing.
[0271] The pre-miRNA hairpin is typically cleaved by the RNase enzyme Dicer. The RNase interacts with 5′ and 3′ ends of the hairpin and cuts away the loop joining the 3′ and 5′ arms, resulting in a miRNA: miRNA duplex about 22 nucleotides in length. Overall hairpin length and loop size influence the efficiency of Dicer processing. Although either strand of the duplex may potentially act as a functional miRNA, only one strand is generally incorporated into the RNA-induced silencing complex (RISC) where the miRNA and its mRNA target interact.
[0272] The present invention provides miR-338-3p, miR-138-5p, and miR-9-5p functional sites that inhibit expression of a transgene to selectively inhibit expression of the transgene in the dorsal root ganglia (DRG).
I. Recombinant Nucleic Acids
[0273] Aspects of the present disclosure provide a recombinant nucleic acid comprising a muscle specific promoter, a sequence encoding microdystrophin, and at least one DRG de-targeting miRNA binding site. In some embodiments, the recombinant nucleic acid can further comprise an intron and/or a Poly A signal. Alternatively, or in addition to, the recombinant nucleic acid can further comprise a 5′ inverted terminal repeat (ITR) and a 3′ ITR.
[0274] Various promoters, including inducible promoters and constitutive promoters, can be used to drive expression from the recombinant nucleic acids described herein. Non-limiting examples of promoters that can be used in the recombinant nucleic acids disclosed herein include a cytomegalovirus (CMV) promoter, a chicken β-actin (CBA) promoter, an alpha-1 antitrypsin (hAAT) promoter, and any promoter derived from an immunoglobulin gene, SV40, or other tissue specific genes.
[0275] In some embodiments, the promoter is tissue-specific such that, in a multi-cellular organism, the promoter drives expression only in a subset of specific cells. For example, the promoter is a muscle specific promoter. Non-limiting examples of muscle specific promoters include a CK8 promoter, a desmin promoter, a Mb promoter, a MCK promoter, a MHCK7 promoter, a skeletal muscle alpha actin promoter, or a TTNI2 promoter.
[0276] In some embodiments, the promoter is a novel promoter described herein. For example, the promoter comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404. In another example, the promoter comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2402. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2403. In some embodiments, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2404.
[0277] Recombinant nucleic acids described herein comprise a sequence encoding microdystrophin (e.g., a sequence encoding human microdystrophin). Upon administration of any of the recombinant nucleic acids described herein, microdystrophin protein (e.g., any biologically active fragment of dystrophin, e.g., ΔR4-R23/ΔCT microdystrophin) is expressed in muscle cells or muscle tissue to produce a therapeutic effect.
[0278] In some embodiments, the microdystrophin protein can comprise or consist of an amino acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to SEQ ID NO: 2439. In some embodiments, the microdystrophin protein comprises or consists of the amino acid sequence of SEQ ID NO: 2439.
[0279] The sequence encoding microdystrophin can be an endogenous sequence encoding micodystrophin or a codon optimized sequence encoding microdystrophin such as the codon optimized sequences encoding microdystrophin provided as SEQ ID NOs: 2405-2407.
[0280] In some embodiments, the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405. In some embodiments, the sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2405.
[0281] In some embodiments, the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406. In some embodiments, the sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2406.
[0282] In some embodiments, the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407. In some embodiments, the sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2407.
[0283] Any of the sequences encoding microdystrophin described herein can produce a microdystrophin protein that

retains its biological activity (e.g., muscle membrane stabilization).

[0284] For example, the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405, and microdystrophin retains biological activity (e.g., muscle membrane stabilization). In another example, the sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2405, and microdystrophin retains biological activity (e.g., muscle membrane stabilization).

[0285] In another example, the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406, and microdystrophin retains biological activity (e.g., muscle membrane stabilization). In another example, the sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2406, and microdystrophin retains biological activity (e.g., muscle membrane stabilization).

[0286] In yet another example, the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407, and microdystrophin retains biological activity (e.g., muscle membrane stabilization). In another example, the sequence encoding microdystrophin comprises or consists of the nucleic acid sequence of SEQ ID NO: 2407, and microdystrophin retains biological activity (e.g., muscle membrane stabilization).

[0287] Recombinant nucleic acids described herein can comprise at least one DRG de-targeting miRNA binding site, e.g., at least one, at least two, at least three, at least four, at least five, or more DRG de-targeting miRNA binding sites.

[0288] Any DRG de-targeting miRNA binding site suitable for binding a DRG de-targeting miRNA can be included in the recombinant nucleic acids described herein. In some embodiments, the DRG de-targeting miRNA binding site can comprise or consist of a binding site for miR-338 (e.g., miR-338-3p), miR-138 (e.g., miR-138-5p), or miR-9 (e.g., miR-9-5p).

[0289] In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401. In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401.

[0290] In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399. In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of the nucleic acid sequence of SEQ ID NOs: 2399.

[0291] In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400. In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of the nucleic acid sequence of SEQ ID NOs: 2400.

[0292] In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401. In some embodiments, the DRG de-targeting miRNA binding site comprises or consists of the nucleic acid sequence of SEQ ID NOs: 2401.

[0293] In some embodiments, when the recombinant nucleic acid comprises more than one DRG de-targeting miRNA binding site, the more than one DRG de-targeting miRNA binding site can have the same sequence or a different sequence.

[0294] For example, when the recombinant nucleic acid comprises a first and a second DRG de-targeting miRNA binding site, each of the DRG de-targeting miRNA binding sites can comprise or consist of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401.

[0295] Alternatively, the first and the second DRG de-targeting miRNA binding sites can comprise or consist of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequence of SEQ ID NO: 2399 and SEQ ID NO: 2400, respectively, or vice versa; the first and the second DRG de-targeting miRNA binding sites can comprise or consist of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequence of SEQ ID NO: 2399 and SEQ ID NO: 2401, respectively, or vice versa; or the first and the second DRG de-targeting miRNA binding sites can comprise or consist of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequence of SEQ ID NO: 2400 and SEQ ID NO: 2401, respectively, or vice versa.

[0296] The at least one DRG de-targeting miRNA binding site can be positioned in the recombinant nucleic acids described herein at any location suitable for binding a DRG de-targeting miRNA. For example, the DRG de-targeting miRNA binding site can be positioned upstream or downstream of the sequence encoding microdystrophin. In another example, when the recombinant nucleic acid comprises more than one DRG de-targeting miRNA binding site, each of the DRG de-targeting miRNA binding sites can be positioned upstream or downstream of the sequence encoding microdystrophin. Alternatively, one or more of the DRG de-targeting miRNA binding sites can be positioned upstream of the sequence encoding microdystrophin and one or more of the DRG de-targeting miRNA binding sites can be positioned downstream of the sequence encoding microdystrophin.

[0297] Recombinant nucleic acids described herein can further comprise an intron to increase overall efficiency and output of gene expression. Preferably, the intron is positioned between the promoter and transgene (e.g., microdystrophin transgene) in a 5′ to 3′ orientation. Non-limiting examples of introns include a simian virus 40 (SV40) intron sequence, a minute virus of mice (MVM) intron, a RK intron, a β-globin intron, and/or a chicken β-actin (CBA) intron, including functional variants or fragments thereof.

[0298] In some embodiments, the intron is a SV40 intron, including functional variants or fragments thereof. In some embodiments, the SV40 intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409. In some embodiments, the SV40 intron comprises or consists of a nucleic acid sequence comprising at least one, two, or three, but no more than four, modifications, e.g., substitutions, relative to the nucleic acid sequence of SEQ ID NO: 2409. In some embodiments, the SV40 intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409.

[0299] Recombinant nucleic acids described herein can further comprise a polyadenylation (PolyA) signal sequence or functional fragment thereof, e.g., a fragment capable of measurably increasing expression as compared to expression in its absence. Non-limiting examples of PolyA signal sequences include a bovine growth hormone polyadenylation (bgh-PolyA) signal, a synthetic PolyA signal, and an SV40 Poly A signal.

[0300] In some embodiments, the Poly A signal comprises a bgh-PolyA signal, and the bgh-PolyA signal comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410. In some embodiments, the PolyA signal comprises a bgh-PolyA signal, and the bgh-PolyA signal comprises or consists of the nucleic acid sequence of SEQ ID NO: 2410.

[0301] Recombinant nucleic acids described herein can comprise one or more Poly A signal sequences or functional fragments thereof. In some embodiments, the recombinant nucleic acid described herein comprises a poly A sequence between the 3′ end of a sequence encoding the microdystrophin transgene and the 5′ end of the 3′ ITR.

[0302] Recombinant nucleic acids described herein can comprise one or more ITRs, e.g., two ITRs, with one upstream and the other downstream of a sequence encoding microdystrophin and/or the other elements discussed above (e.g., a promoter, at least one DRG de-targeting miRNA binding site).

[0303] An ITR sequence may be wild type, or it may comprise one or more mutations, e.g., by the insertion, deletion, or substitution of nucleotides, as long as the sequences retain one or more function of a wild type ITR, such as providing for functional rescue, replication and packaging. In some embodiments, a recombinant nucleic acid as described herein can comprise two ITR sequences, each of which is wild type, variant, or modified ITR sequences. In some embodiments, a recombinant nucleic acid as described herein can comprise two ITR sequences, which are different (e.g., one wild type ITR sequence and one variant or modified ITR sequence). For example, the “left” or 5′ ITR is a modified ITR sequence that allows for production of self-complementary genomes, and the “right” or 3′ ITR is a wild type ITR sequence. In some embodiments, the “right” or 3′ ITR is a modified ITR sequence that allows for the production of self-complementary genomes, and the “left” or 5′ ITR is a wild type ITR sequence.

[0304] In some embodiments, a recombinant nucleic acid can comprise ITR sequences from any AAV serotype suitable for replication and packaging of the virus. Non-limiting examples AAV serotypes include AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV6.2, AAV7, AAV8, AAV9, AAV10, AAV11, AAV12, AAV13, AAVrh, AAVrh10, AAVrh74, AAV-DJ, AAV-DJ/8, Anc80, AAV7m8, or any derivative, hybrid or chimeric serotype thereof.

[0305] In some embodiments, the recombinant nucleic acid comprises one or more ITRs derived from AAV2. The nucleotides in an ITR sequence may be in a forward or reverse orientation. In some embodiments, the ITR sequences are both wild type, variant, or modified AAV2 ITR sequences. In some embodiments, the ITR sequences are different (e.g., one ITR is a wild type AAV2 sequence and one ITR is variant or modified AAV2 ITR sequence). In some embodiments, the “left” or 5′ ITR is a modified AAV2 ITR sequence that allows for production of self-complementary genomes, and the “right” or 3′ ITR is a wild type AAV2 ITR sequence. In some embodiments, the “right” or 3′ ITR is a modified AAV2 ITR sequence that allows for the production of self-complementary genomes, and the “left” or 5′ ITR is a wild type AAV2 ITR sequence. In some embodiments, the 5′ ITR is an AAV2 5′ ITR as described in McCarty, D. M., et al., Gene Ther (2001), which is incorporated by reference in its entirety. In some embodiments, the 3′ ITR is an AAV2 3′ ITR as described in

Rolling, F. & Samulski, R. J. Molecular biotechnology (1995), which is incorporated by reference in its entirety. In some embodiments, the 3′ ITR may comprise a deletion of the terminal resolution site (dTR), which inhibits Rep protein nicking of the single stranded viral genome. The presence of the dTR in the 3′ ITR increases self-complementary binding of the viral genome to itself, which it may do because of its small size that allows for a double-stranded viral genome to be packaged within a viral capsid.

[0306] In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408. In some embodiments, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408.

[0307] In some embodiments, the recombinant nucleic acid comprises a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411. In some embodiments, the recombinant nucleic acid comprises a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

Example Recombinant Nucleic Acids

[0308] Non-limiting examples of recombinant nucleic acids for use in methods described herein are provided below. In some examples, the recombinant nucleic acid comprises, optionally from 5′ to 3′, a muscle specific promoter; a sequence encoding microdystrophin; and at least one DRG de-targeting miRNA binding site.

[0309] For example, the recombinant nucleic acid comprises, optionally from 5′ to 3′, a muscle specific promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401.

[0310] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0311] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0312] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0313] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0314] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0315] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0316] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0317] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0318] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0319] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least

92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0320] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0321] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0322] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0323] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0324] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0325] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0326] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0327] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0328] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0329] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0330] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399.

[0331] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0332] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0333] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400.

[0334] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0335] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0336] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

[0337] In some examples, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0338] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic

acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0339] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0340] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0341] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0342] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0343] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0344] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0345] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405;

[0346] and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0347] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0348] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0349] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0350] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0351] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0352] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0353] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0354] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic

acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0355] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0356] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0357] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0358] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399.

[0359] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0360] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0361] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400.

[0362] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0363] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0364] In another example, the recombinant nucleic acid comprises a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

[0365] In some examples, the recombinant nucleic acid comprises, optionally from 5′ to 3′, a 5′ ITR; a muscle specific promoter; an intron; a sequence encoding microdystrophin; a first DRG de-targeting miRNA binding site; a second DRG de-targeting miRNA binding site; a Poly A signal; and a 3′ ITR.

[0366] For example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0367] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0368] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0369] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0370] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0371] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least

86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0372] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0373] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0374] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least

94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0375] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0376] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399;

a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0377] In another example, the recombinant nucleic acid comprises a 5′ comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0378] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0379] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0380] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0381] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least

97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0382] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0383] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0384] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0385] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0386] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least

90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0387] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0388] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0389] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least

96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0390] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0391] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

[0392] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.
[0393] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.
[0394] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.
[0395] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding

site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0396] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0397] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0398] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0399] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0400] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0401] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0402] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0403] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0404] In another example, the recombinant nucleic acid comprises a 5′ comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0405] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0406] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0407] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0408] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0409] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0410] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of

SEQ ID NO: 2411.

[0411] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0412] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0413] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0414] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0415] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0416] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence of SEQ ID NO: 2411.

[0417] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0418] In another example, the recombinant nucleic acid comprises a 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0419] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0420] In another example, the recombinant nucleic acid comprises a 5′ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; a promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; an intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; a sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; a first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; a PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and a 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.

[0421] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2412. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2412.

[0422] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2413. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2413.

[0423] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2414. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2414.

[0424] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2415. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2415.

[0425] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2416. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2416.

[0426] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2417. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2417.

[0427] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2418. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2418.

[0428] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2419. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2419.

[0429] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least

85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2420. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2420.

[0430] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2421. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2421.

[0431] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2422. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2422.

[0432] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2423. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2423.

[0433] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2424. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2424.

[0434] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2425. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2425.

[0435] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2426. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2426.

[0436] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2427. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2427.

[0437] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2428. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2428.

[0438] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2429. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2429.

[0439] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2430. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2430.

[0440] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2431. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2431.

[0441] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least

94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2432. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2432.

[0442] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2433. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2433.

[0443] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2434. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2434.

[0444] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2435. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2435.

[0445] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2436. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2436.

[0446] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2437. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2437.

[0447] In another example, the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2438. In another example, the recombinant nucleic acid comprises or consists of the nucleic acid sequence of SEQ ID NO: 2438.

II. Recombinant Expression Vectors

[0448] Any of the recombinant nucleic acids described herein can be delivered to a cell (e.g., a muscle cell), a tissue (e.g., muscle tissue), or subject (e.g., human subject) without or with the use of a delivery system (e.g., recombinant expression vector (e.g., viral vector (e.g., AAV (e.g., rAAV))).

(a) Recombinant Expression Vectors

[0449] Any of the recombinant nucleic acids described herein can be included in a recombinant expression vector (e.g., a viral vector (e.g., an AAV vector (e.g., arAAV vector))). Recombinant expression vectors can be delivered into a cell, a tissue, or a subject using any method known in the art or described herein. A recombinant expression vector can be an organic or inorganic molecule. A recombinant expression vector can be a small molecule (i.e., <5 kD) or a macromolecule (i.e., >5 kD). The recombinant expression vector can comprise DNA, RNA, or both DNA and RNA. The recombinant expression vector can be a DNA vector, a circular vector, or a plasmid. The recombinant expression vector can be double stranded or single stranded.

[0450] In some embodiments, a recombinant expression vector (e.g., a viral vector (e.g., an AAV (e.g., a rAAV))) disclosed herein can exhibit higher expression of the recombinant nucleic acid (e.g., expression of microdystrophin) in a specific tissue type (e.g., muscle) as compared to the expression of the same vector in a different tissue type (e.g., liver). For example, the recombinant expression vector can exhibit higher expression of microdystrophin in a skeletal muscle tissue or cell as compared to the expression of the same vector in a non-skeletal muscle tissue (e.g., heart, liver) or cell (e.g., heart cell, liver cell). In some embodiments, the vector exhibits at least about 1.2 fold, at least about 1.5 fold, at least about 1.75 fold, at least about 2 fold, at least about 2.5 fold, at least about 3 fold, at least about 4 fold higher expression of the recombinant nucleic acid (e.g., expression of microdystrophin) in a skeletal muscle tissue or cell as compared to the expression of the same vector in a non-skeletal muscle tissue (e.g., heart, liver) or cell (e.g., heart cell, liver cell).

[0451] In some embodiments, the recombinant expression vector is a viral vector. Non-limiting examples of viral vectors for use in compositions and methods described herein include retroviral vectors, adenoviral vectors, lentiviral vectors, adeno-associated viral (AAVs) vectors, herpes simplex viral (HSV) vectors, alphaviral vectors, pox viral vectors, murine leukemia viral vectors, and hybrid viral vectors.

(b) Adeno Associated Virus Vectors

[0452] AAVs are particularly appropriate viral vectors for delivery of genetic material into mammalian cells. AAVs are not known to cause disease in mammals and cause a very mild immune response. Additionally, AAVs are able to infect cells in

multiple stages whether at rest or in a phase of the cell replication cycle. Advantageously, AAV DNA is not regularly inserted into the host's genome at random sites, reducing the oncogenic properties of this vector.
[0453] AAVs have been engineered to deliver a variety of treatments, especially for genetic disorders caused by single nucleotide polymorphisms ("SNP"). Genetic diseases that have been studied in conjunction with AAV vectors include Cystic fibrosis, hemophilia, arthritis, macular degeneration, muscular dystrophy, Parkinson's disease, congestive heart failure, and Alzheimer's disease. The AAV can be used as a vector to deliver engineered nucleic acid to a host and utilize the host's own ribosomes to transcribe that nucleic acid into the desired proteins. See, e.g., West et al., Virology 160:38-47 (1987); U.S. Pat. No. 4,797,368; WO 93/24641; Kotin, Human Gene Therapy 5:793-801 (1994); and Muzyczka, J. Clin. Invest. 94:1351 (1994). AAVs have some deficiency in their replication and/or pathogenicity and thus can be safer that adenoviral vectors. In some embodiments, the AAV can integrate into a specific site on chromosome 19 of a human cell with no observable side effects. In some embodiments, the capacity of the AAV vector, system thereof, and/or AAV particles can be up to about 4.7 kb. The AAV vector or system thereof can include one or more engineered capsid polynucleotides described herein.
[0454] AAVs are small, replication-defective, nonenveloped viruses that infect humans and other primate species and have a linear single-stranded DNA genome. Naturally occurring AAV serotypes exhibit liver tropism. As a result, transfection of non-liver tissue with traditional AAV vectors is impeded by the virus's natural liver tropism. Moreover, because the liver acts to break down substances delivered to a subject, transfection of non-liver tissue with unmodified AAV vectors requires higher dosing to provide sufficient viral load to overcome the liver and reach non-liver tissue. More than 30 naturally occurring serotypes of AAV are available. Many natural variants in the AAV capsid exist. AAV serotypes include, but are not limited to, AAV serotypes AAV1, AAV2, AAV3, AAV3B, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, AAV12, AAV13. AAVs may be engineered using conventional molecular biology techniques, making it possible to optimize these particles, for example, for cell specific delivery, for minimizing immunogenicity, for tuning stability and particle lifetime, for efficient degradation, for accurate delivery to the nucleus. AAV vectors can be specifically targeted to one or more types of cells by choosing the appropriate combination of AAV serotype, promoter, and delivery method.
[0455] Previous approaches to identify AAV sequences correlated with tropism have relied upon the comparison of highly related extant serotypes with distinct characteristics, random domain swaps between unrelated serotypes, or consideration of higher-order structure, to identify motifs that define liver tropism. For example, mapping determinants of AAV tropism have been carried out by comparing highly related serotypes. One such example is the single-amino acid change (E531K) between AAV1 and AAV6 that improves murine liver transduction in AAV1. See, e.g., Wu et al. (2006) J. Virol., 80 (22): 11393-7, incorporated by reference herein. Another example is a reciprocal domain swap between AAV2 and AAV8 that alters tropism, but fails to define any robust specific tissue-targeting motifs. See, e.g., Raupp et al. (201) J. Virol., 86 (17): 9396-408, incorporated by reference herein. Further, global consideration of structure has only highlighted gross differences between better- or worse-liver-transducers that are more observational than useful in practice. Nam et al (2007) J. Virol., 81 (22): 12260-71.
[0456] AAVs exhibiting modified tissue tropism that may be used with the present invention are described in U.S. Pat. Nos. 9,695,220, 9,719,070; 10,119,125; 10,526,584; U.S. Patent Application Publication No. 2018-0369414; U.S. Patent Application Publication No. 2020-0123504; U.S. Patent Application Publication No. 2020-0318082; PCT International Patent Application Publication No. WO 2015/054653; PCT International Patent Application Publication No. WO 2016/179496; PCT International Patent Application Publication No. WO 2017/100791; and PCT International Patent Application Publication No. WO 2019/217911, the entirety of the contents of each of which are incorporated by reference herein.
[0457] The AAV vector or system thereof may include one or more regulatory molecules, such as promoters, enhancers, repressors and the like. In some embodiments, the AAV vector or system thereof can include one or more polynucleotides that can encode one or more regulatory proteins. In some embodiments, the one or more regulatory proteins can be selected from Rep78, Rep68, Rep52, Rep40, variants thereof, and combinations thereof. In some embodiments, the muscle specific promoter can drive expression of an engineered AAV capsid polynucleotide.
[0458] The AAV vector or system thereof can include one or more polynucleotides that can encode one or more capsid proteins, such as the engineered AAV capsid proteins described elsewhere herein. The engineered capsid proteins can be capable of assembling into a protein shell (an engineered capsid) of the AAV virus particle. The engineered capsid can have a cell-, tissue-, and/or organ-specific tropism.
[0459] The AAV vector or system thereof can be configured to produce AAV particles having a specific serotype. In some embodiments, the serotype can be AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9 or any combinations thereof. In some embodiments, the AAV can be AAV1, AAV2, AAV5, AAV9 or any combination thereof. One can select the AAV of the AAV with regard to the cells to be targeted; e.g., one can select AAV serotypes 1, 2, 5, 9 or a hybrid capsid AAV1, AAV2, AAV5, AAV9 or any combination thereof for targeting brain and/or neuronal cells; and one can select AAV4 for targeting cardiac tissue; and one can select AAV8 for delivery to the liver. Thus, in some embodiments, an AAV vector or system thereof capable of producing AAV particles capable of targeting the brain and/or neuronal cells can be configured to generate AAV particles having serotypes 1, 2, 5 or a hybrid capsid AAV1, AAV2, AAV5 or any combination thereof. In some embodiments, an AAV vector or system thereof capable of producing AAV particles capable of targeting cardiac tissue can be configured to generate an AAV particle having an AAV4 serotype. In some embodiments, an AAV vector or system thereof capable of producing AAV particles capable of targeting the liver can be configured to generate

an AAV having an AAV8 serotype. See also Srivastava. 2017. Curr. Opin. Virol. 21:75-80.

[0460] It will be appreciated that while the different serotypes can provide some level of cell, tissue, and/or organ specificity, each serotype still is multi-tropic and thus can result in tissue-toxicity if using that serotype to target a tissue that the serotype is less efficient in transducing. Thus, in addition to achieving some tissue targeting capacity via selecting an AAV of a particular serotype, it will be appreciated that the tropism of the AAV serotype can be modified by an engineered AAV capsid described herein. As described elsewhere herein, variants of wild type AAV of any serotype can be generated via a method described herein and determined to have a particular cell-specific tropism, which can be the same or different as that of the reference wild type AAV serotype. In some embodiments, the cell, tissue, and/or specificity of the wild type serotype can be enhanced (e.g., made more selective or specific for a particular cell type that the serotype is already biased towards). For example, wild type AAV9 is biased towards muscle and brain in humans (see, e.g., Srivastava. 2017. Curr. Opin. Virol. 21:75-80.) By including an engineered AAV capsid and/or capsid protein variant of wild type AAV9 as described herein, the tropism for nervous cells might be reduced or eliminated and/or the muscle specificity increased such that the nervous specificity appears reduced in comparison, thus enhancing the specificity for muscle as compared to the wild type AAV9. As previously mentioned, inclusion of an engineered capsid and/or capsid protein variant of a wild type AAV serotype can have a different tropism than the wild type reference AAV serotype. For example, an engineered AAV capsid and/or capsid protein variant of AAV9 can have specificity for a tissue other than muscle or brain in humans.

[0461] In some embodiments, the AAV vector is a hybrid AAV vector or system thereof. Hybrid AAVs are AAVs that include genomes with elements from one serotype that are packaged into a capsid derived from at least one different serotype. For example, if it is the rAAV2/5 that is to be produced, and if the production method is based on the helper-free, transient transfection method discussed above, the 1st plasmid and the 3rd plasmid (the adeno helper plasmid) will be the same as discussed for rAAV2 production. However, the 2nd plasmid, the pRepCap will be different. In this plasmid, called pRep2/Cap5, the Rep gene is still derived from AAV2, while the Cap gene is derived from AAV5. The production scheme is the same as the above-mentioned approach for AAV2 production. The resulting rAAV is called rAAV2/5, in which the genome is based on recombinant AAV2, while the capsid is based on AAV5. It is assumed the cell or tissue-tropism displayed by this AAV2/5 hybrid virus should be the same as that of AAV5. It will be appreciated that wild type hybrid AAV particles suffer the same specificity issues as with the non-hybrid wild type serotypes previously discussed.

[0462] Advantages achieved by the wild type based hybrid AAV systems can be combined with the increased and customizable cell-specificity that can be achieved with the engineered AAV capsids can be combined by generating a hybrid AAV that can include an engineered AAV capsid described elsewhere herein. It will be appreciated that hybrid AAVs can contain an engineered AAV capsid containing a genome with elements from a different serotype than the reference wild type serotype that the engineered AAV capsid is a variant of. For example, a hybrid AAV can be produced that includes an engineered AAV capsid that is a variant of an AAV9 serotype that is used to package a genome that contains components (e.g., rep elements) from an AAV2 serotype. As with wild type based hybrid AAVs previously discussed, the tropism of the resulting AAV particle will be that of the engineered AAV capsid.

[0463] In some embodiments, the AAV vector or system thereof is configured as a “gutless” vector, similar to that described in connection with a retroviral vector. In some embodiments, the “gutless” AAV vector or system thereof can have the cis-acting viral DNA elements involved in genome amplification and packaging in linkage with the heterologous sequences of interest (e.g., the engineered AAV capsid polynucleotide(s)).

[0464] The vectors described herein can be constructed using any suitable process or technique. In some embodiments, one or more suitable recombination and/or cloning methods or techniques can be used to the vector(s) described herein. Suitable recombination and/or cloning techniques and/or methods can include, but not limited to, those described in U.S. Application publication No. US 2004-0171156 A1. Other suitable methods and techniques are described elsewhere herein.

[0465] Construction of recombinant AAV vectors are described in a number of publications, including U.S. Pat. No. 5,173,414; Tratschin et al., Mol. Cell. Biol. 5:3251-3260 (1985); Tratschin, et al., Mol. Cell. Biol. 4:2072-2081 (1984); Hermonat & Muzyczka, PNAS 81:6466-6470 (1984); and Samulski et al., J. Virol. 63:03822-3828 (1989). Any of the techniques and/or methods can be used and/or adapted for constructing an AAV or other vector described herein. AAV vectors are discussed elsewhere herein.

[0466] In some embodiments, the vector can have one or more insertion sites, such as a restriction endonuclease recognition sequence (also referred to as a “cloning site”). In some embodiments, one or more insertion sites (e.g., about or more than about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more insertion sites) are located upstream and/or downstream of one or more sequence elements of one or more vectors.

[0467] Delivery vehicles, vectors, particles, nanoparticles, formulations and components thereof for expression of one or more elements of a engineered AAV capsid system described herein are as used in the foregoing documents, such as International Patent Application Publications WO WO 2021/050974 and WO 2021/077000 and PCT International Application No. WO 2022/020616, the contents of which are incorporated by reference herein.

[0468] Additional AAV vectors are described in International Patent Application Publication WO 2019/2071632, the contents of which are incorporated by reference herein.

[0469] Further AAV vectors are described in International Patent Application Publications WO 2020/086881 and WO 2020/235543, the contents of each of which are incorporated by reference herein.

[0470] Further AAV vectors are described in International Patent Application Publications WO 2005/033321; WO

2006/110689; WO 2007/127264; WO 2008/027084; WO 2009/073103; WO 2009/073104; WO 2009/105084; WO 2009/134681; WO 2009/136977; WO 2010/051367; WO 2010/138675; WO 2001/038187; WO 2012/112832; WO 2015/054653; WO 2016/179496; WO 2017/100791; WO 2017/019994; WO 2018/209154; WO 2019/067982; WO 2019/195701; WO 2019/217911; WO 2020/041498; WO 2020/210839; U.S. Pat. Nos. 7,906,111; 9,737,618; 10,265,417; 10,485,883; 10,695,441; 10,722,598; 8,999,678; 10,301,648; 10,626,415; 9,198,984; 10,155,931; 8,524,219; 9,206,238; 8,685,387; 9,359,618; 8,231,880; 8,470,310; 9,597,363; 8,940,290; 9,593,346; 10,501,757; 10,786,568; 10,973,928; 10,519,198; 8,846,031; 9,617,561; 9,884,071; 10,406,173; 9,596,220; 9,719,010; 10,117,125; 10,526,584; 10,881,548; 10,738,087; U.S. Patent Publication No. 2011-023353; U.S. Patent Publication No. 2019-0015527; U.S. Patent Publication No. 2020-155704; U.S. Patent Publication No 2017-0191079; U.S. Patent Publication No. 2019-0218574; U.S. Patent Publication No. 2020-0208176; U.S. Patent Publication No. 2020-0325491; U.S. Patent Publication No. 2019-0055523; U.S. Patent Publication No. 2020-0385689; U.S. Patent Publication No. 2009-0317417; U.S. Patent Publication No. 2016-0051603; U.S. Patent Publication No. 2016-00244783; U.S. Patent Publication No. 2017-0183636; U.S. Patent Publication No. 2020-0263201; U.S. Patent Publication No. 2020-0101099; U.S. Patent Publication No. 2020-0318082; U.S. Patent Publication No. 2018-0369414; U.S. Patent Publication No. 2019-0330278; U.S. Patent Publication No. 2020-0231986, the contents of each of which are incorporated by reference herein.

III. Viral Particles and Capsid Proteins

[0471] Any of the recombinant nucleic acids or recombinant expression vectors (e.g., recombinant expression vector (e.g., viral vector (e.g., AAV (e.g., rAAV))) described herein can be packaged or encapsulated in a viral particle (e.g., AAV particle). Viral particles for use in compositions and methods described herein can include any viral capsid protein (e.g., AAV capsid protein or variant thereof) known in the art or described herein.

[0472] In some embodiments, a viral particle (e.g., an AAV particle (e.g., arAAV particle)) disclosed herein can exhibit higher expression of the recombinant nucleic acid (e.g., expression of microdystrophin) in a specific tissue type as compared to the expression of the same viral particle in a different tissue type. For example, the viral particle can exhibit higher expression of microdystrophin in a skeletal muscle tissue or cell as compared to the expression of the same viral particle in a non-skeletal muscle tissue (e.g., heart, liver) or cell (e.g., heart cell, liver cell). In some embodiments, the viral particle exhibits at least about 1.2 fold, at least about 1.5 fold, at least about 1.75 fold, at least about 2 fold, at least about 2.5 fold, at least about 3 fold, at least about 4 fold higher expression of the recombinant nucleic acid (e.g., expression of microdystrophin) in a skeletal muscle tissue or cell as compared to the expression of the same viral vector in a non-skeletal muscle tissue (e.g., heart, liver) or cell (e.g., heart cell, liver cell).

[0473] Viral particles described herein can include any virus suitable for delivery of a transgene, e.g., retroviruses, adenovirus, lentivirus, AAV, and murine leukemia viruses. In some embodiments, the viral particle is a recombinant adenovirus comprising a recombinant nucleic acid or a recombinant expression vector described herein. In some embodiments, the viral particle is a recombinant AAV comprising a recombinant nucleic acid or a recombinant AAV vector described herein. The AAV particle can be a scAAV or a ssAVV.

[0474] AAV particles described herein can include one or more AAV capsid proteins. In some embodiments, the one or more AAV capsid proteins can be from one or more AAV serotypes. Non-limiting examples of AAV serotypes include AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, AAV12, AAVrh8, AAVfh10, AAV-DJ, AAV-DJ/8, AAV-PHP.B, AAV-PHP.B2, AAV-PHP.B3, AAV-PHP.A, AAV-PHP.eB, AAV-PHP.S, and functional variants of any AAV serotypes.

[0475] In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence of formula (I) X.sub.1NX.sub.2X.sub.3X.sub.4RGDRX.sub.5X.sub.6L (SEQ ID NO: 1), wherein each of X.sub.1-X.sub.6 is any amino acid. In some embodiments, X.sub.1 is A, I, F, G, H, L, M, Q, S, T, or V. In some embodiments, X.sub.2 is A, G, S, T, or Y. In some embodiments, X.sub.3 is S, N, G, or P. In some embodiments, X.sub.4 is A, G, H, I, M, S, T, or V. In some embodiments, X.sub.5 is A, G, or Q. In some embodiments, X.sub.6 is A, I, L, M, N, S, or Y.

[0476] The amino acid sequence of formula (I) can be in a hypervariable region IV (HVR IV) relative to a wild type AAV9 capsid. For example, relative to a wild type AAV9 capsid, X.sub.1 is a substitution at amino acid 451, X.sub.2 is a substitution at amino acid 453, X.sub.3 is a substitution at amino acid 454, X.sub.4 is a substitution at amino acid 455, and RGDRX.sub.5X.sub.6L (SEQ ID NO: 2441) is inserted after amino acid 455.

[0477] In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence of any one of SEQ ID NOs: 801-1599. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence of any one of SEQ ID NOs: 1600-2398. In some embodiments, the AAV particle comprises or consists of a capsid protein comprising an amino acid sequence of any one of SEQ ID NOs: 2-800. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NOs: 191, 198, 199, 215, 256, 379, 544, 550, or 748. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 191. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 198. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 199. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 215. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 256. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 379. In some embodiments,

the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 544. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 550. In some embodiments, the AAV particle comprises a capsid protein comprising an amino acid sequence comprising or consisting of SEQ ID NO: 748.
[0478] Any of the capsid proteins described herein can further include a deletion. For example, the capsid protein can further comprise a deletion of amino acid G267 relative to a wild type AAV9 vector.
[0479] The capsid protein is the shell or coating of the virus that enables its delivery into the host. Without the protein, the nucleic acids would be destroyed by the host without entering into the host cells and beginning transcription and translation. The capsid protein may be in the natural conformation of a naturally occurring AAV, or it may be modified.
[0480] In certain example embodiments, the AAV capsid protein is an engineered AAV capsid protein having reduced or eliminated uptake in a non-muscle cell as compared to a corresponding wild type AAV capsid polypeptide.
[0481] In some embodiments, the engineered AAV capsid encoding polynucleotide can be included in a polynucleotide that is configured to be an AAV genome donor in an AAV vector system that can be used to generate engineered AAV particles described elsewhere herein. In some embodiments, the engineered AAV capsid encoding polynucleotide can be operably coupled to a poly adenylation tail. In some embodiments, the poly adenylation tail can be an SV40 poly adenylation tail. In some embodiments, the AAV capsid encoding polynucleotide can be operably coupled to a promoter. In some embodiments, the promoter can be a tissue specific promoter. In some embodiments, the tissue specific promoter is specific for muscle (e.g., cardiac, skeletal, and/or smooth muscle), neurons and supporting cells (e.g., astrocytes, glial cells, Schwann cells, etc.), fat, spleen, liver, kidney, immune cells, spinal fluid cells, synovial fluid cells, skin cells, cartilage, tendons, connective tissue, bone, pancreas, adrenal gland, blood cell, bone marrow cells, placenta, endothelial cells, and combinations thereof. In some embodiments, the promoter can be a constitutive promoter. Suitable tissue specific promoters and constitutive promoters are discussed elsewhere herein and are generally known in the art and can be commercially available. Suitable muscle specific promoters include, but are not limited to CK8, MHCK7, Myoglobin promoter (Mb), Desmin promoter, muscle creatine kinase promoter (MCK) and variants thereof, and SPc5-12 synthetic promoter.
[0482] Described herein are various embodiments of engineered viral capsids, such as adeno-associated virus (AAV) capsids, that can be engineered to confer cell-specific tropism, such as muscle specific tropism, to an engineered viral particle. Engineered viral capsids can be lentiviral, retroviral, adenoviral, or AAV capsids. The engineered capsids can be included in an engineered virus particle (e.g., an engineered lentiviral, retroviral, adenoviral, or AAV virus particle), and can confer cell-specific tropism, reduced immunogenicity, or both to the engineered viral particle. The engineered viral capsids described herein can include one or more engineered viral capsid proteins described herein. The engineered viral capsids described herein can include one or more engineered viral capsid proteins described herein that can contain a muscle-specific targeting moiety containing or composed of an n-mer motif described elsewhere herein.
[0483] The engineered viral capsid and/or capsid proteins can be encoded by one or more engineered viral capsid polynucleotides. In some embodiments, the engineered viral capsid polynucleotide is an engineered AAV capsid polynucleotide, engineered lentiviral capsid polynucleotide, engineered retroviral capsid polynucleotide, or engineered adenovirus capsid polynucleotide. In some embodiments, an engineered viral capsid polynucleotide (e.g., an engineered AAV capsid polynucleotide, engineered lentiviral capsid polynucleotide, engineered retroviral capsid polynucleotide, or engineered adenovirus capsid polynucleotide) can include a 3′ polyadenylation signal. The polyadenylation signal can be an SV40 polyadenylation signal.
[0484] The engineered viral capsids can be variants of wild type viral capsid. For example, in some embodiments, the engineered AAV capsids can be variants of wild type AAV capsids. In some embodiments, the wild type AAV capsids can be composed of VP1, VP2, VP3 capsid proteins or a combination thereof. In other words, the engineered AAV capsids can include one or more variants of a wild type VP1, wild type VP2, and/or wild type VP3 capsid proteins. In some embodiments, the serotype of the reference wild type AAV capsid can be AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV8, AAV9 or any combination thereof. In some embodiments, the serotype of the wild type AAV capsid can be AAV9. The engineered AAV capsids can have a different tropism than that of the reference wild type AAV capsid.
[0485] The engineered viral capsid can contain 1-60 engineered capsid proteins. In some embodiments, the engineered viral capsids can contain 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, or 60 engineered capsid proteins. In some embodiments, the engineered viral capsid can contain 0-59 wild type viral capsid proteins. In some embodiments, the engineered viral capsid can contain 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, or 59 wild type viral capsid proteins.
[0486] In some embodiments, the engineered AAV capsid can contain 1-60 engineered capsid proteins. In some embodiments, the engineered AAV capsids can contain 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, or 60 engineered capsid proteins. In some embodiments, the engineered AAV capsid can contain 0-59 wild type AAV capsid proteins. In some embodiments, the engineered AAV capsid can contain 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, or 59 wild type AAV capsid proteins.

[0487] In some embodiments, the engineered viral capsid protein can have an n-mer amino acid motif, where n can be at least 3 amino acids. In some embodiments, n can be 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acids. In some embodiments, an engineered AAV capsid can have a 6-mer or 7-mer amino acid motif. In some embodiments, the n-mer amino acid motif can be inserted between two amino acids in the wild type viral protein (VP) (or capsid protein). In some embodiments, the n-mer motif can be inserted between two amino acids in a variable amino acid region in a viral capsid protein.

[0488] In some embodiments, the n-mer motif can be inserted between two amino acids in a variable amino acid region in an AAV capsid protein. The core of each wild type AAV viral protein contains an eight-stranded beta-barrel motif (betaB to betaI) and an alpha-helix (alphaA) that are conserved in autonomous parvovirus capsids (see, e.g., DiMattia et al. 2012. J. Virol. 86 (12): 6947-6958). Structural variable regions (VRs) occur in the surface loops that connect the beta-strands, which cluster to produce local variations in the capsid surface. AAVs have 12 variable regions (also referred to as hypervariable regions) (see, e.g., Weitzman and Linden. 2011. "Adeno-Associated Virus Biology." In Snyder, R. O., Moullier, P. (eds.) Totowa, NJ: Humana Press). In some embodiments, one or more n-mer motifs can be inserted between two amino acids in one or more of the 12 variable regions in the wild type AVV capsid proteins. In some embodiments, the one or more n-mer motifs can be each be inserted between two amino acids in VR-I, VR-II, VR-III, VR-IV, VR-V, VR-VI, VR-VII, VR-III, VR-IX, VR-X, VR-XI, VR-XII, or a combination thereof. In some embodiments, the n-mer can be inserted between two amino acids in the VR-III of a capsid protein. In some embodiments, the engineered capsid can have an n-mer inserted between any two contiguous amino acids between amino acids 262 and 269, between any two contiguous amino acids between amino acids 327 and 332, between any two contiguous amino acids between amino acids 382 and 386, between any two contiguous amino acids between amino acids 452 and 460, between any two contiguous amino acids between amino acids 488 and 505, between any two contiguous amino acids between amino acids 545 and 558, between any two contiguous amino acids between amino acids 581 and 593, between any two contiguous amino acids between amino acids 704 and 714 of an AAV9 viral protein. In some embodiments, the engineered capsid can have an n-mer inserted between amino acids 588 and 589 of an AAV9 viral protein. In some embodiments, the engineered capsid can have a 7-mer motif inserted between amino acids 588 and 589 of an AAV9 viral protein. In other embodiments, the motif inserted is a 10-mer motif, with replacement of amino acids 586-88 and an insertion before 589. It will be appreciated that n-mers can be inserted in analogous positions in AAV viral proteins of other serotypes. In some embodiments as previously discussed, the n-mer(s) can be inserted between any two contiguous amino acids within the AAV viral protein and in some embodiments the insertion is made in a variable region.

[0489] In some embodiments, the first 1, 2, 3, or 4 amino acids of an n-mer motif can replace 1, 2, 3, or 4 amino acids of a polypeptide into which it is inserted and preceding the insertion site. In some embodiments, the amino acids of the n-mer motif that replace 1 or more amino acids of the polypeptide into which the n-mer motif is inserted come before or immediately before an "RGDG" (SEQ ID NO: 2440) in an n-mer motif. For example, in one or more of the 10-mer inserts, the first three amino acids shown can replace 1-3 amino acids into a polypeptide to which they may be inserted. Using an AAV as another non-limiting example, one or more of the n-mer motifs can be inserted into, e.g., and AAV9 capsid prolylpeptide between amino acids 588 and 589 and the insert can replace amino acids 586, 587, and 588 such that the amino acid immediately preceding the n-mer motif after insertion is residue 585. It will be appreciated that this principle can apply in any other insertion context and is not necessarily limited to insertion between residues 588 and 589 of an AAV9 capsid or equivalent position in another AAV capsid. It will further be appreciated that in some embodiments, no amino acids in the polypeptide into which the n-mer motif is inserted are replaced by the n-mer motif.

[0490] In some embodiments, the AAV capsids or other viral capsids or compositions can be muscle specific. In some embodiments, muscle-specificity of the engineered AAV or other viral capsid or other composition is conferred by a muscle specific n-mer motif incorporated in the engineered AAV or other viral capsid or other composition described herein. While not intending to be bound by theory, it is believed that the n-mer motif confers a 3D structure to or within a domain or region of the engineered AAV capsid or other viral capsid or other composition such that the interaction of the viral particle or other composition containing the engineered AAV capsid or other viral capsid or other composition described herein has increased or improved interactions (e.g., increased affinity) with a cell surface receptor and/or other molecule on the surface of a muscle cell. In some embodiments, the cell surface receptor is AAV receptor (AAVR). In some embodiments, the cell surface receptor is a muscle cell specific AAV receptor. In some embodiments, the cell surface receptor or other molecule is a cell surface receptor or other molecule selectively expressed on the surface of a muscle cell. In some embodiments, the cell surface receptor or molecule is an integrin or dimer thereof. In some embodiments, the cell surface receptor or molecule is an Vb6 integrin heterodimer.

[0491] In some embodiments, a muscle specific engineered viral particle or other composition described herein containing the muscle-specific capsid, n-mer motif, or muscle-specific targeting moiety described herein can have an increased uptake, delivery rate, transduction rate, efficiency, amount, or a combination thereof in a muscle cell as compared to other cells types and/or other virus particles (including but not limited to AAVs) and other compositions that do not contain the muscle-specific n-mer motif of the present invention.

IV. Pharmaceutical Compositions

[0492] Any of the recombinant nucleic acids, the expression vectors (e.g., AAV vectors), or the AAV particles described herein can be included in a pharmaceutical composition comprising a pharmaceutically acceptable carrier.

[0493] Some embodiments of the invention may include any acceptable form of providing the AAV vector to a subject.

For example, the AAV vector may be provided to the subject in the form of a composition or formulation comprising the AAV vector. The expression vector (e.g., AAV vector) of this invention can be formulated and administered to treat a variety of disease states by any means that produces contact of the active ingredient with the agent's site of action in the body of the subject. The compositions, polynucleotides, polypeptides, particles, cells, vector systems and combinations thereof described herein can be contained in a formulation, such as a pharmaceutical formulation. In some embodiments, the formulations can be used to generate polypeptides and other particles that include one or more muscle-specific targeting moieties described herein. In some embodiments, the formulations can be delivered to a subject in need thereof. In some embodiments, component(s) of the engineered AAV capsid system, engineered cells, engineered AAV capsid particles, and/or combinations thereof described herein can be included in a formulation that can be delivered to a subject or a cell. In some embodiments, the formulation is a pharmaceutical formulation. One or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein can be provided to a subject in need thereof or a cell alone or as an active ingredient, such as in a pharmaceutical formulation. As such, also described herein are pharmaceutical formulations containing an amount of one or more of the polypeptides, polynucleotides, vectors, cells, or combinations thereof described herein. In some embodiments, the pharmaceutical formulation can contain an effective amount of the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein. The pharmaceutical formulations described herein can be administered to a subject in need thereof or a cell.
[0494] In some embodiments, the amount of the one or more of the polypeptides, polynucleotides, vectors, cells, virus particles, nanoparticles, other delivery particles, and combinations thereof described herein contained in the pharmaceutical formulation can range from about 1 μg/kg to about 10 mg/kg based upon the bodyweight of the subject in need thereof or average bodyweight of the specific patient population to which the pharmaceutical formulation can be administered. The amount of the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein in the pharmaceutical formulation can range from about 1 μg to about 10 g, from about 10 nL to about 10 ml. In embodiments where the pharmaceutical formulation contains one or more cells, the amount can range from about 1 cell to 1×10.sup.2, 1×10.sup.3, 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10 or more cells. In embodiments where the pharmaceutical formulation contains one or more cells, the amount can range from about 1 cell to 1×10.sup.2, 1×10.sup.3, 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10 or more cells per nL, μL, mL, or L.
[0495] In embodiments, when engineered AAV capsid particles are included in the formulation, the formulation can contain 1 to 1×10.sup.2, 1×10.sup.3, 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10, 1×10.sup.11, 1×10.sup.12, 1×10.sup.13, 1×10.sup.14, 1×10.sup.15, 1×10.sup.16, 1×10.sup.17, 1×10.sup.18, 1×10.sup.19, or 1×10.sup.20 transducing units (TU)/mL of the engineered AAV capsid particles. In some embodiments, the formulation can be 0.1 to 100 mL in volume and can contain 1 to 1×10.sup.2, 1×10.sup.3, 1×10.sup.4, 1×10.sup.5, 1×10.sup.6, 1×10.sup.7, 1×10.sup.8, 1×10.sup.9, 1×10.sup.10, 1×10.sup.11, 1×10.sup.12, 1×10.sup.13, 1×10.sup.14, 1×10.sup.15, 1×10.sup.16, 1×10.sup.17, 1×10.sup.18, 1×10.sup.19, or 1×10.sup.20 transducing units (TU)/mL of the engineered AAV capsid particles.
(a) Pharmaceutically Acceptable Carriers and Auxiliary Ingredients and Agents
[0496] Any recombinant nucleic acids, expression vectors (e.g., AAV vectors), or AAV particles described herein can be included in a pharmaceutical composition comprising a pharmaceutically acceptable carrier.
[0497] In embodiments, the pharmaceutical formulation containing an amount of one or more of the polypeptides, polynucleotides, vectors, cells, virus particles, nanoparticles, other delivery particles, and combinations thereof described herein can further include a pharmaceutically acceptable carrier. Suitable pharmaceutically acceptable carriers include, but are not limited to, water, salt solutions, alcohols, gum arabic, vegetable oils, benzyl alcohols, polyethylene glycols, gelatin, carbohydrates such as lactose, amylose or starch, magnesium stearate, talc, silicic acid, viscous paraffin, perfume oil, fatty acid esters, hydroxy methylcellulose, and polyvinyl pyrrolidone, which do not deleteriously react with the active composition.
[0498] The pharmaceutical formulations can be sterilized, and if desired, mixed with auxiliary agents, such as lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure, buffers, coloring, flavoring and/or aromatic substances, and the like which do not deleteriously react with the active composition.
[0499] In some embodiments, the pharmaceutical formulations described herein may be in a dosage form. The dosage forms can be adapted for administration by any appropriate route. Appropriate routes include, but are not limited to, oral (including buccal or sublingual), rectal, epidural, intracranial, intraocular, inhaled, intranasal, topical (including buccal, sublingual, or transdermal), vaginal, intraurethral, parenteral, intracranial, subcutaneous, intramuscular, intravenous, intraperitoneal, intradermal, intraosseous, intracardiac, intraarticular, intracavernous, intrathecal, intravitreal, intracerebral, gingival, subgingival, intracerebroventricular, and intradermal. Such formulations may be prepared by any method known in the art.
[0500] Dosage forms adapted for oral administration can be discrete dosage units such as capsules, pellets or tablets, powders or granules, solutions, or suspensions in aqueous or non-aqueous liquids; edible foams or whips, or in oil-in-water liquid emulsions or water-in-oil liquid emulsions. In some embodiments, the pharmaceutical formulations adapted for oral administration also include one or more agents which flavor, preserve, color, or help disperse the pharmaceutical formulation. Dosage forms prepared for oral administration can also be in the form of a liquid solution that can be delivered as foam, spray, or liquid solution. In some embodiments, the oral dosage form can contain about 1 ng to 1000 g

of a pharmaceutical formulation containing a therapeutically effective amount or an appropriate fraction thereof of the targeted effector fusion protein and/or complex thereof or composition containing the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein. The oral dosage form can be administered to a subject in need thereof.

[0501] Where appropriate, the dosage forms described herein can be microencapsulated.

[0502] The dosage form can also be prepared to prolong or sustain the release of any ingredient. In some embodiments, the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein can be the ingredient whose release is delayed. In other embodiments, the release of an optionally included auxiliary ingredient is delayed. Suitable methods for delaying the release of an ingredient include, but are not limited to, coating or embedding the ingredients in material in polymers, wax, gels, and the like. Delayed release dosage formulations can be prepared as described in standard references such as “Pharmaceutical dosage form tablets,” eds. Liberman et. al. (New York, Marcel Dekker, Inc., 1989), “Remington—The science and practice of pharmacy”, 20th ed., Lippincott Williams & Wilkins, Baltimore, MD, 2000, and “Pharmaceutical dosage forms and drug delivery systems”, 6th Edition, Ansel et al., (Media, PA: Williams and Wilkins, 1995). These references provide information on excipients, materials, equipment, and processes for preparing tablets and capsules and delayed release dosage forms of tablets and pellets, capsules, and granules. The delayed release can be anywhere from about an hour to about 3 months or more.

[0503] Examples of suitable coating materials include, but are not limited to, cellulose polymers such as cellulose acetate phthalate, hydroxypropyl cellulose, hydroxypropyl methylcellulose, hydroxypropyl methylcellulose phthalate, and hydroxypropyl methylcellulose acetate succinate; polyvinyl acetate phthalate, acrylic acid polymers and copolymers, and methacrylic resins that are commercially available under the trade name EUDRAGIT® (Roth Pharma, Westerstadt, Germany), zein, shellac, and polysaccharides.

[0504] Coatings may be formed with a different ratio of water-soluble polymer, water insoluble polymers, and/or pH dependent polymers, with or without water insoluble/water soluble non-polymeric excipient, to produce the desired release profile. The coating is either performed on the dosage form (matrix or simple) which includes, but is not limited to, tablets (compressed with or without coated beads), capsules (with or without coated beads), beads, particle compositions, “ingredient as is” formulated as, but not limited to, suspension form or as a sprinkle dosage form.

[0505] Dosage forms adapted for topical administration can be formulated as ointments, creams, suspensions, lotions, powders, solutions, pastes, gels, sprays, aerosols, or oils. In some embodiments for treatments of the eye or other external tissues, for example the mouth or the skin, the pharmaceutical formulations are applied as a topical ointment or cream. When formulated in an ointment, the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein can be formulated with a paraffinic or water-miscible ointment base. In some embodiments, the active ingredient can be formulated in a cream with an oil-in-water cream base or a water-in-oil base. Dosage forms adapted for topical administration in the mouth include lozenges, pastilles, and mouth washes.

[0506] Dosage forms adapted for nasal or inhalation administration include aerosols, solutions, suspension drops, gels, or dry powders. In some embodiments, the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein is contained in a dosage form adapted for inhalation is in a particle-size-reduced form that is obtained or obtainable by micronization. In some embodiments, the particle size of the size reduced (e.g., micronized) compound or salt or solvate thereof, is defined by a D50 value of about 0.5 to about 10 microns as measured by an appropriate method known in the art. Dosage forms adapted for administration by inhalation also include particle dusts or mists. Suitable dosage forms wherein the carrier or excipient is a liquid for administration as a nasal spray or drops include aqueous or oil solutions/suspensions of an active ingredient (e.g., the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein and/or auxiliary active agent), which may be generated by various types of metered dose pressurized aerosols, nebulizers, or insufflators.

[0507] In some embodiments, the dosage forms can be aerosol formulations suitable for administration by inhalation. In some of these embodiments, the aerosol formulation can contain a solution or fine suspension of the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein and a pharmaceutically acceptable aqueous or non-aqueous solvent. Aerosol formulations can be presented in single or multi-dose quantities in sterile form in a sealed container. For some of these embodiments, the sealed container is a single dose or multi-dose nasal, or an aerosol dispenser fitted with a metering valve (e.g., metered dose inhaler), which is intended for disposal once the contents of the container have been exhausted.

[0508] Where the aerosol dosage form is contained in an aerosol dispenser, the dispenser contains a suitable propellant under pressure, such as compressed air, carbon dioxide, or an organic propellant, including but not limited to a hydrofluorocarbon. The aerosol formulation dosage forms in other embodiments are contained in a pump-atomizer. The pressurized aerosol formulation can also contain a solution or a suspension of one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein. In further embodiments, the aerosol formulation can also contain co-solvents and/or modifiers incorporated to improve, for example, the stability and/or taste and/or fine particle mass characteristics (amount and/or profile) of the formulation. Administration of the aerosol formulation can be once daily or several times daily, for example 2, 3, 4, or 8 times daily, in which 1, 2, or 3 doses are delivered each time.

[0509] For some dosage forms suitable and/or adapted for inhaled administration, the pharmaceutical formulation is a dry powder inhalable formulation. In addition to the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein, an auxiliary active ingredient, and/or pharmaceutically acceptable salt thereof, such a dosage form can contain a powder base such as lactose, glucose, trehalose, mannitol, and/or starch. In some of these embodiments, the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein is in a particle-size reduced form. In further embodiments, a performance modifier, such as L-leucine or another amino acid, cellobiose octaacetate, and/or metals salts of stearic acid, such as magnesium or calcium stearate.

[0510] In some embodiments, the aerosol dosage forms can be arranged so that each metered dose of aerosol contains a predetermined amount of an active ingredient, such as the one or more of the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein.

[0511] Dosage forms adapted for vaginal administration can be presented as pessaries, tampons, creams, gels, pastes, foams, or spray formulations. Dosage forms adapted for rectal administration include suppositories or enemas.

[0512] Dosage forms adapted for parenteral administration and/or adapted for any type of injection (e.g., intravenous, intraperitoneal, subcutaneous, intramuscular, intradermal, intraosseous, epidural, intracardiac, intraarticular, intracavernous, gingival, subgingival, intrathecal, intravitreal, intracerebral, and intracerebroventricular) can include aqueous and/or non-aqueous sterile injection solutions, which can contain anti-oxidants, buffers, bacteriostats, solutes that render the composition isotonic with the blood of the subject, and aqueous and non-aqueous sterile suspensions, which can include suspending agents and thickening agents. The dosage forms adapted for parenteral administration can be presented in a single-unit dose or multi-unit dose containers, including but not limited to sealed ampoules or vials. The doses can be lyophilized and resuspended in a sterile carrier to reconstitute the dose prior to administration. Extemporaneous injection solutions and suspensions can be prepared in some embodiments, from sterile powders, granules, and tablets.

[0513] Dosage forms adapted for ocular administration can include aqueous and/or nonaqueous sterile solutions that can optionally be adapted for injection, and which can optionally contain antioxidants, buffers, bacteriostats, solutes that render the composition isotonic with the eye or fluid contained therein or around the eye of the subject, and aqueous and nonaqueous sterile suspensions, which can include suspending agents and thickening agents.

[0514] For some embodiments, the dosage form contains a predetermined amount of the one or more of the polypeptides, polynucleotides, vectors, cells, and combinations thereof described herein per unit dose. In some embodiments, the predetermined amount of the Such unit doses may therefore be administered once or more than once a day. Such pharmaceutical formulations may be prepared by any of the methods well known in the art.

V. Methods of Treatment and Use

[0515] Aspects of the present disclosure provide methods for treating DMD in a subject in need thereof using any of the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), the AAV particles, or pharmaceutical compositions thereof described herein.

(a) Subjects

[0516] A subject can be any subject for whom diagnosis, treatment, or therapy is desired. A subject can be a human or a non-human primate. A subject can be any age. In some embodiments, the subject is a human subject that is any age (e.g., a human subject that is between the ages of 2 and 5).

[0517] A subject (e.g., human subject) to be treated by the methods described herein can be a subject having, suspected of having, or at risk for having DMD. A subject (e.g., human subject) suspected of having DMD might show one or more symptoms of DMD, e.g., muscle weakness, muscle wasting, muscle pain, enlarged muscles, walking abnormalities, or difficulty breathing. A subject (e.g., human subject) at risk for DMD can be a subject having one or more risk factors for DMD, e.g., one or more mutations in a dystrophin gene. A subject (e.g., human subject) who needs the treatment described herein can be identified by routine medical examination, e.g., laboratory tests (e.g., serum creatine kinase (CK) blood tests), genetic testing (e.g., testing for one or more mutations in a dystrophin gene), or muscle biopsy.

[0518] Any of the methods described herein can further comprise a step of identifying a subject (e.g., a human subject) for treatment based on identifying one or more mutations in a dystrophin gene. In some embodiments, presence of one or more mutations in a dystrophin gene is determined in a biological sample (e.g., a blood sample) obtained from a subject (e.g., a human subject).

(b) Administration

[0519] Methods described herein encompass administering an effective amount of a recombinant nucleic acid, an expression vector (e.g., an AAV vector), an AAV particle, or a pharmaceutical composition thereof to a subject, e.g., a subject in need thereof, for example, a human subject, by a variety of methods. The route of administration can be intramuscular administration, subcutaneous injection, intravenous injection, or intravenous (IV) administration.

[0520] An effective amount refers to the amount of a recombinant nucleic acid, an expression vector (e.g., an AAV vector), an AAV particle, or a pharmaceutical composition thereof needed to prevent or alleviate at least one or more signs or symptoms of DMD, and relates to a sufficient amount of a recombinant nucleic acid, an expression vector (e.g., an AAV vector), an AAV particle, or pharmaceutical compositions thereof that provides the desired effect, e.g., to treat a subject (e.g., human subject) having DMD. An effective amount also can include an amount sufficient to prevent or delay the development of a symptom of DMD, alter the course of a symptom of DMD (e.g., slow the progression of a symptom of DMD), or reverse a symptom of DMD.

[0521] Methods described herein comprise administering (e.g., intramuscularly or intravenously) to a subject (e.g., a human subject) a recombinant nucleic acid, an expression vector (e.g., an AAV vector), an AAV particle, or a pharmaceutical composition described herein one or more times. In some embodiments, an effective amount of a recombinant nucleic acid, an expression vector (e.g., an AAV vector), an AAV particle, or a pharmaceutical composition thereof is administered to a subject (e.g., a human subject) no more than once or more than once (e.g., administered twice, administered three times, or administered four times).

[0522] Any of the methods for treating DMD described herein can be used in combination therapies. For example, any of the methods for treating DMD described herein can be co-used with other therapeutic agents for treating DMD, or for enhancing efficacy of the compositions described herein (e.g., the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), or pharmaceutical compositions) and/or reducing side effects of the compositions described herein (e.g., the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), or pharmaceutical compositions).

[0523] Any of the recombinant nucleic acids, the expression vectors (e.g., the AAV vectors), or pharmaceutical compositions described herein can be used as a medicament or in the manufacture of a medicament for the treatment of DMD, or ameliorating one or more symptoms thereof in a subject.

VI. Kits for Therapeutic Use

[0524] The present disclosure provides kits for treating DMD. Such kits can include any of the recombinant nucleic acids, AAV vectors, AAV particles, or pharmaceutical compositions thereof described herein. In some embodiments, the kit can additionally comprise instructions for practicing any of the methods described herein. The instructions may include information as to dosage, dosing schedule, and route of administration. Instructions supplied in the kits of the disclosure can be written instructions on a label or a package insert.

OTHER EMBODIMENTS

[0525] Without further elaboration, it is believed that one skilled in the art can, based on the above description, utilize the present disclosure to its fullest extent. The specific embodiments herein are to be construed as merely illustrative, and not limitative of the remainder of the disclosure in any way whatsoever. The recitation of a listing of elements in any definition of a variable herein includes definitions of that variable as any single element or combination (or sub-combination) of listed elements. The recitation of an embodiment herein includes that embodiment as any single embodiment or in combination with any other embodiments or portions thereof.

[0526] In any of the aspects and embodiments described herein, the nucleic acid sequences contemplated can be DNA, RNA, or modified versions thereof. Modified nucleic acids can be distinguished from naturally occurring nucleic acids by modifications to the backbone of the polynucleotide chain, for example, peptide nucleic acids (PNA), morpholinos, locked nucleic acids (LNA), glycol nucleic acids (GNA) and threose nucleic acid (TNA). Modified nucleic acids can also include analogs with modifications to the four nucleobases. In some embodiments, the nucleic acids are PNAs. In some embodiments, the nucleic acids are LNAs. In some embodiments, the nucleic acids are morpholinos. In some embodiments, the nucleic acids are in a single-stranded form. In some embodiments, the nucleic acids are in double-stranded form. In some embodiments, the nucleic acids are linear. In some embodiments, the nucleic acids are circular. In some embodiments, the nucleic acids are plasmids.

[0527] Non-limiting embodiments of the present disclosure include:

[0528] Embodiment 1 is a composition comprising an adeno-associated virus (AAV) particle comprising an engineered capsid protein comprising the amino acid sequence RGDR (SEQ ID NO: 2440); and a nucleic acid molecule comprising a promoter; a sequence having at least 95% sequence identity with a sequence selected from SEQ ID NOs: 2405-2407; and a dorsal root ganglia (DRG) de-targeting miRNA binding site.

[0529] Embodiment 2 is the composition of Embodiment 1, wherein the nucleic acid molecule comprises a sequence having the sequence identity with a sequence selected from SEQ ID NOs: 2405-2407.

[0530] Embodiment 3 is the composition of Embodiment 2, wherein the DRG de-targeting miRNA binding site comprises a sequence selected from among SEQ ID NOs: 2399-2401.

[0531] Embodiment 4 is the composition of Embodiment 3, wherein the promoter comprises the sequence having at least 95% sequence identity a sequence selected from among SEQ ID NOs: 2402-2404.

[0532] Embodiment 5 is the composition of Embodiment 4, wherein the nucleic acid molecule comprises in order: a first inverted terminal repeats (ITR) region; a promoter; an intron; the sequence having at least 95% sequence identity with a sequence selected from SEQ ID NOs: 2405-2407; a first DRG de-targeting miRNA binding site; a second DRG de-targeting miRNA binding site; a poly adenylation signal; and a second ITR region.

[0533] Embodiment 6 is the composition of Embodiment 5, wherein when administered to a subject with a mutation in the DMD gene, resulting in lack of dystrophin protein expression, the composition is effective at restoring a healthy phenotype.

[0534] Embodiment 7 is the composition of Embodiment 6, wherein the nucleic acid molecule comprises a sequence selected from among of SEQ ID NOs: 2412-2438.

[0535] Embodiment 8 is the composition of Embodiment 1, wherein the capsid protein comprises an amino acid sequence selected from Table 1a in hypervariable region IV (HVR IV) relative to wild type AAV9, wherein the capsid protein comprises substitutions at amino acids 451-455 relative to a wild type AAV9 vector capsid selected from column 1 of Table 1b; and a 7-mer insert selected from column 2 of Table 1b inserted after amino acid 455 relative to a wild type AAV9 vector.

[0536] Embodiment 9 is the composition of Embodiment 8, wherein the capsid protein further comprises a deletion of G267 relative to wild type AAV9 capsid or equivalent position in another AAV capsid.

[0537] Embodiment 10 is the composition of Embodiment 9, wherein the AAV vector exhibits reduced liver tropism and increased muscle tropism as compared to a wild type AAV vector.

[0538] Embodiment 11 is a method of treating a subject suffering from DMD, the method comprising providing to the subject a composition comprising an adeno-associated virus (AAV) particle comprising an engineered capsid protein comprising the amino acid sequence RGDR (SEQ ID NO: 2440); and a nucleic acid molecule comprising a promoter; a sequence having at least 95% sequence identity with a sequence selected from SEQ ID NOs: 2405-2407; and a dorsal root ganglia (DRG) de-targeting miRNA binding site.

[0539] Embodiment 12 is the method of Embodiment 11, wherein the nucleic acid molecule comprises a sequence having the sequence identity with a sequence selected from SEQ ID NOs: 2405-2407.

[0540] Embodiment 13 is the method of Embodiment 12, wherein the DRG de-targeting miRNA binding site comprises a sequence selected from among SEQ ID NOs: 2399-2401.

[0541] Embodiment 14 is the method of Embodiment 13, wherein the promoter comprises the sequence having at least 95% sequence identity a sequence selected from among SEQ ID NOs: 2402-2404.

[0542] Embodiment 15 is the method of Embodiment 14, wherein the nucleic acid molecule comprises in order: a first inverted terminal repeats (ITR) region; a promoter; an intron; the sequence having at least 95% sequence identity with a sequence selected from SEQ ID NOs: 2405-2407; a first DRG de-targeting miRNA binding site; a second DRG de-targeting miRNA binding site; a poly adenylation signal; and a second ITR region.

[0543] Embodiment 16 is the method of Embodiment 15, wherein when administered to a subject with a mutation in the DMD gene, resulting in lack of dystrophin protein expression, the method is effective at restoring a healthy phenotype.

[0544] Embodiment 17 is the method of Embodiment 16, wherein the nucleic acid molecule comprises a sequence selected from among of SEQ ID NOs: 2412-2438

[0545] Embodiment 18 is the method of Embodiment 11, wherein the capsid protein comprises an amino acid sequence selected from Table 1a in hypervariable region IV (HVR IV) relative to wild type AAV9, wherein the capsid protein comprises substitutions at amino acids 451-455 relative to a wild type AAV9 vector capsid selected from column 1 of Table 1b; and a 7-mer insert selected from column 2 of Table 1b inserted after amino acid 455 relative to a wild type AAV9 vector.

[0546] Embodiment 19 is the method of Embodiment 18, wherein the capsid protein further comprises a deletion of G267 relative to wild type AAV9 capsid or equivalent position in another AAV capsid.

[0547] Embodiment 20 is the method of Embodiment 19, wherein the AAV vector exhibits reduced liver tropism and increased muscle tropism as compared to a wild type AAV vector. Embodiment 21 is a composition comprising a nucleic acid molecule comprising a sequence having at least 95% sequence identity with a sequence selected from any one of SEQ ID NOs: 2405-2407.

[0548] Embodiment 22 is the composition of Embodiment 21, wherein the nucleic acid molecule comprises a sequence having the sequence identity of SEQ ID NO: 2405.

[0549] Embodiment 23 is the composition of Embodiment 21, wherein the nucleic acid molecule comprises a sequence having the sequence identity of SEQ ID NO: 2406.

[0550] Embodiment 24 is the composition of Embodiment 21, wherein the nucleic acid molecule comprises a sequence having the sequence identity of SEQ ID NO: 2407.

[0551] Embodiment 25 is the composition of Embodiment 21, wherein the nucleic acid molecule is packaged in a vector.

[0552] Embodiment 26 is the composition of Embodiment 25, wherein the vector is an adeno-associated viral (AAV) vector.

[0553] Embodiment 27 is the composition of Embodiment 26, wherein the nucleic acid molecule comprises, in order, an inverted terminal repeats (ITR) region; a promoter; an intron; the sequence having at least 95% sequence identity with a sequence selected from any one of SEQ ID NOs: 2405-2407; a poly adenylation signal; and an ITR region.

[0554] Embodiment 28 is the composition of Embodiment 26, wherein each AAV vector comprises a capsid protein having at least one modification that results in reduced liver-tropism of the AAV vector and/or preferential targeting of the AAV vector to muscle tissue.

[0555] Embodiment 29 is the composition of Embodiment 21, wherein, when administered to a subject with a mutation in the DMD gene, which results in lack of dystrophin protein expression, the composition is effective at restoring a healthy phenotype.

[0556] Embodiment 30 is the composition of Embodiment 21, wherein a microdystrophin protein expressed by the nucleic acid molecule is smaller than 137.5 kilodaltons.

[0557] Embodiment 31 is a method of treating a subject suffering from DMD, the method comprising providing to the subject a composition comprising a nucleic acid molecule comprising a sequence having at least 95% sequence identity with a sequence selected from any one of SEQ ID NOs: 2405-2407.

[0558] Embodiment 32 is the method of Embodiment 31, wherein the nucleic acid molecule comprises a sequence having the sequence identity of SEQ ID NO: 2405.

[0559] Embodiment 33 is the method of Embodiment 31, wherein the nucleic acid molecule comprises a sequence having the sequence identity of SEQ ID NO: 2406.

[0560] Embodiment 34 is the method of Embodiment 31, wherein the nucleic acid molecule comprises a sequence having the sequence identity of SEQ ID NO: 2407.

[0561] Embodiment 35 is the method of Embodiment 31, wherein the nucleic acid molecule is packaged is in a vector.

[0562] Embodiment 36 is the method of Embodiment 35, wherein the vector is an adeno-associated viral (AAV) vector.
[0563] Embodiment 37 is the method of Embodiment 36, wherein the nucleic acid molecule comprises, in order, an inverted terminal repeats (ITR) region; a promoter; an intron; the sequence having at least 95% sequence identity with a sequence selected from SEQ ID NO: 2405-2407; a poly adenylation signal; and an ITR region.
[0564] Embodiment 38 is the method of Embodiment 36, wherein each AAV vector comprises a capsid protein having at least one modification that results in reduced liver-tropism of the AAV vector and/or preferential targeting of the AAV vector to muscle tissue.
[0565] Embodiment 39 is the method of Embodiment 31, wherein the providing step comprises providing to a subject having a mutation in the DMD gene, which results in lack of dystrophin protein expression, the composition is effective at restoring a healthy phenotype. Embodiment 40 is the method of Embodiment 31, wherein the protein expressed by the nucleic acid molecule is smaller than 137.5 kilodaltons.

EXAMPLES

[0566] In order that the invention described may be more fully understood, the following examples are set forth. The examples described in this application are offered to illustrate the methods and compositions provided herein and are not to be construed in any way as limiting their scope.

Example 1: Exemplary Recombinant Nucleic Acids and AAV Particles

[0567] Exemplary AAV particles were engineered comprising the capsid protein MyoAAV-LD B comprising an RGDR (SEQ ID NO: 2440) motif and a recombinant nucleic acid (also referred to as cargo) described herein.
[0568] The capsid protein MyoAAV-LD A can be described relative to a wild type AAV9, specifically the capsid protein MyoAAV-LD A includes the amino acid sequence of LNASM (SEQ ID NO: 998) at positions 451-455, the amino acid sequence of RGDRQSL (SEQ ID NO: 1797) inserted after amino acid 455, a deletion of the amino acid G267 relative to the wild type AAV9 capsid protein, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0569] The capsid protein MyoAAV-LD B can be described relative to a wild type AAV9, specifically the capsid protein MyoAAV-LD B includes the amino acid sequence of LNAST (SEQ ID NO: 1014) at positions 451-455, the amino acid sequence of RGDRQYL (SEQ ID NO: 1854) inserted after amino acid 455, a deletion of the amino acid G267 relative to the wild type AAV9 capsid protein, and the remainder of the capsid is otherwise identical to wild type AAV9.
[0570] The capsid encapsidated a cargo nucleic acid comprising a novel promoter comprising a sequence selected from any one of SEQ ID NO: 2402-2404, a codon optimized microdystrophin comprising a sequence selected from any one of SEQ ID NO: 2405-2407, and a first and a second DRG binding sites comprising a sequence selected from any one of SEQ ID NOs: 2399-2401.
[0571] Exemplary recombinant nucleic acids are described below:
TABLE-US-00007 Nucleic acid 1 (SEQ ID NO: 2412)
GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTTTGGTCGCCCGGCCTCAGT
GAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAACTCCATCACTAGGGGTTCCTTAGTTATTAATAGTAATCAATT
ACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGA
CCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTTCCAT
TGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACG
CCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCT
ACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGCTAGACTAGCATGCTGCCCATGTAAGGAGGCAA
GGCCTGGGGACACCCGAGATGCCTGGTTATAATTAACCCAGACATGTGGCTGCCCCCCCCCCCCAACACCTGCTG
CCTCTAAAAATAACCCTGCATGCCATGTTCCCGGCGAAGGGCCAGCTGTCCCCCGCCAGCTAGACTCAGCTCTCA
GGGGCCCCTCCCTGGGGACAGCCCCTCCTGGCTAGTCACACCCTGTAGGCTCCTCTATATAACCCAGGGGCACAG
GGGCTGCCCTCATTCTACCACCACCTCCACAGCACAGACAGACACTCAGGAGCCAGCCAGCCAGCCCAGGTACCT
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GCTCTCAACAACGTGAACAAGGCCCTGAGAGTGCTGCAGAACAACAACGTGGACCTTGTGAACATCGGAAGCACC
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CCCGACAAGAAGAGCATCCTGATGTACATCACCAGCCTGTTCCAGGTGCTGCCCCAGCAAGTGAGCATCGAGGCC
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CACTACAGCCAGCAGATCACCGTGAGCCTGGCCCAGGGCTACGAGAGAACCAGCAGCCCCAAGCCCAGATTCAAG
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TABLE-US-00008 Nucleic acid cargo 1 annotated Sequence AAV2 ITR

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CTAGGGGTTCCT (SEQ ID NO: 2408) Novel
TAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTC promoter
CGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCC
CATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTTCCATTG
ACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTAT
CATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATT
ATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGT
CATCGCTATTACCATGCTAGACTAGCATGCTGCCCATGTAAGGAGGCAAGGCCTGGGG
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GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGTCGA
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TGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon
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TTCAGAACCAAGAGATACTTCGCTAAGCACCCCAGAATGGGCTACCTGCCCGTGCAGA
CCGTGCTGGAGGGCGATAACATGGAGACCGACACCATGTGA (SEQ ID NO: 2405) First DRG de-
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) de-targeting miRNA binding site Short polyA
AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2
ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTGA
GGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGTGAGC

GAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00009 Nucleic acid 2 (SEQ ID NO: 2413)

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TABLE-US-00010 Nucleic acid cargo 2 annotated Sequence AAV2 ITR
GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTT
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CACTAGGGGTTCCT (SEQ ID NO: 2408) Novel
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTG promoter
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CCGGAGACTGAAGACA (SEQ ID NO: 2403) Intron
GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGTCG
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TTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon
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CAGACCGTGCTGGAGGGCGATAACATGGAGACCGACACCATGTGA (SEQ ID NO: 2405) First DRG de-
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) de-targeting miRNA binding site Short polyA
AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2
ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTG
AGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGTGA
GCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00011 Nucleic acid 3 (SEQ ID NO: 2414)

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TABLE-US-00012 Nucleic acid cargo 3 annotated Sequence AAV2 ITR
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ID NO: 2404) Intron
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CCGTGCTGGAGGGCGATAACATGGAGACCGACACCATGTGA (SEQ ID NO: 2405) First DRG de-
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) de-targeting miRNA binding site Short polyA
AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2
ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTGA
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GAGCGAGCGCGC (SEQ ID NO: 2411)
TABLE-US-00013 Nucleic acid 4 (SEQ ID NO: 2415)
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TABLE-US-00014 Nucleic acid 4 cargo annotated Sequence AAV2 ITR
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CAGACCGTGCTGGAGGGCGATAACATGGAGACCGACACCATGTGA (SEQ ID NO: 2405) First DRG de-
CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site Second DRG
CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) de-targeting miRNA binding site Short polyA
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TABLE-US-00016 Nucleic acid cargo 5 annotated Sequence AAV2 ITR
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CAGACCGTGCTGGAGGGCGATAACATGGAGACCGACACCATGTGA (SEQ ID NO: 2405) First DRG de-
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2400) targeting miRNA binding site Second DRG
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2400) de-targeting miRNA binding site Short polyA
AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2

ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTG
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TABLE-US-00017 Nucleic acid 6 (SEQ ID NO: 2417)
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TABLE-US-00018 Nucleic acid cargo 6 annotated Sequence AAV2 ITR
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CACTAGGGGTTCCT (SEQ ID NO: 2408) Novel
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTG promoter
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TABLE-US-00019 Nucleic acid 7 (SEQ ID NO: 2418)

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TABLE-US-00020 Nucleic acid cargo 7 annotated Sequence AAV2 ITR

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TGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon
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CCTGTTCAGCGACCTGCAGGACGGCAGAAGACTGCTGGACCTGCTGGAGGGCCTGACC in coding

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CCGTGCTGGAGGGCGATAACATGGAGACCGACACCATGTGA (SEQ ID NO: 2405) First DRG de-
TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Second DRG
TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) de-targeting miRNA binding site Short polyA

AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTGA GGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGTGAGC GAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00021 Nucleic acid 8 (SEQ ID NO: 2419)

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TABLE-US-00022 Nucleic acid cargo 8 annotated Sequence AAV2 ITR

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CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTGG promoter
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TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Second DRG
TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) de-targeting miRNA binding site Short polyA
AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2
ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTGA
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GAGCGAGCGCGC (SEQ ID NO: 2411)
TABLE-US-00023 Nucleic acid 9 (SEQ ID NO: 2420)
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CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTGG promoter
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ID NO: 2404) Intron
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TGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon
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GGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGTGAGC
GAGCGAGCGCGC (SEQ ID NO: 2411)
TABLE-US-00025 Nucleic acid 10 (SEQ ID NO: 2421)
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TABLE-US-00026 Nucleic acid cargo 10 annotated Sequence AAV2 ITR
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CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) de-targeting miRNA binding site Short polyA
AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2
ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTG
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TABLE-US-00027 Nucleic acid 11 (SEQ ID NO: 2422)
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TCTAAACTCTGGCACATTCCAGCCCCTCTTTGGGAGAAAGAACATCTCTTCTTGCCCAGGCCCTAGTGAGACATG
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TGAGTTGGGCTATTTTGGTATCTGGGGTGGGCACCCGCAGGGCTAAGGTTACCTTGGTATGTTAAGGGCTCCCTG
GGGCAGGACTATATAACCCCAGAGGGACTGCCCCATGCTGACTCCTTGCTTCTTTCCTGCTCGAGCCGCTGCCTT
GCCCCCCGGAGACTGAAGACACAGCCCAGGTACCTGGGCAGGTAAGTATCAAGGTTACAAGACAGGTTTAAGGAG
ACCAATAGAAACTGGGCTTGTCGAGACAGAGAAGACTCTTGCGTTTCTGATAGGCACCTATTGGTCTTACTGACA
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AGCGGACCCTGGAGCGCCTGCAGGAGCTCCAGGAGGCCACCGACGAACTGGATCTGAAGCTGAGGCAGGCCGAGG
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CAACTTGCTGGGATCACCCCAAGATGACAGAACTCTATCAGTCACTCGCCGATCTCAATAATGTACGGTTCTCTG
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AACACTTCAATTATGACATCTGCCAGTCATGCTTCTTCAGCGGACGCGTGGCCAAGGGACACAAGATGCATTATC
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ACATGGAAACCGATACAATGTGATAGATTCCAACAAAATCACTGATGCTGGAAGATCAGCAACAAAATCACTGAT
GCTGGAAGACGCAGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTGAGGAACCCCTA
GTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGACGC
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TABLE-US-00028 Nucleic acid cargo 11 annotated Sequence AAV2 ITR
GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTTT
GGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAACTCCATCA
CTAGGGGTTCCT (SEQ ID NO: 2408) Novel
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTGG promoter
CCTATTCTCTGAGAGGACGCTGGGCTAATAATAGGAGGGGAGATTAGGGAGGGATTTG
ACTGCATTGTTATCTCACAGGAATGGCGTTTGAGAGCCTTGCTTCCAAGCTCTTCCAA
GAGTCATGCCACCCCCCTCCAATCTCTGTGGAAGTGCCCACTGCCTTTCCCGCTTCGT
TCTCTTCTCTCTAAATCTGAGCCCACCCCACTCAGGGGTACAGGATCTGGTTCATTTA
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TCTTTGGGAGAAAGAACATCTCTTCTTGCCCAGGCCCTAGTGAGACATGCATCGCAAG
ACATGCTCCCCCCTTCCTCCCCGCTGGGCTCCAGCTGCTGCCAGCCCTCCAAGAAAGG
AGAGGCCCTGAGTTGGGCTATTTTGGTATCTGGGGTGGGCACCCGCAGGGCTAAGGTT
ACCTTGGTATGTTAAGGGCTCCCTGGGGCAGGACTATATAACCCCAGAGGGACTGCCC
CATGCTGACTCCTTGCTTCTTTCCTGCTCGAGCCGCTGCCTTGCCCCCCGGAGACTGA AGACA (SEQ ID NO: 2403) Intron GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGTCGA
GACAGAGAAGACTCTTGCGTTTCTGATAGGCACCTATTGGTCTTACTGACATCCACTT
TGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon
ATGCTGTGGTGGGAAGAGGTCGAGGACTGCTACGAGCGCGAAGATGTGCAGAAGAAAA optimized
CTTTCACTAAATGGGTGAATGCCCAGTTCTCTAAGTTCGGAAAGCAGCACATCGAAAA microdystroph
CCTGTTTTCCGATCTGCAAGATGGCCGGCGGCTGCTGGACCTGCTGGAGGGGCTGACC in coding
GGACAGAAGCTGCCTAAGGAGAAAGGATCTACCAGGGTCCATGCTCTGAATAACGTGA sequence
ACAAGGCTCTGAGGGTGCTGCAGAACAATAACGTGGACCTGGTGAACATTGGTTCCAC

AGACATCGTGGATGGAAACCACAAACTGACACTGGGACTGATCTGGAACATCATTCTG
CATTGGCAAGTGAAGAACGTGATGAAGAACATCATGGCCGGACTGCAGCAGACGAACT
CCGAGAAGATCCTGCTTAGCTGGGTGAGACAGAGTACCCGGAATTACCCACAGGTGAA
CGTGATCAATTTCACAACCAGCTGGTCAGACGGCCTGGCTCTGAATGCCCTGATTCAT
AGCCACCGGCCAGATCTGTTTGACTGGAACTCTGTCGTGTGTCAGCAGTCAGCCACCC
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ACAAGCCTGTTCCAGGTGCTGCCACAGCAGGTCTCCATCGAGGCCATTCAAGAAGTGG
AGATGCTGCCCAGGCCTCCTAAGGTCACCAAAGAGGAGCACTTTCAGCTCCATCATCA
GATGCACTACAGCCAGCAAATTACCGTGTCCCTGGCCCAGGGCTACGAGAGGACAAGT
TCCCCCAAGCCTAGATTTAAGAGCTATGCCTATACCCAGGCCGCCTACGTGACTACCT
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CTTTGGCAGCAGCCTGATGGAATCTGAGGTGAACCTGGACAGATACCAGACTGCGCTG
GAGGAGGTGCTGTCCTGGCTGCTGAGCGCCGAGGACACCCTGCAGGCTCAGGGCGAGA
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AAGCACGCCCAGGAAGAGCTGCCGCCCCCACCACCCCAGAAAAAGCGGACCCTGGAGC
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GGTGATTAAGGGATCCTGGCAGCCTGTGGGAGACCTGCTGATCGACAGCCTGCAGGAT
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GCCATATTATATCAATCATGAGACTCAGACAACTTGCTGGGATCACCCCAAGATGACA
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TCTTCAGCGGACGCGTGGCCAAGGGACACAAGATGCATTATCCTATGGTGGAATACTG
CACTCCAACTACTTCCGGCGAGGACGTGAGAGATTTCGCTAAAGTTCTTAAAAACAAA
TTTCGGACTAAGAGGTACTTCGCCAAGCACCCCCGCATGGGCTATCTGCCTGTGCAGA
CAGTGCTTGAAGGAGACAACATGGAAACCGATACAATGTGA (SEQ ID NO: 2406) First DRG de-
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG
CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) de-targeting miRNA binding site Short polyA
AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2
ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTGA

GGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGTGAGC
GAGCGAGCGCGC (SEQ ID NO: 2411)
TABLE-US-00029 Nucleic acid 12 (SEQ ID NO: 2423)
GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTTTGGTCGCCCGGCCTCAGT
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GGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTGGCCTATTCTCTGAGAGGACGCTGGGCTAATAATAGGAGGG
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AAGAGTCATGCCACCCCCCTCCAATCTCTGTGGAAGTGCCCACTGCCTTTCCCGCTTCGTTCTCTTCTCTCGGTC
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CCAGCTGGTCAGACGGCCTGGCTCTGAATGCCCTGATTCATAGCCACCGGCCAGATCTGTTTGACTGGAACTCTG
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CCTACCGGACAGCTATGAAGCTGCGGAGGCTGCAGAAGGCACTCTGTCTGGACCTGCTGAGTCTCTCTGCGGCCT
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TABLE-US-00030 Nucleic acid cargo 12 annotated Sequence AAV2 ITR
GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTT
TGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAACTCCAT
CACTAGGGGTTCCT (SEQ ID NO: 2408) Novel
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGTG promoter
GCCTATTCTCTGAGAGGACGCTGGGCTAATAATAGGAGGGGAGATTAGGGAGGGATT
TGACTGCATTGTTATCTCACAGGAATGGCGTTTGAGAGCCTTGCTTCCAAGCTCTTC
CAAGAGTCATGCCACCCCCCTCCAATCTCTGTGGAAGTGCCCACTGCCTTTCCCGCT
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CAACGTTAGTAGAGGA (SEQ ID NO: 2404) Intron
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TTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon
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AACCTGTTTTCCGATCTGCAAGATGGCCGGCGGCTGCTGGACCTGCTGGAGGGGCTG in coding
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CAGGTGAACGTGATCAATTTCACAACCAGCTGGTCAGACGGCCTGGCTCTGAATGCC
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CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) de-targeting miRNA binding site Short polyA
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CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site Second DRG
CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) de-targeting miRNA binding site Short polyA
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TABLE-US-00034 Nucleic acid cargo 14 annotated Sequence AAV2 ITR
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CCATCACTAGGGGTTCCT (SEQ ID NO: 2408) Novel promoter
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTG
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CAGGACTATATAACCCCAGAGGGACTGCCCCATGCTGACTCCTTGCTTCTTTCCT
GCTCGAGCCGCTGCCTTGCCCCCCGGAGACTGAAGACA (SEQ ID NO: 2403) Intron
GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGT
CGAGACAGAGAAGACTCTTGCGTTTCTGATAGGCACCTATTGGTCTTACTGACAT
CCACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon optimized
ATGCTGTGGTGGGAAGAGGTCGAGGACTGCTACGAGCGCGAAGATGTGCAGAAGA microdystrophin
AAACTTTCACTAAATGGGTGAATGCCCAGTTCTCTAAGTTCGGAAAGCAGCACAT coding sequence
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GTTCTTAAAAACAAATTTCGGACTAAGAGGTACTTCGCCAAGCACCCCCGCATGG
GCTATCTGCCTGTGCAGACAGTGCTTGAAGGAGACAACATGGAAACCGATACAAT GTGA (SEQ ID NO: 2406) First DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2400) targeting miRNA binding site Second DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2400) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR
AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCAC
TGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCA
GTGAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00035 Nucleic acid 15 (SEQ ID NO: 2426)

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TABLE-US-00036 Nucleic acid cargo 15 annotated Sequence AAV2 ITR
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CCATCACTAGGGGTTCCT (SEQ ID NO: 2408) Novel promoter
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AGACCCCCAGGTCTACTTAGAGCAACGTTAGTAGAGGA (SEQ ID NO: 2404) Intron
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CGAGACAGAGAAGACTCTTGCGTTTCTGATAGGCACCTATTGGTCTTACTGACAT
CCACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon optimized
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GTTCTTAAAAACAAATTTCGGACTAAGAGGTACTTCGCCAAGCACCCCCGCATGG
GCTATCTGCCTGTGCAGACAGTGCTTGAAGGAGACAACATGGAAACCGATACAAT GTGA (SEQ ID NO: 2406) First DRG de- CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site Second DRG de- CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR
AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCAC
TGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCA
GTGAGCGAGCGAGCGCGC (SEQ ID NO: 2411)
TABLE-US-00037 Nucleic acid 16 (SEQ ID NO: 2427)
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TABLE-US-00038 Nucleic acid cargo 16 annotated Sequence AAV2 ITR
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(SEQ ID NO: 2402) Intron
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CCTGTGCAGACAGTGCTTGAAGGAGACAACATGGAAACCGATACAATGTGA (SEQ ID NO: 2406) First DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Second DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACT
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GAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00039 Nucleic acid 17 (SEQ ID NO: 2428)

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GTTCTTAAAAACAAATTTCGGACTAAGAGGTACTTCGCCAAGCACCCCCGCATGG
GCTATCTGCCTGTGCAGACAGTGCTTGAAGGAGACAACATGGAAACCGATACAAT GTGA (SEQ ID NO: 2406) First DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Second DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR
AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCAC
TGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCA
GTGAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00041 Nucleic acid 18 (SEQ ID NO: 2429)

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AACACTTCAATTATGACATCTGCCAGTCATGCTTCTTCAGCGGACGCGTGGCCAAGGGACACAAGATGCATTATC
CTATGGTGGAATACTGCACTCCAACTACTTCCGGCGAGGACGTGAGAGATTTCGCTAAAGTTCTTAAAAACAAAT
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ACCAAAGAAGACGCAGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTGAGGAACCCC
TAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGAC
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TABLE-US-00042 Nucleic acid cargo 18 annotated Sequence AAV2 ITR

GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACC TTTGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAACT CCATCACTAGGGGTTCCT (SEQ ID NO: 2408)	Novel promoter
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTG TGGCCTATTCTCTGAGAGGACGCTGGGCTAATAATAGGAGGGGAGATTAGGGAGG GATTTGACTGCATTGTTATCTCACAGGAATGGCGTTTGAGAGCCTTGCTTCCAAG CTCTTCCAAGAGTCATGCCACCCCCCTCCAATCTCTGTGGAAGTGCCCACTGCCT TTCCCGCTTCGTTCTCTTCTCTCGGTCCCATCCCTGACCCACTAGCCTGGTGCCG GGGGGAGACCAATTTCTGGAGGAAGCTCTGGGTACCCAGTCTTGCCCTAAGCCTG CCTCTGAACAGGCTGTTCTTGTTTATGACCCCCGTCCCAGCAGAATAGTGATTGC TCATCCCTGCATGTGATTTGAGAGGGCCAGTCTTTCAGGCCTGGAGGAAACTCAA GCCCCACCAGCCGCTTGGCATCGGAATGCTCAGCTGGTCCCCTCCTGGGGCTCAT GTGACCAAGATCTAGCTATAAATATCAGAGGGGCCTCAGACCAAGACAGAATTCG GGCACCAGGAGAAGGAAGCCAACAGGATCCGACCCGGTGTTTTGTGACAAAGGCA AGACCCCCAGGTCTACTTAGAGCAACGTTAGTAGAGGA (SEQ ID NO: 2404)	Intron
GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGT CGAGACAGAGAAGACTCTTGCGTTTCTGATAGGCACCTATTGGTCTTACTGACAT CCACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409)	Codon optimized
ATGCTGTGGTGGGAAGAGGTCGAGGACTGCTACGAGCGCGAAGATGTGCAGAAGA AAACTTTCACTAAATGGGTGAATGCCCAGTTCTCTAAGTTCGGAAAGCAGCACAT CGAAAACCTGTTTTCCGATCTGCAAGATGGCCGGCGGCTGCTGGACCTGCTGGAG GGGCTGACCGGACAGAAGCTGCCTAAGGAGAAAGGATCTACCAGGGTCCATGCTC TGAATAACGTGAACAAGGCTCTGAGGGTGCTGCAGAACAATAACGTGGACCTGGT GAACATTGGTTCCACAGACATCGTGGATGGAAACCACAAACTGACACTGGGACTG ATCTGGAACATCATTCTGCATTGGCAAGTGAAGAACGTGATGAAGAACATCATGG CCGGACTGCAGCAGACGAACTCCGAGAAGATCCTGCTTAGCTGGGTGAGACAGAG TACCCGGAATTACCCACAGGTGAACGTGATCAATTTCACAACCAGCTGGTCAGAC GGCCTGGCTCTGAATGCCCTGATTCATAGCCACCGGCCAGATCTGTTTGACTGGA ACTCTGTCGTGTGTCAGCAGTCAGCCACCCAGAGGCTGGAGCACGCCTTCAATAT TGCCCGCTATCAGCTGGGGATTGAGAAACTGCTGGACCCCGAGGATGTGGACACA ACATACCCTGACAAGAAGAGCATACTTATGTATATCACAAGCCTGTTCCAGGTGC TGCCACAGCAGGTCTCCATCGAGGCCATTCAAGAAGTGGAGATGCTGCCCAGGCC TCCTAAGGTCACCAAAGAGGAGCACTTTCAGCTCCATCATCAGATGCACTACAGC CAGCAAATTACCGTGTCCCTGGCCCAGGGCTACGAGAGGACAAGTTCCCCCAAGC CTAGATTTAAGAGCTATGCCTATACCCAGGCCGCCTACGTGACTACCTCCGATCC CACCAGGAGCCCATTTCCCAGCCAGCACCTGGAGGCTCCCGAAGATAAGAGCTTT GGCAGCAGCCTGATGGAATCTGAGGTGAACCTGGACAGATACCAGACTGCGCTGG AGGAGGTGCTGTCCTGGCTGCTGAGCGCCGAGGACACCCTGCAGGCTCAGGGCGA GATTAGTAATGATGTGGAGGTGGTGAAGGATCAGTTTCACACTCACGAAGGCTAC	microdystrophin coding sequence

ATGATGGACCTGACCGCCCACCAGGGCAGGGTGGGGAACATTCTGCAGCTCGGGA
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GTTCTTAAAAACAAATTTCGGACTAAGAGGTACTTCGCCAAGCACCCCCGCATGG
GCTATCTGCCTGTGCAGACAGTGCTTGAAGGAGACAACATGGAAACCGATACAAT GTGA (SEQ ID NO: 2406) First DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Second DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR
AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCAC
TGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCA
GTGAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00043 Nucleic acid 19 (SEQ ID NO: 2430)

GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTTTGGTCGCCCGGCCTCAGT
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AGCGCGC

TABLE-US-00044 Nucleic acid cargo 19 annotated Sequence AAV2 ITR
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TTGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAACTCC
ATCACTAGGGGTTCCT (SEQ ID NO: 2408) Novel promoter
TAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGT

TCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCC
CGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTT
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CTACGTATTAGTCATCGCTATTACCATGCTAGACTAGCATGCTGCCCATGTAAGGA
GGCAAGGCCTGGGGACACCCGAGATGCCTGGTTATAATTAACCCAGACATGTGGCT
GCCCCCCCCCCCCAACACCTGCTGCCTCTAAAAATAACCCTGCATGCCATGTTCCC
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GCACAGGGGCTGCCCTCATTCTACCACCACCTCCACAGCACAGACAGACACTCAGG AGCCAGCCAGC
(SEQ ID NO: 2402) Intron
GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGTC
GAGACAGAGAAGACTCTTGCGTTTCTGATAGGCACCTATTGGTCTTACTGACATCC
ACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon optimized
ATGCTGTGGTGGGAGGAGGTGGAGGATTGCTACGAGAGAGAGGACGTGCAGAAGAA microdystrophin
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CCAGTGCAGACCGTGCTGGAGGGGGACAATATGGAGACAGACACAATGTGA (SEQ ID NO: 2407) First DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACT
GAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGT
GAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00045 Nucleic acid 20 (SEQ ID NO: 2431)

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TABLE-US-00046 Nucleic acid Sequence cargo 20 annotated AAV2 ITR
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CCATCACTAGGGGTTCCT (SEQ ID NO: 2408) Novel promoter
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTG
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CAGGACTATATAACCCCAGAGGGACTGCCCCATGCTGACTCCTTGCTTCTTTCCT
GCTCGAGCCGCTGCCTTGCCCCCCGGAGACTGAAGACA (SEQ ID NO: 2403) Intron
GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGT
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ATGCTGTGGTGGGAGGAGGTGGAGGATTGCTACGAGAGAGAGGACGTGCAGAAGA microdystrophin
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GCTACCTGCCAGTGCAGACCGTGCTGGAGGGGGACAATATGGAGACAGACACAAT GTGA (SEQ ID NO: 2407) First DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCAC
TGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCA
GTGAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00047 Nucleic acid 21 (SEQ ID NO: 2432)

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TABLE-US-00048 Nucleic acid cargo 21 annotated Sequence AAV2 ITR
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ATCACTAGGGGTTCCT (SEQ ID NO: 2408) Novel promoter
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGT
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TCTACTTAGAGCAACGTTAGTAGAGGA (SEQ ID NO: 2404) Intron
GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGTC
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ACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon optimized
ATGCTGTGGTGGGAGGAGGTGGAGGATTGCTACGAGAGAGAGGACGTGCAGAAGAA microdystrophin
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CCAGTGCAGACCGTGCTGGAGGGGGACAATATGGAGACAGACACAATGTGA (SEQ ID NO: 2407) First DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Second DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2399) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACT
GAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGT
GAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00049 Nucleic acid 22 (SEQ ID NO: 2433)

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CGAGCGCGC

TABLE-US-00050 Nucleic acid cargo 22 annotated Sequence AAV2 ITR
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(SEQ ID NO: 2402) Intron
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ACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon optimized
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CCAGTGCAGACCGTGCTGGAGGGGGACAATATGGAGACAGACACAATGTGA (SEQ ID NO: 2407) First DRG de- CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site Second DRG de- CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACT
GAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGT

GAGCGAGCGAGCGCGC (SEQ ID NO: 2411)
TABLE-US-00051 Nucleic acid 23 (SEQ ID NO: 2434)
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TABLE-US-00052 Nucleic acid cargo 23 annotated Sequence AAV2 ITR
GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCT
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ATCACTAGGGGTTCCT (SEQ ID NO: 2408) Novel promoter
CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGT
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TGCCTTGCCCCCCGGAGACTGAAGACA (SEQ ID NO: 2403) Intron
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ACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon optimized
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AGAACAAGTTCCGGACAAAGAGATACTTTGCAAAACACCCAAGAATGGGCTACCTG
CCAGTGCAGACCGTGCTGGAGGGGGACAATATGGAGACAGACACAATGTGA (SEQ ID NO: 2407) First DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2400) targeting miRNA binding site Second DRG de- CAACAAAATCACTGATGCTGGA (SEQ ID NO: 2400) targeting miRNA binding site Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACT
GAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGT
GAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00053 Nucleic acid 24 (SEQ ID NO: 2435)

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TABLE-US-00054

Nucleic acid cargo	24 annotated Sequence
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Novel promoter	CGTCTGTCTAATTCCAGCCAAGGGCATGGGGGAAAGGACAGAGCCCAGGGACCTGT GGCCTATTCTCTGAGAGGACGCTGGGCTAATAATAGGAGGGGAGATTAGGGAGGGA TTTGACTGCATTGTTATCTCACAGGAATGGCGTTTGAGAGCCTTGCTTCCAAGCTC TTCCAAGAGTCATGCCACCCCCCTCCAATCTCTGTGGAAGTGCCCACTGCCTTTCC CGCTTCGTTCTCTTCTCTCGGTCCCATCCCTGACCCACTAGCCTGGTGCCGGGGGG AGACCAATTTCTGGAGGAAGCTCTGGGTACCCAGTCTTGCCCTAAGCCTGCCTCTG AACAGGCTGTTCTTGTTTATGACCCCCGTCCCAGCAGAATAGTGATTGCTCATCCC TGCATGTGATTTGAGAGGGCCAGTCTTTCAGGCCTGGAGGAAACTCAAGCCCCACC AGCCGCTTGGCATCGGAATGCTCAGCTGGTCCCCTCCTGGGGCTCATGTGACCAAG ATCTAGCTATAAATATCAGAGGGGCCTCAGACCAAGACAGAATTCGGGCACCAGGA GAAGGAAGCCAACAGGATCCGACCCGGTGTTTTGTGACAAAGGCAAGACCCCCAGG TCTACTTAGAGCAACGTTAGTAGAGGA (SEQ ID NO: 2404)
Intron	GTAAGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCTTGTC GAGACAGAGAAGACTCTTGCGTTTCTGATAGGCACCTATTGGTCTTACTGACATCC ACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409)
Codon optimized microdystrophin coding sequence	ATGCTGTGGTGGGAGGAGGTGGAGGATTGCTACGAGAGAGAGGACGTGCAGAAGAA AACCTTCACCAAATGGGTGAACGCCCAGTTCAGCAAGTTTGGCAAGCAGCACATCG AGAACCTGTTCAGCGATCTGCAGGACGGCCGCAGGCTGCTGGATCTGCTGGAGGGT CTGACCGGCCAGAAACTGCCCAAAGAGAAGGGATCAACCAGGGTGCACGCCCTGAA TAATGTGAACAAGGCCCTGAGAGTGCTGCAAAATAATAATGTGGATCTGGTGAACA

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AGAACAAGTTCCGGACAAAGAGATACTTTGCAAAACACCCAAGAATGGGCTACCTG
CCAGTGCAGACCGTGCTGGAGGGGGACAATATGGAGACAGACACAATGTGA (SEQ ID NO: 2407) First DRG de- CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site
Second DRG de- CGGCCTGATTCACAACACCAGCT (SEQ ID NO: 2400) targeting miRNA binding site

Short polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410) AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACT GAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGT GAGCGAGCGAGCGCGC (SEQ ID NO: 2411)

TABLE-US-00055 Nucleic acid 25 (SEQ ID NO: 2436)

GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTTTGGTCGCCCGGCCTCAGT
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CGAGCGCGC

TABLE-US-00056 Nucleic acid cargo 25 annotated Sequence AAV2 ITR
GCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCT
TTGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAACTCC
ATCACTAGGGGTTCCT (SEQ ID NO: 2408) Novel promoter
TAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGT
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CGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTT
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(SEQ ID NO: 2402) Intron
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ACTTTGCCTTTCTCTCCACAG (SEQ ID NO: 2409) Codon optimized
ATGCTGTGGTGGGAGGAGGTGGAGGATTGCTACGAGAGAGAGGACGTGCAGAAGAA microdystrophin
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AGAACAAGTTCCGGACAAAGAGATACTTTGCAAAACACCCAAGAATGGGCTACCTG
CCAGTGCAGACCGTGCTGGAGGGGGACAATATGGAGACAGACACAATGTGA (SEQ ID NO: 2407) First
DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Second
DRG de- TCATACAGCTAGATAACCAAAGA (SEQ ID NO: 2401) targeting miRNA binding site Short
polyA AGGAATAAAATATCTTTATTTTCATTACATCTGTGTGTTGGTTTTTTGTGTG (SEQ ID NO: 2410)
AAV2 ITR AGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTCGCTCACT
GAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGGCCTCAGT
GAGCGAGCGAGCGCGC (SEQ ID NO: 2411)
TABLE-US-00057 Nucleic acid 26 (SEQ ID NO: 2437)
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TABLE-US-00058 Nucleic acid cargo 26 annotated Sequence AAV2 ITR
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TABLE-US-00059 Nucleic acid 27 (SEQ ID NO: 2438)

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Example 2: Materials and Methods

[0572] The following Materials and Methods were used in experiments described herein.

[0573] Mice: The following mice were purchased from the Jackson laboratories: B10-mdx (JAX, #001801) and DBA/2J-mdx (JAX, 013141). Mice were injected retro-orbitally with Composition 1 at the indicated dose.

[0574] Cynomolgus Macaques: Juvenile male non-human primates were injected intravenously with Composition 1 at the indicated dose.

[0575] Microdystrophin Expression: Microdystrophin mRNA was quantified using a standard taqman assay. nhpRPL30 mRNA was used as the housekeeping control for quantification of microdystrophin expression. Microdystrophin protein was quantified using an automated capillary-based immunoassay (JESS™ automated western blotting). Protein was extracted from tissues and cultured cells. The mDys-FLAG protein expression was evaluated using an anti-dystrophin antibody at 1:10 dilution (Cat #MANHINGE2A, Developmental Studies Hybridoma Bank, University of Iowa). The amount of the mDys-FLAG expressed in tissues was calculated by interpolation to the full-length dystrophin standard curve derived from a pool of normal human heart or skeletal muscle samples that were diluted in dystrophin-deficient mdx mouse skeletal muscle to achieve levels of 100, 75, 50, 25, 0% of normal dystrophin. The relative intensity of the full-length and mDys-FLAG protein was normalized to alpha actinin-2 (ACTN2) using an anti-ACTN2 antibody at 1:800 dilution, which was used as a loading control (Abcam Cat #ab68167).

[0576] Vector Genome Quantification: For the vector genome quantification experiments, total DNA was extracted from tissue samples and DNA concentration was determined by Qubit dsDNA high-sensitivity assay kit (Thermo Fisher Scientific) using the QUBIT™ 4.0 Fluorometer (ThemoFisher Scientific). DNA was diluted to 0.5 ng/μL.

[0577] The VCN was quantified by digital droplet polymerase chain reaction (ddPCR) using ddPCR Supermix for Probes (no dUTP) (Bio-Rad Laboratories, Inc.) and an assay targeting a region in the viral genome. The VCN was normalized by the number of diploid genomes using the endogenous NHP telomerase gene (nhpTERT). All primers and probes were synthesized by Thermo Fisher Scientific. The ddPCR was performed using the QX200 ddPCR system (Bio-Rad Laboratories, Inc).

[0578] Immunofluorescence: Cryosections (10 mm in thickness) were sectioned from isopentane-frozen tissue and used for immunofluorescence staining to assess the level of the mDys-FLAG in a subset set of skeletal and heart tissues. The mDys-FLAG protein expression was examined with a rabbit monoclonal antibody against the FLAG peptide (Cat #14793S, 1:100, Cell Signaling Technology) and was visualized using Alexa Fluor 546 conjugated goat anti-rabbit IgG (H+L) antibody (Catalog #A-11035, 1:100, Thermo Fisher Scientific). The percentage of mDys-positive cells was quantified from digitalized FLAG immunostaining images using Biodock, an AI imaging software platform.

[0579] Mouse Muscle Physiology Analysis: The mechanical properties of the EDL muscle were assessed using an ex-vivo assay to evaluate the isometric, eccentric, and passive properties. The assay was performed using an isolated muscle force system (Aurora Scientific) according to published protocols, except that the EDL muscle was strained at 15% of the optimal muscle length during the eccentric contraction cycle (Hakim, Li, and Duan 2011; Hakim, Grange, and Duan 2011; Hakim, Wasala, and Duan 2013).

[0580] Statistical Analysis: Data are presented as means±standard deviation (SD). Statistical significance from the mdx-vehicle group was determined with one-way ANOVA analysis for the grip strength, ex-vivo assay (anatomical properties, isometric force, and viscous properties of the EDL muscle), and morphological analysis, or Two-way ANOVA analysis for the eccentric contraction profile and elastic properties. All statistical analyses were performed using GraphPad PRISM software version 10.2.3. $P<0.05$ was considered statistically significant.

Example 3: Administration of Composition 1 to B10-Mdx Mice Resulted in Widespread Microdystrophin Expression

[0581] A composition comprising a vector and transgene of the invention, referred to as Composition 1 was engineered, comprising the AAV capsid MyoAAV-LD B and a cargo comprising the novel promoter, codon optimized microdystrophin, a FLAG peptide tag, and two DRG de-targeting miRNA binding sites (two binding sites for miR-338-3p). The cargo nucleic acid sequence for Composition 1 is provided as SEQ ID NO: 2430 (SEQ ID NO: 2430 does not include the FLAG tag sequence). The FLAG tag in Composition 1 is fused to the C-terminal of microdystrophin and is used to detect FLAG tagged microdystrophin. Composition 1 includes the cargo nucleic acid sequence (SEQ ID NO: 2430) and the AAV capsid protein MyoAAV-LD B, which is described above in Example 1.

[0582] The Reference AAV composition comprised an AAVrh74 capsid, an MHCK7 promoter, microdystrophin, and a FLAG peptide tag. The Reference AAV composition is a surrogate of SRP-9001, which is a microdystrophin gene replacement therapy approved by the Food and Drug Administration (FDA). Compositions were administered into 8 weeks old B10-mdx mice, which is a mouse model of DMD. B10-mdx mice were injected with vehicle (n=8) as a negative control. Mice were necropsied 6 weeks post-injection. Microdystrophin expression was detected and quantified using whole-tissue fluorescent imaging and a capillary-based immunoassay.

[0583] Administration is detailed in the table below:

TABLE-US-00061 Test Article No. of Animals Dose Composition 1 8 4E+13 vg/kg Reference AAV composition 8 1.33E+14 vg/kg

[0584] FIG. **1**A shows images of FLAG immunofluorescence of microdystrophin expression in B10-mdx mouse quadriceps following administration of Composition 1 or the Reference AAV composition (AAVrh74-MHCK7-uDys-FLAG). FIG. **1**B shows results from capillary electrophoresis quantifying microdystrophin expression in mouse quadriceps following administration of Composition 1 or the Reference AAV composition.

[0585] Administration of Composition 1 resulted in 8 times higher microdystrophin protein expression in quadriceps of dystrophic mice compared to the Reference AAV composition (FIGS. **1**A-**1**B).

[0586] FIG. **2**A shows images of FLAG immunofluorescence of microdystrophin expression in B10-mdx mouse triceps following administration of Composition 1 or the Reference AAV composition. FIG. **2**B shows results from capillary electrophoresis quantifying microdystrophin expression in B10-mdx mouse triceps following administration of Composition 1 or the Reference AAV composition.

[0587] Administration of Composition 1 resulted in 8 times higher microdystrophin protein expression in triceps of dystrophic mice compared to the Reference AAV composition (FIGS. **2**A-**2**B).

[0588] FIG. **3**A shows images of FLAG immunofluorescence of microdystrophin expression in B10-mdx mouse heart following administration of Composition 1 or the Reference AAV composition. FIG. **3**B shows results from capillary electrophoresis quantifying microdystrophin expression in B10-mdx mouse hearts following administration of Composition 1 or the Reference AAV composition.

[0589] Administration of Composition 1 resulted in 3 times higher microdystrophin protein expression in heart muscle of dystrophic mice compared to the Reference AAV composition (FIGS. **3**A-**3**B).

[0590] Taken together, these results demonstrate that Composition 1-injected mice had greater expression of microdystrophin in each muscle group tested compared to Reference AAV composition-injected mice. This observed increase in microdystrophin expression could be achieved using a lower dose of Composition 1 (4E+13 vg/kg) than Reference AAV composition (1.33E+14 vg/kg).

Example 4: Administration of Composition 1 to B10-Mdx Mice Resulted in Restoration of Muscle Function

[0591] Muscle physiology of mice injected with Composition 1 or Reference AAV composition was analyzed. Specific force (FIG. **4**A), protection of muscle against eccentric contraction (FIG. **4**B), and microdystrophin expression (FIG. **4**C) of extensor digitorum longus (EDL) muscle was analyzed. Protection of dystrophic skeletal muscle against eccentric damage was shown to be dependent on microdystrophin expression levels (FIGS. **4**A-**4**C). EDL muscle was analyzed in mice 6 weeks after injection.

[0592] FIG. **4**A shows a graph of specific force following administration of Composition 1 or the Reference AAV composition. FIG. **4**B shows a graph of protection against eccentric damage following administration of Composition 1 or the Reference AAV composition. FIG. **4**C shows results from capillary electrophoresis from EDL muscle following administration of Composition 1 or the Reference AAV composition. The level of microdystrophin expression was determined by interpolation using a dystrophin standard curve derived from normal human skeletal muscle samples.

[0593] Composition 1 outperformed the Reference AAV composition, reverting specific force back to wild type levels (FIG. **4**A). Composition 1 administration resulted in protection of the EDL muscle of B10-mdx mice against eccentric contraction (FIG. **4**B). Notably, the Reference AAV failed to outperform the vehicle control in protecting against eccentric damage (FIG. **4**B). Microdystrophin expression in EDL muscles of mice administered Composition 1 was confirmed (FIG. **4**C), while little to no microdystrophin expression was detected in mice administered Reference AAV composition (FIG. **4**C).

[0594] Taken together, these results demonstrate that muscles from Composition 1-injected mice had higher levels of microdystrophin expression, recovered significantly greater specific force, and were more protected from damage compared to muscles from Reference AAV composition-injected animals. Notably, the observed increase in microdystrophin expression and restoration of muscle function could be achieved using a lower dose of Composition 1 (4E+13 vg/kg) than the dose of Reference AAV composition (1.33E+14 vg/kg).

Example 5: Administration of Composition 1 to Non-Human Primates (NHPs) Resulted in Effective Microdystrophin Expression in Heart and Skeletal Muscle

[0595] Composition 1 was further benchmarked against Reference AAV composition in NHPs, specifically cynomolgus macaques. Tissues were collected 28 days post-injection, and microdystrophin expression was detected on the sarcolemma using immunofluorescence and quantified using a capillary-based immunoassay. The level of microdystrophin expression was determined by interpolation using a dystrophin standard curve derived from normal human heart or skeletal muscle samples.

[0596] Administration is detailed in the table below:

TABLE-US-00062 Test Article No. of Animals Dose Composition 1 3 4E+13 vg/kg Reference AAV composition 3 1.33E+14 vg/kg

NHP Heart:

[0597] FIG. **5**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP heart left ventricles following administration of Composition 1 or the Reference AAV composition. FIG. **5**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP heart left ventricles following administration of Composition 1 or the Reference AAV composition.

[0598] Administration of Composition 1 to NHPs resulted in ˜60% microdystrophin protein expression in NHP heart left ventricles as compared to normal human heart dystrophin expression (FIG. **5**B). The microdystrophin protein expression levels after injection of Reference AAV composition were below the limit of detection in NHP heart left ventricles (FIGS. **5**A-**5**B).

[0599] FIG. **6**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP heart right ventricles following administration of Composition 1 or the Reference AAV composition. FIG. **6**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP heart right ventricles following administration of Composition 1 or the Reference AAV composition.

[0600] Administration of Composition 1 to NHPs resulted in ˜80% microdystrophin protein expression in NHP heart right ventricles as compared to normal dystrophin expression (FIG. **6**B, see FIG. **5**B for human heart dystrophin standard). The microdystrophin protein levels after injection of Reference AAV composition was below the limit of detection in NHP heart right ventricles (FIGS. **6**A-**6**B).

[0601] FIG. **7**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP heart interventricular septum following administration of Composition 1 or the Reference AAV composition. FIG. **7**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP heart interventricular septum following administration of Composition 1 or the Reference AAV composition.

[0602] Administration of Composition 1 to NHPs resulted in ˜70% microdystrophin protein expression in NHP heart interventricular septum as compared to normal dystrophin expression (FIG. **7**B, see FIG. **5**B for human heart dystrophin standard). The microdystrophin protein levels after injection of Reference AAV composition was below the limit of detection in NHP heart interventricular septum (FIGS. **7**A-**7**B).

[0603] Taken together, these results demonstrate that Composition 1-injected NHPs had greater expression of microdystrophin in each heart muscle group tested as compared to Reference AAV composition-injected NHPs. This observed increase in microdystrophin expression could be achieved using a lower dose of Composition 1 (4E+13 vg/kg) than Reference AAV composition (1.33E+14 vg/kg).

NHP Skeletal Muscle:

[0604] FIG. **8**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP gastrocnemius following administration of Composition 1 or the Reference AAV composition. FIG. **8**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP gastrocnemius following administration of Composition 1 or the Reference AAV composition.

[0605] Administration of Composition 1 to NHPs resulted in ˜55% microdystrophin protein expression in NHP gastrocnemius as compared to normal dystrophin expression (FIG. **8**B). The microdystrophin protein levels after injection of Reference AAV composition was below the limit of detection in NHP gastrocnemius muscle (FIGS. **8**A-**8**B).

[0606] FIG. **9**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP pectoralis major following administration of Composition 1 or the Reference AAV composition. FIG. **9**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP pectoralis major following administration of Composition 1 or the Reference AAV composition.

[0607] Administration of Composition 1 resulted in ˜100% microdystrophin protein expression as compared to normal dystrophin expression (FIG. **9**B, see FIG. **8**B for human skeletal muscle dystrophin standard). The microdystrophin protein levels after injection of Reference AAV composition were below the limit of detection in NHP pectoralis major muscle (FIGS. **9**A-**9**B).

[0608] FIG. **10**A shows images of FLAG immunofluorescence of microdystrophin expression in NHP diaphragm following administration of Composition 1 or the Reference AAV composition. FIG. **10**B shows results from capillary electrophoresis quantifying microdystrophin expression in NHP diaphragm following administration of Composition 1 or the Reference AAV composition.

[0609] Administration of Composition 1 resulted ˜120% transgene protein expression as compared to normal dystrophin expression (FIG. **10**B, see FIG. **8**B for human skeletal muscle dystrophin standard). The Reference AAV composition results resulted in less than 5% transgene expression in NHP diaphragm (FIGS. **10**A-**10**B).

[0610] Taken together, these results demonstrate that Composition 1-injected NHPs had greater expression of microdystrophin in each skeletal muscle group tested as compared to Reference AAV composition-injected NHPs. This observed increase in microdystrophin expression in skeletal muscle could be achieved using a lower dose of Composition 1 (4E+13 vg/kg) than Reference AAV composition (1.33E+14 vg/kg).

Liver Transduction

[0611] The liver is a major site of transduction by naturally occurring AAV capsids and a target of toxicity associated with administration of AAV vectors. Therefore, the abundance of liver genomes in the Composition 1-injected NHPs and the Reference AAV composition-injected NHPs was analyzed. FIG. **11** shows a graph of vector genome detected in the liver of NHPs injected with Composition 1 or Reference AAV composition.

[0612] Composition 1-injected NHPs had a 27 times lower vector genome per diploid genome in the liver compared to the Reference AAV composition-injected NHPs (FIG. **11**).

[0613] Taken together, these results demonstrate that treatment with Composition 1 can achieve high levels of microdystrophin expression in heart and skeletal muscles of NHPs with lower transduction of liver cells.

Example 6: Systemic Administration of Composition 1 Results in Dose-Dependent Improvement of Muscle Function in a Severe DMD Mouse Model and Results in High and Uniform Microdystrophin Expression in the Heart and Skeletal Muscles of Non-Human Primates

[0614] Duchenne muscular dystrophy (DMD) is a fatal disease, and the most common muscular dystrophy, with 20,000 boys diagnosed worldwide each year. DMD is caused by loss of the dystrophin protein, an essential component of the dystrophin-glycoprotein complex that preserves myofiber membrane integrity during contraction. Adeno-associated virus (AAV) mediated microdystrophin (mDys) gene replacement therapy is a promising approach to treat DMD. Unfortunately, clinical stage AAV-mDys therapies produce low, nonuniform mDys expression in skeletal muscles, with no clinical data supporting mDys expression in the heart, leading to suboptimal clinical efficacy and limited potential to address cardiomyopathy, a major cause of mortality in DMD. Engineered capsids together with regulatory element technologies of were used to develop the lead candidates of the invention, a novel AAV-mDys therapy with the potential to be safer and more effective. The lead candidate expresses a codon-optimized mDys selectively in cardiac and skeletal muscles by using a skeletal muscle/heart-tropic and liver de-targeted capsid (MyoAAV-LD B) and novel regulatory elements. A FLAG tag was added at the C-terminus of the mDys protein to facilitate immunofluorescent detection. To evaluate safety, efficacy, and mDys expression, 8-week-old male D2-mdx mice were injected intravenously with the lead candidate at 1E13, 2E13, and 4E13 vg/kg, and were evaluated 7 weeks later. The lead candidate produced a significant dose-dependent improvement in muscle pathology and fibrosis, grip strength, and mechanical properties of the extensor digitorum longus muscle ex vivo. Maximal efficacy occurred at 4E13 vg/kg, associated with mDys expression in nearly all myofibers at protein levels between 62 and 83% of normal human dystrophin levels. Furthermore, the lead candidate was well tolerated and did not induce toxicity in any treated mice.

[0615] To translate these findings to large animals and to guide human dose selection, biodistribution and safety of the lead candidate was evaluated in juvenile male non-human primates (NHPs). Animals received a single intravenous dose of 4E13 vg/kg, and tissues were collected 28- and 90-days post-dose. The lead candidate was well tolerated, had no effects on chemistry, hematology, or histopathology up to 90 days, and produced robust, uniform mDys expression in almost all skeletal muscle fibers and cardiomyocytes. The average levels of mDys protein in heart and skeletal muscles relative to normal human dystrophin were 72% and 66%, respectively. Further, the expression of the mDys transgene was highly selective for skeletal muscle and heart, with no transgene expression detected in off-target tissues.

[0616] In summary, the results illustrate the need for high mDys expression in the majority of myofibers to produce maximal efficacy, and for the first time demonstrate quantitative expression of mDys protein in NHP muscle at levels that are associated with maximal efficacy in a DMD mouse model. These results provide compelling preclinical data to support further development of Composition 1 to treat DMD.

Example 7: Administration of Composition 1 Results in Dose-Dependent Microdystrophin Expression in a Severe DMD Mouse Model

[0617] To evaluate safety, efficacy, and microdystrophin expression of Composition 1 (described in Example 3), 8-week-old male D2-mdx mice, which is a severe mouse model of DMD, were injected intravenously with Composition 1 at 1E13, 2E13, or 4E13 vg/kg. Mice were evaluated 7 weeks and 22 weeks post-injection. WT mice and D2-mdx mice were injected with vehicle as a control. Microdystrophin expression was detected by FLAG immunofluorescence and quantified using capillary-based immunoassay.

[0618] Treatment with Composition 1 did not induce cardiotoxicity or hepatotoxicity in D2-mdx mice (data not shown). This demonstrates that Composition 1 can be administered with little to no damage or dysfunction to the heart and the liver.

[0619] FIG. **12** shows images of FLAG immunofluorescence of microdystrophin expression in D2-mdx mouse diaphragm, quadriceps, heart, and extensor digitorum longus (EDL) tissue at 7 weeks post injection following administration of Composition 1.

[0620] Immunofluorescence analysis of mouse tissues showed that Composition 1 treatment resulted in dose-dependent expression of microdystrophin in each of the muscles examined (EDL, diaphragm, quadriceps, and heart) (FIG. **12**).

[0621] FIGS. **13**A-**13**B shows results from capillary electrophoresis quantifying microdystrophin expression in D2-mdx mouse heart tissue at 7 weeks (FIG. **13**A) and 22 weeks (FIG. **13**B) following administration of Composition 1.

[0622] The examined heart tissues showed supra-physiological microdystrophin protein levels, exceeding 300% of normal in the low-dose group, and up to 535% in the high-dose group at 7 weeks post injection (FIG. **13**A). At 22 weeks post dosing, the microdystrophin protein level in the heart was ˜2.2-fold higher than the 7-week time point across each of the tested doses (FIGS. **13**A-**13**B).

[0623] FIGS. **14**A-**14**B shows results from capillary electrophoresis quantifying microdystrophin expression in D2-mdx mouse skeletal muscle tissue at 7 weeks (FIG. **14**A) and 22 weeks (FIG. **14**B) following administration of Composition 1.

[0624] In skeletal muscle, microdystrophin protein level was, on average, 70% of normal in the low-dose group and was, on average, 83% of normal in the mid and high-dose groups at 7 weeks post-dosing (FIG. **14**A). By 22 weeks, the microdystrophin protein level in skeletal muscles increased minimally by 1.2-fold across all doses (FIGS. **14**A-**14**B). At 22 weeks post-injection, the percent of microdystrophin-FLAG positive myofibers in EDL and quadriceps was similar, with 76%, 88%, and 93% expression in the low, mid, and high doses, respectively (FIG. **14**B).

[0625] Taken together, these results demonstrate that treatment with Composition 1 restored microdystrophin protein levels in heart and skeletal muscles in a severe DMD mouse model, without inducing cardiotoxicity or hepatotoxicity.

Example 8: Administration of Composition 1 Increased Body and Tibialis Anterior (TA) Muscle Weights in a Severe DMD Mouse Model

[0626] Composition 1 (described in Example 3) was intravenously injected into 8-week-old male D2-mdx mice, a severe DMD mouse model, at a dose of 1E13, 2E13, or 4E13 vg/kg. WT mice and D2-mdx mice were injected with vehicle as a control. Body weight of mice was measured pre-dose and monitored twice weekly throughout the 7 weeks and 22 weeks study cohorts (FIGS. **15**A-**15**B). Body weight and TA muscle weight were measured at 7 weeks and 22 weeks post-injection (FIGS. **16**A-**16**B).

[0627] FIGS. **15**A-**15**B show graphs of body weight of D2-mdx mouse over 7 weeks (FIG. **15**A) and 22 weeks (FIG. **15**B) following administration of Composition 1. After about 16 weeks, body weights of mice administered the high dose of Composition 1 were comparable to wild type mice (FIGS. **15**A-**15**B).

[0628] FIGS. **16**A-**16**B show graphs of body weight and TA weight of D2-mdx mouse at 7 weeks (FIG. **16**A) and at 22 weeks (FIG. **16**B) following administration of Composition 1.

[0629] Mice injected with the mid or high dose of Composition 1 showed a significant increase in body weight at 22 weeks post-dosing (FIG. **16**B, left panel). Composition 1-injected mice at each of the tested doses showed a significant increase in TA weight after 7 weeks (FIG. **16**A, right panel) and 22 weeks. (FIG. **16**B, right panel).

[0630] Taken together, these results demonstrate that treatment with Composition 1 increased body and TA muscle weights in a severe DMD mouse model. The observed increase in body and TA muscle weights could be achieved using low (1E13 vg/kg), middle (2E13 vg/kg), and high (4E13 vg/kg) doses of Composition 1.

Example 9: Administration of Composition 1 Improved the Mechanical Properties of Extensor Digitorum Longus (EDL) Muscle in a Severe DMD Mouse Model

[0631] Composition 1 (described in Example 3) was intravenously injected into 8-week-old male D2-mdx mice, a severe DMD mouse model, at a dose of 1E13, 2E13, or 4E13 vg/kg. WT mice and D2-mdx mice were injected with vehicle as a control. The anatomical properties (FIG. **17**A and FIG. **18**A), muscle contractility (FIG. **17**B and FIG. **18**B), protection against eccentric contraction (FIG. **17**C and FIG. **18**C), elastic properties (FIG. **17**D and FIG. **18**D), and viscous properties (FIG. **17**E and FIG. **18**E) of EDL muscle was analyzed at 7 weeks and 22 weeks post-injection.

[0632] FIG. **17**A and FIG. **18**A show graphs of anatomical measurements of EDL muscle, including cross-section area (CSA) and optimal length (Lo), in D2-mdx mouse at 7 weeks (FIG. **17**A) or 22 weeks (FIG. **18**A) post-injection of

Composition 1.
[0633] FIG. **17**B and FIG. **18**B show graphs of quantitative measurements of muscle contractility in EDL muscle, including specific twitch force (Pt) and tetanic force (Po), in D2-mdx mouse at 7 weeks (FIG. **17**B) or 12 weeks (FIG. **18**B) post-injection of Composition 1.
[0634] FIG. **17**C and FIG. **18**C show a graph of eccentric contraction of EDL muscle in D2-mdx mice at 7 weeks (FIG. **17**C) or 12 weeks (FIG. **18**C) post-injection of Composition 1.
[0635] FIG. **17**D and FIG. **18**D show a graph of the stress-strain relationship, evaluating elasticity, of EDL muscle in D2-mdx mice at 7 weeks (FIG. **17**D) or 12 weeks (FIG. **18**D) post-injection of Composition 1.
[0636] FIG. **17**E and FIG. **18**E shows a graph of the stress relaxation rate, evaluation the viscous properties of muscle, of EDL muscle in D2-mdx mice at 7 weeks (FIG. **17**E) or 12 weeks (FIG. **18**E) post-injection of Composition 1.
[0637] Taken together, these results demonstrate that treatment of D2-mdx mice with Composition 1 improved the contractile properties of the EDL muscle in a dose-dependent manner and provided long-term protection of muscle function up to 22 weeks post-dosing. The treatment with Composition 1 was fully efficacious at the mid-dose of 2E13 vg/kg. However, further improvements in many mechanical parameters (e.g., EDL weight, optimal length, specific twitch force) were noted in the high-dose group (4E13 vg/kg).
Example 10: Administration of Composition 1 Resulted in High and Selective Microdystrophin Expression in Heart and Skeletal Muscles of Non-Human Primates (NHPs)
[0638] Biodistribution and safety of the Composition 1 was evaluated in juvenile male non-human primates (NHPs). Animals received a single intravenous dose of 4E13 vg/kg of Composition 1, and tissues were collected 28- and 90-days post-injection. Microdystrophin-FLAG mRNA levels were assessed in heart, skeletal muscles, and off-target tissues. Expression of microdystrophin was analyzed by FLAG immunostaining.
[0639] Composition 1 was safe and well tolerated, with no treatment-related effects in clinical chemistry, hematology, and histopathology, including liver and DRG up to 90 days post-injection (data not shown).
[0640] FIG. **19**A-**19**C show graphs of results quantifying transgene mRNA levels in NHP heart, skeletal muscle, and off-target tissues 28 days post administration of Composition 1. The transcript copy was normalized to nhpRPL30 (n=3/group). These results demonstrate that treatment of NHPs with Composition 1 results in high and selective expression of microdystrophin in NHP heart and skeletal muscles, with little to no expression in off-target tissues.
[0641] FIG. **20** shows FLAG immunofluorescence images of microdystrophin expression in NHP heart and skeletal muscles 28 days post administration of Composition 1. FIGS. **21**A-**21**B show graphs of results from capillary electrophoresis quantifying microdystrophin expression in NHP heart and skeletal muscles at 28 days and 90 days post administration of Composition 1. The amount of the microdystrophin expressed in tissues was calculated by interpolation to the full-length dystrophin standard curve derived from normal human heart or skeletal muscle samples.
[0642] Treatment with Composition 1 resulted in high microdystrophin protein expression in skeletal muscles and the cardiomyocytes. After 28 days, treatment with Composition 1 resulted in robust microdystrophin protein expression levels in heart and skeletal muscles that were, on average, 72% and 66% of normal full-length dystrophin, respectively (FIGS. **21**A-**21**B). Microdystrophin expression levels increased after 90 days post-injection, with microdystrophin protein expression levels in heart and skeletal muscles that were, on average, 136% and 96% of normal full-length dystrophin, respectively (FIGS. **21**A-**21**B).
[0643] Taken together, these results demonstrate that a single intravenous dose of Composition 1 was safe and well tolerated in NHPs up to 90 days post injection and resulted in high microdystrophin transcript and protein expression in skeletal muscles and the heart.
[0644] The Examples described herein use Composition 1 as an example composition for treating DMD using the recombinant nucleic acids, vectors, and viral particles described herein.
Example 11: Codon Optimization Increased Expression of Microdystrophin
[0645] Three different codon optimized microdystrophin sequences (referred to as v1, v2, and v3) were generated and the expression of microdystrophin from the codon optimized sequences was tested in C2C12 derived myotubes and human primary myotubes. Microdystrophin expression was detected and quantified using a capillary-based immunoassay. The level of microdystrophin expressed from codon optimized microdystrophin sequences was compared to the level of microdystrophin expressed from an endogenous coding sequence of dystrophin. Vinculin was used as a loading control.
[0646] FIGS. **22**A-**22**B show results from capillary electrophoresis analysis of microdystrophin expression from an endogenous coding sequence or a codon optimized microdystrophin sequence (v1, v2, or v3) in C2C12 derived myotubes.
[0647] FIGS. **23**A-**23**B show results from capillary electrophoresis analysis of microdystrophin expression from an endogenous coding sequence or a codon optimized microdystrophin sequence (v1, v2, or v3) in human primary myotubes.
[0648] In both C2C12 derived myotubes (FIGS. **22**A-**22**B) and human primary myotubes (FIGS. **23**A-**23**B), the level of expression of microdystrophin from each of the codon optimized microdystrophin sequences (v1, v2, v3) was significantly greater than the level of microdystrophin expressed from the endogenous coding sequence.
[0649] These results demonstrate that microdystrophin expression can be significantly increased using a codon optimized microdystrophin sequence (v1, v2, or v3) compared to the endogenous microdystrophin sequence.
Example 12: Administration of MyoAAV-LD a Resulted in Higher Microdystrophin Expression Compared to AAV9 in NHPs
[0650] To investigate the muscle transduction profile of MyoAAV-LD A after systemic delivery, NHPs were injected with

either MyoAAV-LD A- or AAV9-CBh-uDys-FLAG, and microdystrophin transgene expression in different muscle tissues of the injected NHPs was examined. CBh-uDys-FLAG includes a C-terminal FLAG tagged microdystrophin (uDys) under the control of a hybrid CBA promoter (CBh). Gastrocnemius muscle from NHPs injected with vehicle (saline) was used as a negative control.

[0651] Microdystrophin expression was detected using immunofluorescence and quantified using a capillary-based immunoassay. Microdystrophin mRNA was quantified using a standard taqman assay. nhpRPL30 mRNA was used as the housekeeping control for quantification of microdystrophin expression. Microdystrophin expression was assessed in skeletal muscles including gastrocnemius (FIGS. **24**A-**24**C), triceps (FIGS. **25**A-**25**C), biceps brachii (FIGS. **26**A-**26**C), and pectoralis major (FIGS. **27**A-**27**C).

[0652] Fluorescence imaging of harvested tissues revealed far greater fluorescence intensity in muscles of NHP injected with MyoAAV-LD A—as compared to AAV9-injected NHP (FIG. **24**A, FIG. **25**A, FIG. **26**A, and FIG. **27**A). Strong microdystrophin transgene expression in muscle fibers of MyoAAV-LD A-injected NHP was further confirmed by capillary electrophoresis (FIG. **24**B, FIG. **25**B, FIG. **26**B, and FIG. **27**B). Quantification of microdystrophin mRNA in different skeletal muscles of NHP revealed 15 to 40 times higher microdystrophin transgene expression in muscles of MyoAAV-LD A-injected NHPs compared to AAV9-injected NHPs (FIG. **24**C, FIG. **25**C, FIG. **26**C, and FIG. **27**C).

[0653] Taken together, these results demonstrate that MyoAAV-LD A can transduce skeletal muscles with significantly higher efficiency that wild type AAV9.

Example 13: Administration of DMD Tool Compound Restores Muscle Force and Protects the Muscle Against Eccentric Damage in B10-Mdx Mice

[0654] Muscle physiology, including specific force (FIG. **28**A) and protection of muscle against eccentric contraction (FIG. **28**B), was analyzed in mice injected with a DMD tool compound that includes MyoAAV-LD B-uDys under the control of a muscle specific promoter. 8 weeks old B10-mdx mice were injected with 2E+13 vg/kg of the DMD tool compound, and EDL muscle was analyzed 6 weeks after injection.

[0655] Administration of the DMD tool compound reverted specific force back to wild type levels (FIG. **28**A) and resulted in protection of the EDL muscle against eccentric contraction (FIG. **28**B). These results demonstrate that delivery of microdystrophin to muscle tissues via MyoAAV-LD B can restore muscle strength and protect muscle against damage in a mouse model of DMD.

CONCLUSION

[0656] AAV compositions of the invention (combining an engineered liver de-targeted MyoAAV capsid comprising the amino acid sequence RGDR (SEQ ID NO: 2440), a novel promoter of the invention, and a codon optimized microdystrophin sequence of the invention) result in expression of microdystrophin in mice and NHPs at rates far superior to a Reference AAVrh74 capsid comprising a muscle specific promoter (MHCK7) and a transgene encoding microdystrophin, while also reducing the presence of AAV genomes in the liver. Notably, the Reference AAV capsid resulted in very low microdystrophin expression in NHP skeletal muscles and undetectable expression in the NHP heart, while compositions of the invention resulted in uniform microdystrophin expression throughout NHP heart and skeletal muscle tissues with levels comparable to endogenous dystrophin levels.

INCORPORATION BY REFERENCE

[0657] References and citations to other documents, such as patents, patent applications, patent publications, journals, books, papers, web contents, have been made throughout this disclosure. All such documents are hereby incorporated herein by reference in their entirety for all purposes.

EQUIVALENTS

[0658] Various modifications of the invention and many further embodiments thereof, in addition to those shown and described herein, will become apparent to those skilled in the art from the full contents of this document, including references to the scientific and patent literature cited herein. The subject matter herein contains important information, exemplification and guidance that can be adapted to the practice of this invention in its various embodiments and equivalents thereof.

Claims

1. A recombinant nucleic acid comprising: a muscle specific promoter; a sequence encoding microdystrophin; and at least one dorsal root ganglia (DRG) de-targeting miRNA binding site.

2. The recombinant nucleic acid of claim 1, wherein the muscle specific promoter comprises a CK8 promoter, a desmin promoter, a Mb promoter, a MCK promoter, a MHCK7 promoter, a skeletal muscle alpha actin promoter, or a TTNI2 promoter.

3. The recombinant nucleic acid of claim 1, wherein the promoter comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404.

4.-**6**. (canceled)

7. The recombinant nucleic acid of claim 1, wherein the sequence encoding microdystrophin encodes a microdystrophin protein comprising or consisting of an amino acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at

least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to SEQ ID NO: 2439.
8. (canceled)
9. The recombinant nucleic acid of claim 1, wherein the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407.
10.-**12**. (canceled)
13. The recombinant nucleic acid of claim 1, wherein the at least one DRG de-targeting miRNA binding site comprises a binding site for miR-338, miR-138, or miR-9.
14. The recombinant nucleic acid of claim 1, wherein the at least one DRG de-targeting miRNA binding site comprises a binding site for miR-338-3p, miR-138-5p, or miR-9-5p.
15. The recombinant nucleic acid of claim 1, wherein the at least one DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401.
16.-**20**. (canceled)
21. The recombinant nucleic acid of claim 1, wherein the recombinant nucleic acid comprises, optionally from 5′ to 3′: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401.
22. The recombinant nucleic acid of claim 1, wherein the recombinant nucleic acid is selected from: (a) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (b) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (c) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (d) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least

85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (e) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (f) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (g) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; (h) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; (i) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; (j) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least

97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (k) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (l) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (m) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (n) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (o) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (p) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; (q) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; (r) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; (s) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (t) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (u) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; (v) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (w) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (x) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; (y) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; (z) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; and (aa) a recombinant nucleic acid comprising: the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401.

23. The recombinant nucleic acid of claim 1, wherein the recombinant nucleic acid is selected from: (a) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (b) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (c) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (d) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (e) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (f) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (g) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; (h) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; (i) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; (j) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (k) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (l) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (m) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (n) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (o) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (p) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; (q) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; (r) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; (s) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (t) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (u) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; (v) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (w) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (x) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; (y) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; (z) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ

ID NO: 2401; and (aa) a recombinant nucleic acid comprising: the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; and the at least one DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401.

24. The recombinant nucleic acid of claim 1, further comprising an intron.

25.-**27**. (canceled)

28. The recombinant nucleic acid of claim 1, further comprising a polyadenylation (PolyA) signal.

29.-**31**. (canceled)

32. The recombinant nucleic acid of claim 1, further comprising a 5′ inverted terminal repeat (ITR) and a 3′ ITR.

33.-**37**. (canceled)

38. The recombinant nucleic acid of claim 1, wherein the recombinant nucleic acid comprises, optionally from 5′ to 3′: a 5′ ITR; a promoter; an intron; a sequence encoding microdystrophin; a first DRG de-targeting miRNA binding site; a second DRG de-targeting miRNA binding site; a PolyA signal; and a 3′ ITR.

39. The recombinant nucleic acid of claim 38, wherein: the 5′ ITR comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404; the intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407; the first DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401; the second DRG de-targeting miRNA binding site comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2399-2401; the PolyA signal comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

40. The recombinant nucleic acid of claim 38, wherein the recombinant nucleic acid is selected from: (a) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; and the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (b) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (c) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (d) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (e) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (f) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (g) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (h) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (i) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (j) a recombinant nucleic acid comprising: the 5′ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (k) a recombinant nucleic acid comprising: the 5′ comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (l) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least

96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (m) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (n) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (o) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (p) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (q) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (r) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (s) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (t) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (u) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (v) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of

SEQ ID NO: 2411; (w) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (x) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (y) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%

identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; (z) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411; and (aa) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of a nucleic acid sequence that is at least least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of a nucleic acid sequence that is at least 85%, 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of SEQ ID NO: 2411.

41. The recombinant nucleic acid of claim 38, wherein the recombinant nucleic acid is selected from: (a) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (b) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (c) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (d) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (e) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (f) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (g) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (h) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (i) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2405; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (j) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO:

2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprises or consists of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (k) a recombinant nucleic acid comprising: the 5′ comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (l) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (m) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (n) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (o) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (p) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (q) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (r) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2406; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (s) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (t) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (u) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2399; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (v) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (w) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (x) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2400; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (y) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2402; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411; (z) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2403; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid

sequence of SEQ ID NO: 2411; and (aa) a recombinant nucleic acid comprising: the 5′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2408; the promoter comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2404; the intron comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2409; the sequence encoding microdystrophin comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2407; the first DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the second DRG de-targeting miRNA binding site comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2401; the PolyA signal comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2410; and the 3′ ITR comprising or consisting of the nucleic acid sequence of SEQ ID NO: 2411.
42. The recombinant nucleic acid of claim 1, wherein the recombinant nucleic acid comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2412-2438.
43. (canceled)
44. A recombinant nucleic acid comprising a promoter, wherein the promoter comprises or consists of a nucleic acid sequence that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2402-2404.
45.-**46**. (canceled)
47. A recombinant nucleic acid comprising a codon optimized sequence encoding microdystrophin that is at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the nucleic acid sequence of any one of SEQ ID NOs: 2405-2407.
48.-**49**. (canceled)
50. An adeno-associated virus (AAV) particle comprising the recombinant nucleic acid of claim 1 and a capsid protein.
51. The AAV particle of claim 50, wherein the capsid protein is derived from a serotype selected from AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV6.2, AAV7, AAV8, AAV9, AAV10, AAV11, AAV12, AAV13, AAVrh, AAVrh10, AAVrh74, AAV-DJ, AAV-DJ/8, Anc80, and AAV7m8, or a derivative, hybrid or chimeric serotype thereof.
52.-**53**. (canceled)
54. The AAV particle of claim 50, wherein the capsid protein comprises an amino acid sequence of formula (I): TABLE-US-00063 (I) X.sub.1NX.sub.2X.sub.3X.sub.4RGDRX.sub.5X.sub.6L, wherein each of X.sub.1-X.sub.6 is any amino acid.
55. (canceled)
56. The AAV particle of claim 54, wherein, relative to a wild type AAV9 capsid, X.sub.1 is a substitution at amino acid 451, X.sub.2 is a substitution at amino acid 453, X.sub.3 is a substitution at amino acid 454, X.sub.4 is a substitution at amino acid 455, and RGDRX.sub.5X.sub.6L is inserted after amino acid 455.
57.-**59**. (canceled)
60. The AAV particle of claim 54, wherein the amino acid sequence of formula (I) comprises or consists of any one of SEQ ID NOs: 2-800.
61. (canceled)
62. The AAV particle of claim 54, wherein the capsid protein further comprises a deletion of amino acid G267 relative to a wild type AAV9 vector.
63.-**64**. (canceled)
65. A method of treating Duchenne muscular dystrophy (DMD) in a subject in need thereof, the method comprising administering to the subject an effective amount of the recombinant nucleic acid of claim 1.
66. The method of claim 65, wherein the subject is a human subject.
67. The method of claim 65, wherein the subject has at least one mutation in the dystrophin gene.
68.-**72**. (canceled)